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Discoveries

Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer drive macro-evolutionary adaptation across viral, bacterial, and eukaryotic genomes by assembling mosaic chromosomes from divergent lineages. Trees cannot describe hybrids. Existing detection tools remain trapped between computationally prohibitive phylogenetic changepoint algorithms that explore combinatorial partition spaces in cubic time and isolated triplet scans that suffer severe false positive rates under realistic mutational rate variation. We introduce `RhizAeon`, an open-source framework that maps multiple sequence alignments into continuous metric spaces, tracking sequence affinities along the chromosome as continuous trajectories without navigating discrete trees or enumerating triplets. By coupling rotary phase embeddings with an unpolarized ancestral root token that absorbs autapomorphic noise, `RhizAeon` deconvolves reticulation through physical force field dynamics, isolating mosaic breakpoints in linear time. Trajectories replace combinatorial tree searches. Across five thousand synthetic alignments incorporating extreme rate heterogeneity, the trajectory engine maintained a zero percent false positive rate, while empirical analyses resolved complex retroviral mosaics, coronaviral chimeras, and multi-megabase bacterial genomes in fractions of a second. Compiled to native Rust binaries and zero-install client-side WebAssembly, `RhizAeon` executes real-time reticulation surveillance within modern web browsers without transmitting sequence data across external networks.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Introduction

Mutation, recombination, and selection drive evolutionary diversification across all domains of life. While point mutations introduce novel genetic variants, homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages in sudden macro-evolutionary leaps (43; 13). Reticulate exchanges reshape viral, bacterial, and eukaryotic genomes by shuffling adaptive alleles and breaking linkage disequilibrium. Mosaicism alters pathogen transmission dynamics. In viral and bacterial epidemics, genetic exchange shifts antigenic presentation and accelerates antimicrobial resistance, as seen across circulating recombinant forms of HIV-1, SARS-CoV-2, and multi-drug resistant bacteria (2; 13). Beyond pathogen surveillance, meiotic crossover and adaptive introgression govern eukaryotic evolution (38; 33). They purge deleterious mutational load, drive ecological speciation, and assemble complex hybrid phenotypes across plant and animal taxa. Resolving physical crossover boundaries remains foundational.

At the sequence level, genetic exchange physically splices distinct evolutionary ancestries into a single continuous molecule. Trees cannot describe hybrids. Instead of hierarchical branches, the true evolutionary genealogy forms a reticulating network or ancestral recombination graph (23; 22). Graphs are notoriously difficult. Reconstructing ancestral recombination graphs requires exploring an astronomical combinatorial space of coalescent and reticulation events, scaling exponentially with sample size and leaving internal graph topologies heavily underdetermined by contemporary sequence data. Yet genetic exchange leaves an unmistakable physical footprint across multiple sequence alignments. Mutations across the chromosome generate phylogenetic incompatibility, where site patterns cannot be mapped onto any common tree without invoking parallel mutations or reversals (6; 36). Most intuitively, the nearest neighbor of a recombinant changes depending on where one looks along the sequence. Within one genomic span, the mosaic chromosome clusters tightly with its primary parental clade, but across a crossover junction, its evolutionary affinity abruptly pivots toward a divergent donor. Tracking spatial handovers in sequence neighborhood reveals the physical boundaries of genetic exchange.

Computational tools designed to track this reticulate footprint remain divided between two incompatible algorithmic paradigms (36; 31). Phylogenetic changepoint algorithms, exemplified by GARD (25), formulate recombination detection as statistical model selection across

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partitioned trees. Partitioning strategies fit independent continuous-time Markov substitution processes to candidate genomic segments, optimizing partition boundaries through likelihood criteria. Because the joint combinatorial search space of partition boundaries and tree topologies scales exponentially with breakpoint count and cubically with cohort size ($\mathcal{O}(L^K \cdot N^3)$), exploring multiple breakpoints across hundreds of genomes requires days of cluster compute. Trees are expensive. Isolated triplet scans take the opposite extreme. Methods such as 3Seq (27) and RDP5 (32) evaluate sequence triplets for clustering of informative polymorphisms. Triplet enumeration scales combinatorially as $\binom{N}{3} \sim \mathcal{O}(N^3)$, becoming computationally intractable on modern genomic cohorts. Evaluating triplets in isolation discards the global phylogenetic reference frame. When substitution rates vary across sites, localized mutational clustering mimics reciprocal genetic exchange. Under realistic Gamma rate heterogeneity, isolated triplet scans mistake hyper-variable patches for mosaicism, inflating false positive rates beyond thirty percent (37; 25). When recombinant segments descend from unsampled ancestral lineages, triplet scans fail entirely, placing arbitrary boundaries inside flat, uninformative genomic intervals.

Overcoming this computational divide requires an analytical framework that replaces discrete tree searches with continuous sequence geometry. Here we introduce `RhizAeon`, named after the botanical and philosophical concept of the *rhizome*—a reticulate, non-hierarchical network of subterranean roots that defies strictly bifurcating trees (11). `RhizAeon` maps multiple sequence alignments into a continuous metric manifold, using rotary position embeddings and substitution priors to parameterize cross-sequence attention along the chromosome. As alignment coordinates advance, attention weights trace continuous trajectory curves that measure the instantaneous affinity of each genome to candidate parental lineages. Trajectories replace trees. To prevent private substitution bursts from mimicking recombination, the architecture anchors an unpolarized ancestral root token at the metric origin, which acts as a sink that absorbs autapomorphic noise and heterotachy. A true crossover produces a localized force imbalance, deflecting the cumulative trajectory from its home corridor toward a donor lineage. By evaluating trajectory inflections through continuous force fields rather than combinatorial tree partitions, `RhizAeon` detects reticulation in linear time, screening genomic cohorts in milliseconds.

We evaluated `RhizAeon` across standardized simulation batteries and real-world viral and bacterial cohorts. Across five thousand synthetic alignments incorporating extreme Gamma rate variation and episodic heterotachy, the manifold engine maintained a zero percent false positive rate while retaining high detection power. Speed does not sacrifice sensitivity. Head-to-head benchmarks against 3Seq confirm identical breakpoint resolution alongside a hundred-fold runtime reduction on moderate cohorts. On large cohorts, computational gains exceed three orders of magnitude. Across empirical retroviral and coronaviral datasets, the continuous framework resolved complex multi-way mosaicism in HIV-1 KAL153 and accurately mapped circulating recombinant breakpoints in SARS-CoV-2 XBB lineages. For multi-megabase bacterial chromosomes, parsimony-informative site compression shrinks memory footprints by over two hundred fold. Whole-genome recombination scans of *Streptococcus pneumoniae* PMEN1 execute in under one second on standard personal laptops (9). The software is distributed as a high-performance native Rust binary and an interactive client-side WebAssembly interface that runs locally within modern web browsers without transmitting sequence data across external networks.

Geometric Foundations and Continuous Trajectory Theory

Architectural Overview and Method Pipeline

Recombination and horizontal gene transfer violate the fundamental tree-like assumption of phylogenetic inference, generating mosaic genomes that inherit disparate segments from divergent ancestral lineages. Historically, inferring these reticulate exchanges has confronted a severe computational dilemma: algorithms either slice alignments into sliding windows to infer local phylogenetic trees across moving intervals ($\mathcal{O}(L \cdot M^3)$) (31; 25), or exhaustively evaluate all possible sequence triplets for non-random clustering of polymorphic substitutions ($\mathcal{O}(M^3 \cdot L)$) (5; 27). Both paradigms exhibit cubic scaling with taxon cohort size (M), rendering real-time surveillance across thousands of viral or bacterial genomes intractable. Furthermore, discrete triplet scans suffer elevated false positive rates under biological rate heterogeneity (heterotachy) and cannot resolve multi-parental mosaics that combine genetic material from three or more lineages simultaneously.

`RhizAeon` resolves this impasse by reformulating recombination detection as continuous dynamical motion across low-dimensional sequence metric manifolds (Figure 1). Discrete tree traversals and triplet enumerations are replaced by a four-stage pipeline. The workflow executes in strictly linear time with alignment length ($\mathcal{O}(L)$) and taxon count ($\mathcal{O}(N \cdot M \cdot L)$).

In Stage 1 (metric manifold embedding), `RhizAeon` collapses quadratic distance scaling by selecting a sparse basis of $N \ll M$ diverse reference landmarks ($N \approx 16$) using sequential Farthest-Point Minimax Sampling in $\mathcal{O}(M \cdot L)$ operations (19). Classical multidimensional scaling (MDS) of the landmark evolutionary distance matrix constructs an invariant d-dimensional coordinate frame ($\mathbb{R}^d$, $d \in [2, 4]$) where orthogonal axes correspond to fundamental phylogenetic bipartitions (Buneman coordinates) (7). Gower out-of-sample projection then interpolates any contemporary isolate into this invariant reference frame, yielding a continuous spatial coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ across all genomic sites $u \in [1, L]$ via a single matrix-vector multiplication without sliding window re-embedding (Figure 1, Stage 1).

In Stage 2 (directed attention and root sink decoupling), `RhizAeon` evaluates directed parentage at each site by transforming alignment columns into a position-specific taxa-attention tensor $A_u(i \to j)$. Tree-RoPE rotary phase embeddings parameterize attention rotations from Buneman metric coordinates, attenuating affinities between phylogenetically divergent lineages without tree traversal (45). A continuous-time Markov substitution prior dynamically discounts invariant positions while elevating shared derived synapomorphies. An unpolarized ancestral root sink token (node zero) anchored at the metric origin absorbs private autapomorphies and heterotachy rate bursts ($A_u(\text{Root})$), preventing lineage-specific rate spikes from leaking into candidate donor affinity. Cumulative summation across the chromosome yields continuous trajectory curves $C(u)$ and differential trajectory flux $Y(u) = C_{\text{Donor}}(u) - C_{\text{Home}}(u)$, measuring net parental balance along the genome (Figure 1, Stage 2).

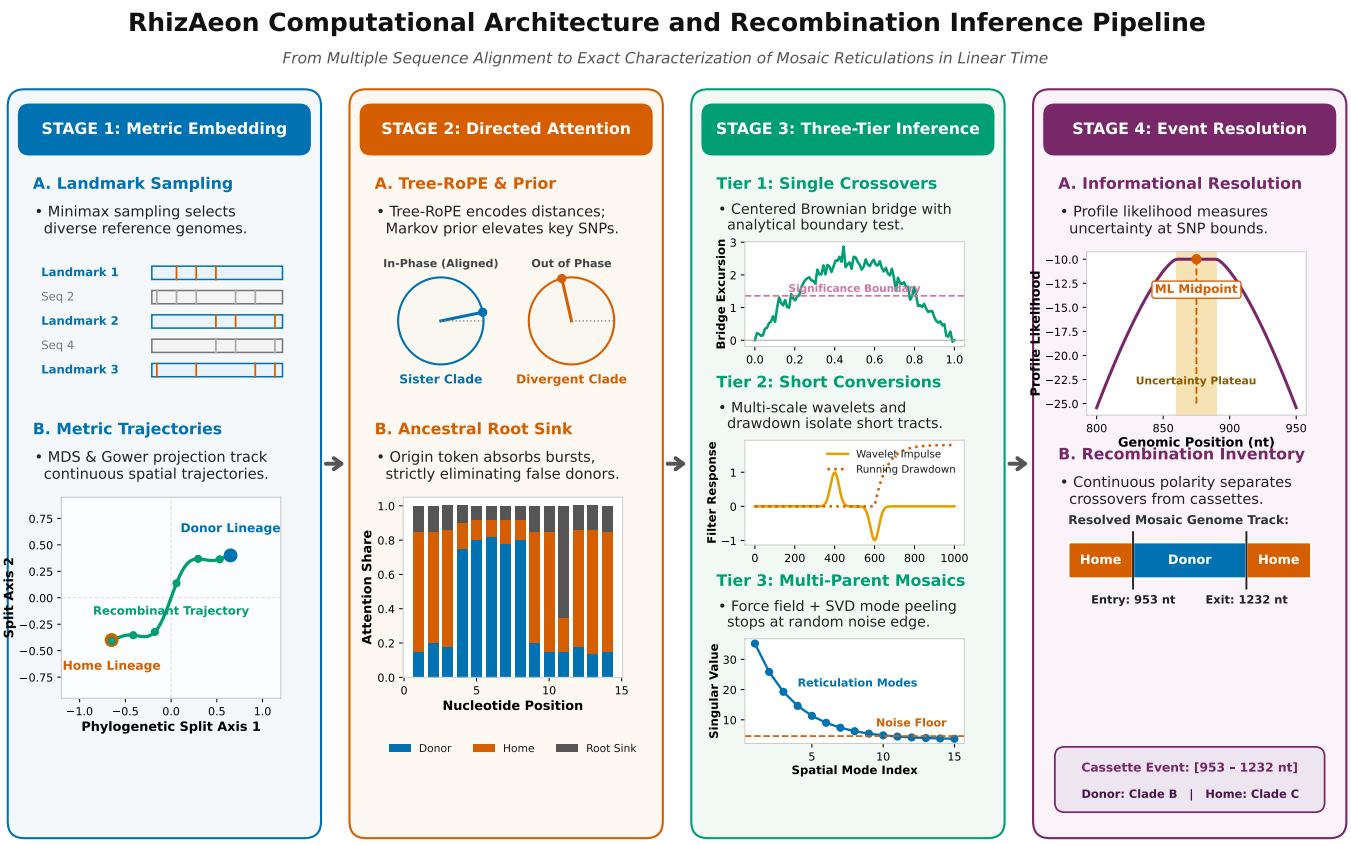


Figure 1 Computational architecture and recombination inference pipeline of `RhizAeon`**.** The four-stage processing workflow transforms raw multiple sequence alignments into exact, model-calibrated characterizations of reticulate evolution in linear time without discrete tree reconstruction or combinatorial triplet enumeration. **Stage 1 (Metric Manifold Embedding):** Sequential Farthest-Point Minimax Sampling extracts $N \ll M$ diverse reference landmarks from M genomes in $\mathcal{O}(M \cdot L)$ operations. Classical multidimensional scaling (MDS) embeds landmark distance matrices into a fixed d-dimensional Euclidean manifold ($\mathbb{R}^d$, $d \in [2, 4]$) where orthogonal axes represent fundamental Buneman phylogenetic bipartitions. Gower out-of-sample projection continuously tracks any isolate as a spatial trajectory $\mathbf{z}_i(u) \in \mathbb{R}^d$ across all positions $u \in [1, L]$ without sliding window re-embedding. **Stage 2 (Directed Attention & Root Sink Decoupling):** Tree-RoPE rotary phase embeddings encode Buneman metric distances into complex attention rotations, coupled with a continuous-time Markov substitution prior that discounts invariant vertical sites and elevates rare shared synapomorphies. An ancestral Root Sink token (node zero) anchored at the metric origin absorbs lineage-specific mutational bursts (heterotachy, APOBEC editing), preventing spurious donor affinity. The resulting position-specific attention tensor $A_u(i \to j)$ integrates into a cumulative differential flux curve $Y(u) = C_{\text{Donor}}(u) - C_{\text{Home}}(u)$. **Stage 3 (Three-Tier Hierarchical Statistical Inference):** Inference addresses the three evolutionary regimes of recombination. First, Tier 1 resolves macroscopic single crossovers via centered Brownian bridges $B(u)$ with closed-form Kolmogorov extreme-value boundaries. Second, Tier 2 isolates short gene conversions (50–300 nt) via a multi-scale bank of zero-mean bipolar dyadic wavelets h_τ convolved continuously across all $u \in [1, L]$ in $\mathcal{O}(L)$ time, paired with running drawdown $\text{DD}(u)$ to capture momentum collapse at exit. Third, Tier 3 resolves multi-parental mosaics via multidimensional restoring force vectors $\mathbf{f}(u) \in \mathbb{R}^d$, singular value decomposition (SVD) mode peeling, and parameter-free halting at the Marchenko-Pastur bulk spectral noise edge (s_{bulk}). **Stage 4 (Profile Polishing & Output Inventory):** Candidate breakpoints are refined using profile maximum likelihood polishing under the fitted nucleotide substitution model, identifying the uninformative gap and biological uncertainty bounds. Continuous spatial mode polarity ratios (p_{mono}) objectively classify reticulation topologies, outputting an authenticated event roster and interactive WebAssembly dashboards.

In Stage 3 (three-tier hierarchical statistical inference), `RhizAeon` addresses the reality that recombination in nature spans diverse evolutionary and structural regimes. Exchange geometries vary. They range from isolated single crossovers that divide an entire chromosome, to localized micro-conversions (50–300 nucleotides) embedded within extensive clonal flanks, to complex multi-parental mosaics that combine genetic material from three or more lineages simultaneously. Because each regime presents distinct mathematical challenges, no single statistical operator can resolve the entire evolutionary spectrum without sacrificing computational speed or detection power. To achieve mathematical optimality and linear-time scaling, the engine structures statistical changepoint inference into a three-tier hierarchy. For isolated macroscopic crossovers (Tier 1), detrending whole-chromosome linear drift yields centered continuous Brownian bridges $B(u)$ with closed-form Kolmogorov boundaries under Donsker's invariance principle (15). For localized micro-conversions and short cassettes (Tier 2), differential trajectory flux is filtered through a multi-scale bank of zero-mean bipolar dyadic wavelets h_τ paired with running drawdown $\text{DD}(u)$ to capture momentum collapse at cassette exit. For complex multi-parental mosaics (Tier 3), the engine models ancestry as continuous vector restoring force fields $\mathbf{f}(u) \in \mathbb{R}^d$,

iteratively peeling reticulation modes via singular value decomposition (SVD) until residual singular values merge into the Marchenko-Pastur bulk spectral noise edge (s_{bulk}) (30) (Figure 1, Stage 3).

In Stage 4 (profile polishing and event inventory), candidate breakpoint boundaries are refined to single-nucleotide resolution using profile maximum likelihood polishing under the fitted nucleotide substitution model. `RhizAeon` explicitly characterizes the uninformative interval between flanking polymorphisms where likelihood forms an exact mathematical plateau, reporting the informational uncertainty interval alongside the ML midpoint. Evaluating continuous spatial mode polarity ratios (p_{mono}) objectively classifies reticulation topologies (monopolar crossovers versus bipolar cassettes). The pipeline adjudicates candidate reticulation graphs to eliminate reciprocal parental cycles, resolves transitive donor chains across co-recombinant isolates, and collapses shared ancestral events into consensus envelopes, producing an authenticated event inventory and interactive client-side WebAssembly dashboards (Figure 1, Stage 4).

In the following subsections, we formulate the mathematical principles, geometric theorems, and statistical properties that underpin each stage of the `RhizAeon` engine.

Sequence Metric Space and Buneman Coordinates

To track recombination across large genomic cohorts ($M \gg 1,000$) without prohibitive all-to-all quadratic distance scaling ($O(M^2L)$), `RhizAeon` extracts a sparse basis of N reference landmarks using sequential Farthest-Point Minimax Sampling (19). The algorithm seeds the initial reference with the sequence exhibiting maximal non-gap coverage, computing pairwise distances to all M isolates in linear time. Each subsequent landmark is selected by maximizing the minimum evolutionary distance to the existing reference set, updating running distance minima in $O(ML)$ operations (see Materials and Methods, Section 5.1). Selecting N landmarks collapses distance computations to strictly $N \times M$—dropping distance evaluations for 10,000 genomes with $N = 16$ from fifty million to 160,000, a 300-fold empirical reduction. From these N landmarks, classical multidimensional scaling of pairwise evolutionary distances constructs an invariant coordinate basis $\mathbf{x}_i \in \mathbb{R}^d$ (47; 20). By Buneman's four-point metric theorem (7), additive evolutionary tree metrics embed into Euclidean space such that orthogonal axes correspond to fundamental phylogenetic bipartitions (splits); we term these embedding dimensions Buneman coordinates. Quartet geometry and spectral decay strictly bound $d \in [2, 4]$: four unrooted lineages $((A, B), (C, D))$ require the vertices of a regular tetrahedron in $\mathbb{R}^3$ to preserve equidistant tree symmetry without planar distortion, while steep eigenvalue decay ensures the leading four dimensions capture 75%–85% of macro-clade variance. In our retroviral benchmark of HIV-1 circulating recombinant forms CRF07_BC and CRF08_BC ($N = 22$, *pol* gene) (46), the primary Buneman axis Z_1 captures the deepest bifurcation (50.8% of variance), cleanly segregating Subtype B from Subtype C (Figure 2A). The secondary axis Z_2 (21.4% of variance) provides the orthogonal freedom needed to resolve recombinant trajectories into open orbital loops, while restricting coordinates to low dimensions acts as an intrinsic low-pass filter that retains clade splits and projects stochastic mutational noise into the null space.

Informational Capacity and the Landmark Identifiability Assumption.

The geometric validity of the Buneman metric frame relies on an implicit information-theoretic condition: the global distance matrix $\mathbf{D}_{\text{glob}}$ must reflect structured, tree-like phylogenetic divergence across at least a core subset of the cohort. Farthest-Point Minimax Sampling reliably extracts landmarks that span these principal cladogenetic divides when clonal backbones exist. For cohorts with $M \leq 64$ genomes, `RhizAeon` retains the full cohort ($N = M$) to prevent downsampling loss. For larger cohorts ($M > 64$), `RhizAeon` couples landmark selection with two-pass clonal backbone re-anchoring, purging candidate recombinant swarms from the landmark pool to eliminate self-masking (see Materials and Methods, Section 5.1). However, if a population has undergone pervasive, fine-scale homologous recombination such that the recombination rate vastly outpaces mutation ($r/\mu \gg 1$, as in panmictic bacterial populations or dense viral quasispecies), every sampled genome is itself a mosaic of mosaics. In this hyper-recombinant regime, global pairwise distances collapse into uniform spatial averages of scrambled ancestries, and the leading MDS eigenvalues fail to separate into discrete bipartition axes, flattening into an isotropic high-dimensional sphere. Under these conditions, discrete parental attribution is fundamentally ill-posed: when unadmixed parental lineages no longer exist in the contemporary sample, resolving discrete "donor" and "home" backbones exceeds the informational capacity of the alignment. Instead of forcing parental assignments onto scrambled backgrounds, `RhizAeon` diagnoses this regime through the eigenvalue decay of the landmark Gram matrix $\mathbf{B}$ and halts at the Marchenko–Pastur random bulk noise edge (s_{bulk}), distinguishing coherent reticulate introgression from panmictic genetic shuffling.

To track sequence affinity along the chromosome, `RhizAeon` projects instantaneous local distance vectors into this fixed reference basis using Gower's analytical out-of-sample interpolation (21) (see Materials and Methods, Section 5.2). At each genomic coordinate $u \in [1, L]$, sequence i presents an evolutionary distance vector to the N reference landmarks, mapping to a continuous spatial coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ via a single matrix-vector multiplication. Anchoring every site to an invariant metric basis establishes a stationary geometric reference frame across the entire chromosome, ensuring coordinates remain globally comparable without axis drift. When applied to the HIV-1 benchmark, the resulting coordinate trajectories reveal the physical dynamics of genetic exchange. Non-recombinant Subtype B and Subtype C genomes remain stationary within their respective corridors across the entire gene, where point mutations generate harmless local flutter around parental medoids without breaching the inter-subtype divide ($Z_1 = 0$, Figure 2B). In contrast, the recombinant CRF07_BC trajectory tracks within Subtype C territory through Protease, vaults across the zero-axis into Subtype B space at nucleotide 953 in Reverse Transcriptase, and plunges back into Subtype C at nucleotide 1,232 (Figure 2C). CRF08_BC enters the Subtype B cassette approximately 60 nucleotides upstream at nucleotide 894. In two-dimensional metric phase space, the two chimeras trace distinct, open orbital loops, resolving competing recombinant structures in linear time without triplet enumeration (Figure 2D).

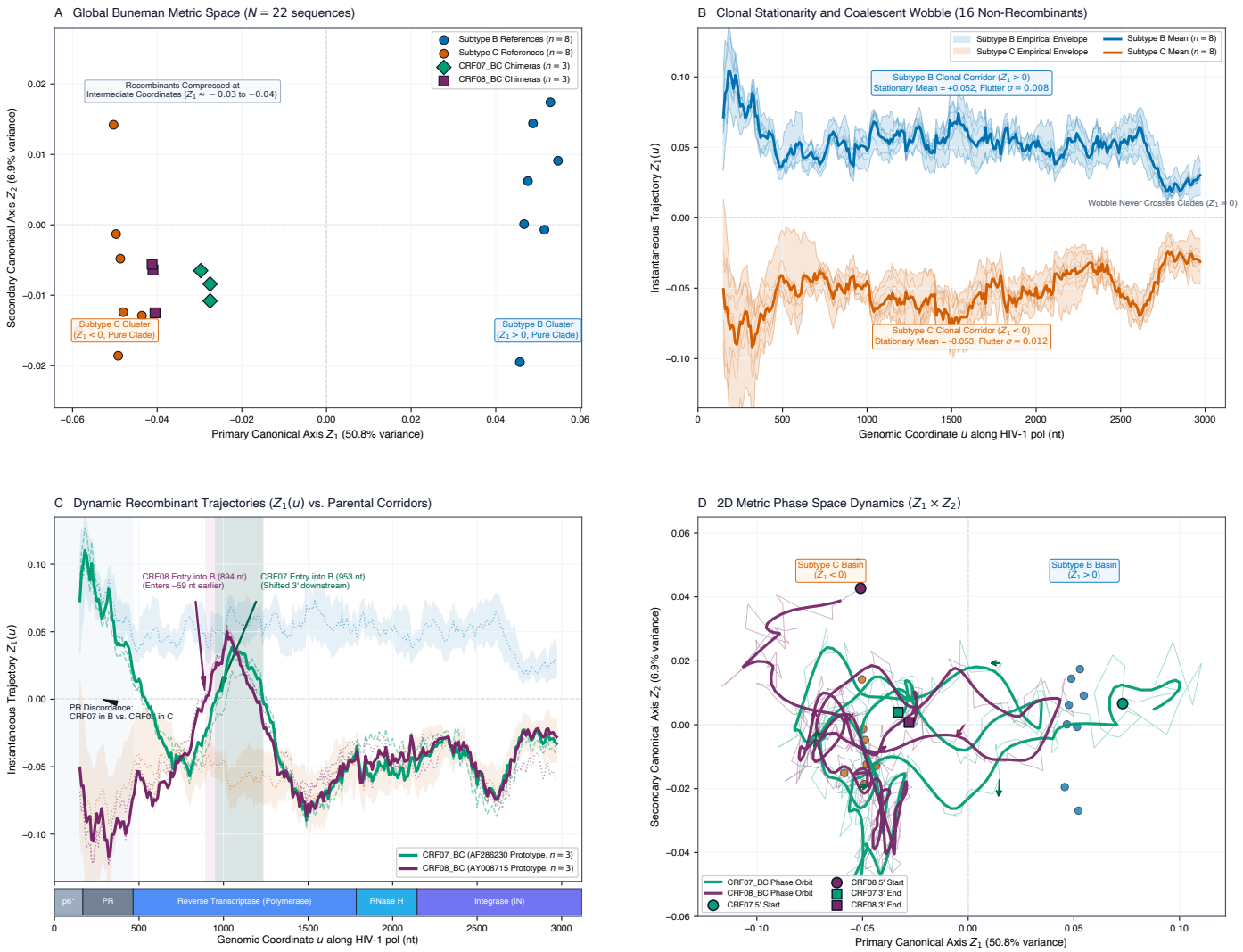


Figure 2 Continuous sequence metric manifolds resolve competing retroviral chimeras in linear time without triplet enumeration. (A) Canonical Sequence Metric Space: Classical multidimensional scaling (MDS) embedding of the invariant Gram matrix derived from global pairwise Tamura-Nei distances ($N = 22$). The primary axis Z_1 (50.8% variance) cleanly separates Subtype B references ($n = 8$, blue circles) from Subtype C references ($n = 8$, vermilion circles). Recombinant forms CRF07_BC ($n = 3$, green diamonds) and CRF08_BC ($n = 3$, purple squares) sit compressed at intermediate coordinates ($Z_1 \approx -0.03$ to -0.04). Whole-genome distances collapse mosaic structures into static averages, obscuring reticulate ancestry. **(B)** Clonal Stationarity and Coalescent Wobble: Instantaneous Gower out-of-sample projected trajectory $Z_1(u)$ ($W = 300$ nt, stride 10 nt) along the 3,120-nucleotide *pol* gene for the 16 non-recombinant references. Both clades remain strictly stationary within their empirical envelopes (Subtype B mean $= +0.052$, flutter $\sigma = 0.008$; Subtype C mean $= -0.053$, flutter $\sigma = 0.012$). Point mutations generate local coalescent flutter, but unreticulated trajectories never cross the inter-clade divide ($Z_1 = 0$). **(C)** Dynamic Recombinant Trajectories and Resolved Breakpoints: Continuous $Z_1(u)$ trajectories for CRF07_BC (prototype AF286230, green) and CRF08_BC (prototype AY008715, purple) plotted against shaded parental reference corridors. Aligned beneath is the HIV-1 *pol* functional domain track ($p6^*$, PR, RT, RNase H, IN). In Protease (PR), CRF07 tracks within Subtype B while CRF08 tracks within Subtype C. In Reverse Transcriptase, both lineages enter a Subtype B cassette. CRF08 crosses the zero-axis at nucleotide 894, entering approximately 60 nucleotides upstream of CRF07 (nucleotide 953). Both exit back into Subtype C at nucleotide $\sim 1,230$ and remain in Subtype C through Integrase. **(D)** Metric Phase Space Dynamics: Continuous two-dimensional dynamical orbits in $(Z_1 \times Z_2)$ space. Non-recombinant parental lineages form stationary point basins at $(+0.05, 0.00)$ and $(-0.05, 0.04)$. Recombinant lineages trace smooth, open orbital loops that sweep across the inter-clade divide from $5'$ start positions (circles) to $3'$ terminal positions (squares) with tangent flow arrows. Continuous manifold geometry resolves distinct mosaic architectures in $O(N)$ time without combinatorial explosion.

Rotary Phase Embeddings and Markov Substitution Priors

While continuous metric trajectories reveal where a sequence resides in macroscopic space, identifying recombination requires determining which specific lineages contributed genetic material to a chimeric genome. Recombination is fundamentally a directed process: at each genomic position, a query isolate inherits from either its ancestral home background or an introgressing donor. Standard phylogenetic approaches evaluate this inheritance by slicing alignments into sliding windows and reconstructing discrete local trees—a combinatorial process that scales poorly across large cohorts. Cross-attention mechanisms offer a continuous alternative: instead of rebuilding trees, the model evaluates directed affinity between a query genome and candidate reference taxa at each site. Standard transformer attention, however, is phylogenetically blind;

it treats all candidate taxa symmetrically, unable to differentiate whether sequence identity reflects expected clonal inheritance from a close sister isolate or genuine reticulate exchange with a divergent lineage.

To give attention a continuous evolutionary coordinate system, `RhizAeon` anchors taxa attention in the Buneman metric space established in Section 2.2. In natural language processing, Rotary Position Embeddings (RoPE) rotate token representations across two-dimensional planes by angles proportional to word index along a one-dimensional sentence (45). `RhizAeon` generalizes rotary embeddings to branching evolutionary space via Tree-RoPE (see Materials and Methods, Section 5.3). Each taxon's d-dimensional Buneman coordinate $\mathbf{x}_i \in \mathbb{R}^d$ parameterizes a set of rotation angles across orthogonal planes. When two sequences share close evolutionary ancestry, their phase angles remain aligned; as phylogenetic distance in Buneman space increases, their representations rotate out of phase, naturally attenuating pairwise attention weights without traversing discrete tree topologies (see Materials and Methods, Section 5.3).

Yet geometric distance alone cannot determine whether sequence identity reflects recent reticulate exchange or baseline neutral conservation. Two divergent lineages frequently share identical nucleotides at highly conserved or invariant sites simply because those positions have not undergone mutation. A shared base at an invariant position provides no evidence of recombination, whereas a shared derived polymorphism that contradicts the species tree provides decisive signal. To calibrate attention against neutral drift, `RhizAeon` augments the Tree-RoPE attention logits with a continuous-time Markov substitution prior (see Materials and Methods, Section 5.3). The substitution prior dynamically weights sequence identity by evolutionary likelihood: invariant sites expected under neutral vertical evolution are discounted, while rare, shared mutations that require horizontal transfer to explain observed identity are elevated. Combining continuous Buneman metric geometry with Markov substitution priors transforms the alignment into a directed, position-specific taxa-attention tensor $A_u(i \to j)$ at every nucleotide along the chromosome (Figure 3A). At each position, directed attention evaluates how strongly the query sequence aligns with candidate donor and parental home clades, alongside an ancestral root sink token (Figure 3A,C).

The Ancestral Root Sink Token

Standard attention architectures enforce a closed-world normalization where attention weights across all candidate sequences must sum strictly to unity. In sequence analysis, this normalization creates a vulnerability to lineage-specific rate variation. When an isolated lineage experiences an episodic burst of private substitutions—whether through host-mediated editing such as APOBEC cytidine deamination, error-prone replication, or lineage-specific rate acceleration (heterotachy)—these private variants have no homologous counterpart among sampled contemporary genomes (37). Because standard softmax normalization forces all attention mass to be distributed exclusively among contemporary leaves, a query genome with private mutations is compelled to allocate attention to whichever leaf happens to present the least mismatch, misinterpreting mutational noise as reticulate exchange.

To mitigate this closed-world vulnerability, `RhizAeon` incorporates an ancestral root sink token, designated as node zero and anchored at the origin of the metric coordinate space ($\mathbf{0} \in \mathbb{R}^d$) with a baseline neutral substitution prior. Total attention probability is strictly conserved across all N contemporary reference leaves plus this neutral root node ($\sum_{j=1}^{N} A_u(i \to j) + A_u(i \to \text{root}) = 1.0$; see Materials and Methods, Section 5.4, Equation (23)). Because the root node resides at the metric origin without clade-specific phase rotation ($\boldsymbol{\theta}_{\text{root}} = \mathbf{0}$) and incurs no tree-discordance penalty under the substitution prior, private autapomorphies naturally route their attention mass into node zero without inflating affinity to a sister taxon (Figure 3B). Conversely, when genuine genetic exchange occurs, attention shifts coherently toward the introgressing donor clade while the root sink remains quiescent (Figure 3B,C). Providing an explicit destination for non-homologous private substitutions isolates lineage-specific rate bursts from shared phylogenetic signal, substantially reducing false positive detections under extreme heterotachy.

Brownian Bridges and Invariance Principles for Macroscopic Crossovers (Tier 1)

For isolated macroscopic crossovers (Tier 1 of the inference hierarchy; Section 2.1), inference operates on continuous cumulative trajectories of differential attention $Y(u) \in \mathbb{R}$. At individual polymorphic sites, instantaneous attention weights fluctuate with local mutational noise. To extract macroscopic evolutionary structure, `RhizAeon` integrates attention across the chromosome into continuous cumulative trajectory curves (see Materials and Methods, Section 5.5). The reference lineage accumulating the greatest integrated affinity across the entire sequence defines the query's home clade, while the difference between candidate donor and home affinity traces a continuous differential trajectory, $Y(u) = C_{\text{Donor}}(u) - C_{\text{Home}}(u)$, measuring net phylogenetic balance up to coordinate u.

Under clonal evolution (the null hypothesis H_0), the query genome evolves strictly within its parental background. Because home attention consistently exceeds donor attention at nearly all positions, $Y(u)$ exhibits a steady, monotonic negative drift throughout the chromosome. Horizontal transfer breaks this background drift: within an introgressed donor segment, sequence affinities invert. Attention to the donor surges while home attention collapses, driving $Y(u)$ upward into a prominent excursion that crests at the exchange boundary before descending back into negative drift once home sequence resumes (Figure 3D). Single crossovers trace asymmetric global arches spanning half the chromosome, whereas intercalary cassette insertions trace localized pulses that return cleanly to baseline on both flanks.

To formulate an exact statistical changepoint test on this trajectory, `RhizAeon` transforms $Y(u)$ into a centered Brownian bridge by subtracting the chromosome-wide linear drift (see Materials and Methods, Section 5.5). Because sequence alignments span a finite genomic window from coordinate 0 to L, both the start and end of the cumulative trajectory are fixed by observation. Subtracting the overall linear slope clamps both endpoints to zero, so that under clonal evolution, the detrended path resembles an elastic string pulled taut at both ends, fluctuating stably around zero. When recombination occurs, the localized surge in donor affinity violates the uniform-rate assumption, forcing the bridge to bow upward into a coherent arch whose peak precisely identifies the recombination boundary.

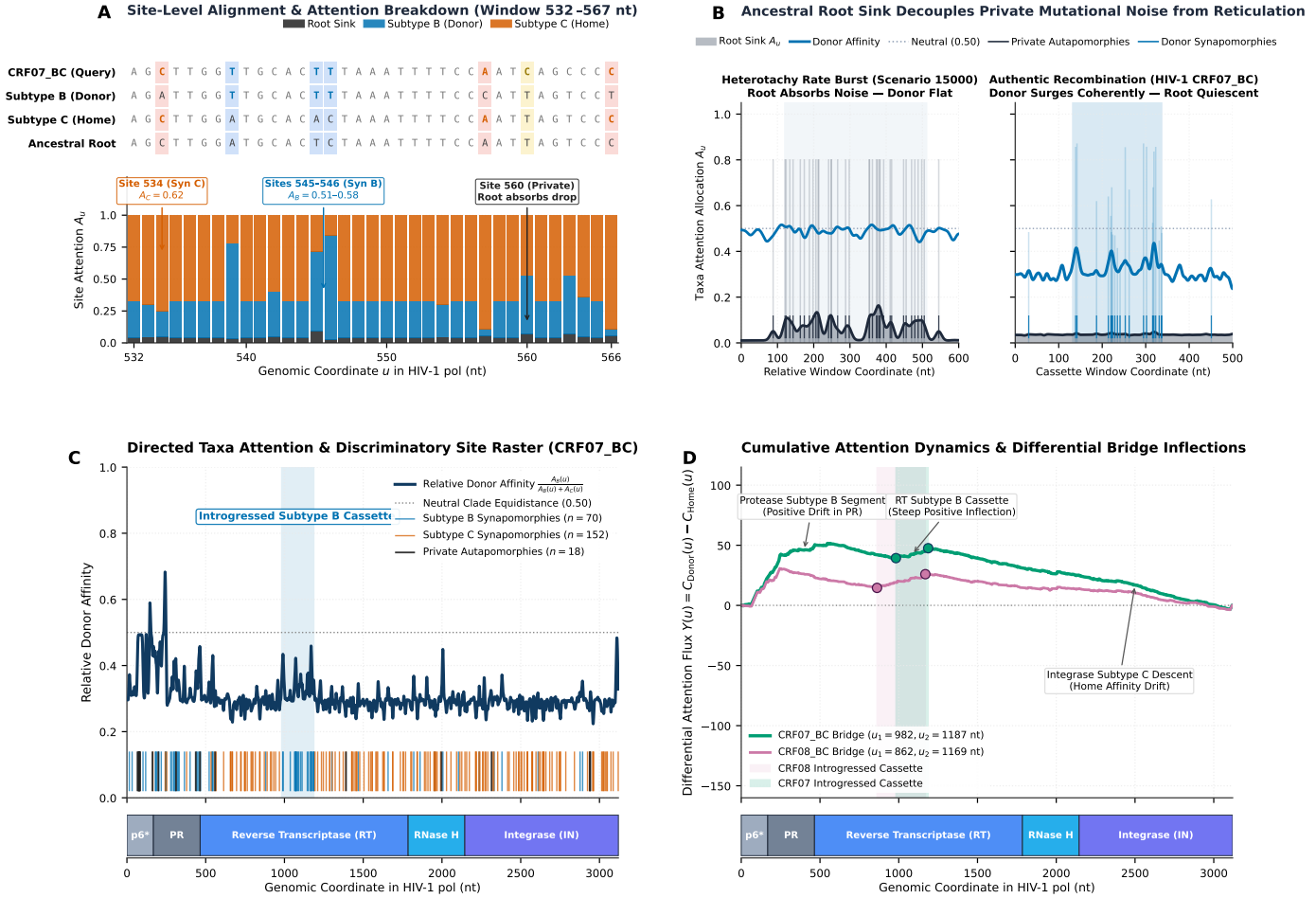


Figure 3 Site-level directed attention, ancestral root sink decoupling, and cumulative Brownian bridge dynamics along recombinant genomes. (A) Site-Level Alignment and Position-Specific Attention Breakdown: Microscopic view across a 35-nucleotide window ($u = 532$–567 nt) in HIV-1 *pol*. Query sequence CRF07_BC is aligned against candidate donor Subtype B, parental home Subtype C, and the ancestral root consensus. The stacked bar chart displays instantaneous directed attention $[A_u(\text{Root}), A_u(\text{Donor}), A_u(\text{Home})]$ summing strictly to 1.0 at each nucleotide. Shared invariant sites allocate baseline attention to home Subtype C. Donor synapomorphies (sites 545–546) elevate Subtype B attention ($A_B = 0.51$–0.58). Private autapomorphies (site 560) route into the ancestral Root Sink ($A_{\text{Root}} = 0.52$). Routing private autapomorphies to node zero prevents spurious leakage into candidate donor affinity. **(B)** Ancestral Root Sink Decouples Private Mutational Noise from Reticulation. Under an episodic private mutational burst (Darren heterotachy trap, benchmark scenario 15000, $4.9\times$ rate acceleration across nucleotides $1,020$–$1,412$), private autapomorphies ($n = 36$) spike Root Sink attention $A_u(\text{Root})$ (grey impulse bars and smoothed curve). Relative donor affinity (blue curve) remains flat at neutral baseline (0.50, dotted line). Under authentic genetic recombination (HIV-1 CRF07_BC RT cassette, nucleotides 982–$1,187$), donor affinity surges coherently across the 205-nucleotide tract driven by donor synapomorphies ($n = 24$). Root Sink attention remains quiescent (≈ 0.038). **(C)** Directed Taxa Attention and Discriminatory Site Raster: Instantaneous relative donor affinity $A_B(u)/(A_B(u) + A_C(u))$ for CRF07_BC along the full 3,120-nucleotide *pol* gene, plotted above the discriminatory site raster (Subtype B synapomorphies, blue ticks; Subtype C synapomorphies, vermilion ticks; private autapomorphies, black ticks) and aligned protein domain track. Neutral expectation sits at 0.50 (dotted line). In the introgressed Subtype B cassette (nucleotides 982–1,187, light blue shading), donor synapomorphies drive a sustained surge in relative affinity. **(D)** Cumulative Attention Dynamics and Differential Bridge Inflections: Continuous differential attention flux $Y(u) = C_{\text{Donor}}(u) - C_{\text{Home}}(u)$ for CRF07_BC (green curve, breakpoints at 982 and $1,187$ nt) and CRF08_BC (purple curve, breakpoints at 862 and $1,169$ nt). In Protease, shared B-like mutations impart positive drift; across the RT cassette, donor affinity surges, producing a steep positive inflection; upon exiting into Integrase, negative drift resumes as home Subtype C affinity steadily dominates.

To evaluate whether an excursion is statistically significant without relying on computationally demanding permutations or simulations, `RhizAeon` invokes Donsker's invariance principle (15). As the functional extension of the central limit theorem from individual sums to continuous paths, Donsker's theorem establishes that as sequence length grows, the standardized cumulative trajectory of any stationary random walk with finite-variance increments converges in distribution to a standard continuous Brownian bridge. This limiting behavior is invariant to whether individual site contributions arise from discrete matches or continuous attention weights, allowing the maximum excursion along the chromosome to be evaluated against the analytical two-sided Kolmogorov distribution. Correcting this extreme-value boundary across all candidate parental combinations yields exact, closed-form family-wise error rate control in microseconds, bypassing both empirical bootstrap permutations and computationally demanding coalescent simulations.

Continuous trajectory tracking connects with and generalizes classical non-parametric recombination detection algorithms such as 3SEQ (5; 27). In 3SEQ, mosaic structure in sequence triplets (C, P_1, P_2) is inferred by tracking partial-sum random walks across filtered polymorphic sites, where positions matching parent P_1 increment the walk by $+1$ and positions matching P_2 decrement by -1. `RhizAeon` generalizes this discrete walk in three fundamental respects: First, instead of restricting steps to discrete binary transitions (± 1) on pre-filtered biallelic columns, `RhizAeon` computes continuous, model-calibrated attention increments simultaneously across all M reference lineages in $\mathcal{O}(N \cdot M \cdot U)$ time, eliminating the combinatorial $\mathcal{O}(N^3)$ triplet bottleneck. Second, while discrete walks discard invariant positions and uninformative sites, the continuous attention framework incorporates all genomic sites: invariant sites allocate baseline weight according to the Markov substitution prior, synapomorphic positions drive strong directional drift, and private autapomorphies pool harmlessly into the root sink. Third, by mapping cumulative attention into continuous functional space, Donsker's invariance principle grounds changepoint significance in continuous extreme-value theory, replacing discrete random-walk tables and permutation tests with exact analytical Kolmogorov boundaries.

Multi-Scale Dyadic Filters and Running Drawdown for Localized Conversions (Tier 2)

While the centered Brownian bridge of Tier 1 provides an exact analytical test for macroscopic crossovers spanning large fractions of the genome, it confronts a severe mathematical limitation when applied to short conversion tracts (50–300 nucleotides) embedded within extensive non-recombinant chromosomes. This limitation stems from the phenomenon of *whole-chromosome signal dilution*. In bacterial transformation, viral antigenic cassettes, and eukaryotic gene conversion, genetic exchanges frequently take the form of localized micro-conversions. Under the functional central limit theorem (Donsker's theorem), the cumulative Brownian bridge excursion is normalized globally by the whole-chromosome scale $\sigma\sqrt{L}$. Across an extensive non-recombinant background, the localized affinity surge of a short conversion tract is heavily diluted: the excursion is compressed into a narrow spike whose maximum deviation fails to breach the chromosome-wide extreme-value boundary (Figure 4B, top). Consequently, an authentic micro-conversion with intense local divergence can remain statistically undetectable under global bridge tests.

To resolve localized micro-conversions without whole-chromosome signal dilution (Tier 2), `RhizAeon` remains within the one-dimensional scalar domain of directed attention, but shifts from cumulative trajectory integration $Y(u)$ to instantaneous differential trajectory flux $\Delta Y(u) = Y(u) - Y(u-1)$ (derived from the site-level attention differences established in Section 2.5). `RhizAeon` processes this flux through a multi-scale bank of zero-mean bipolar dyadic wavelets h_τ spanning scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nucleotides (see Materials and Methods, Section 5.6). The bipolar wavelet template consists of an inverted block followed by an upright block, summing strictly to zero ($\int h_\tau(s)ds = 0$). Because the genomic location of an introgressed tract is unknown *a priori*, this filtering is not applied selectively to pre-defined candidate windows. Instead, the continuous convolution $(\Delta Y * h_\tau)(u) = \sum_s \Delta Y(u-s)h_\tau(s)$ is evaluated continuously across *all* genomic positions $u \in [1, L]$ as a linear-time array operation ($\mathcal{O}(L)$ operations per scale). Across non-recombinant background sequence, neutral mutational fluctuations cancel out under the zero-mean kernel, keeping the filter response near zero. However, wherever an authentic conversion tract matching scale τ resides along the chromosome, the opposing lobes align with the entry and exit boundaries, causing the convolution response to resonate into a sharp, prominent impulse peak (Figure 4B, middle). The position of the maximum resonant peak across dyadic scales automatically localizes the candidate conversion tract without requiring prior boundary knowledge.

Concurrently, `RhizAeon` performs a continuous linear prefix scan across the entire chromosome to track the running high-water mark $M(u) = \max_{v \le u} Y(v)$ and running drawdown $\mathrm{DD}(u) = M(u) - Y(u)$ (see Materials and Methods, Section 5.6). Originally developed in quantitative finance to measure the decline of an asset price from its historical peak, running drawdown in evolutionary sequence analysis captures the sudden collapse in donor momentum that occurs the exact instant the introgressed cassette terminates and home sequence resumes, pinpointing the exit boundary u_2 with single-nucleotide precision (Figure 4B, bottom).

Dynamical Metric Field Deconvolution of Multi-Parental Mosaics (Tier 3)

While Tiers 1 and 2 resolve pairwise recombination events by projecting sequence divergence onto a one-dimensional scalar axis of directed attention ($Y(u)$ and $\Delta Y(u)$), this scalar reduction fundamentally breaks down when dealing with complex multi-parental mosaics. In both natural viral pandemics (such as circulating recombinant forms of HIV-1) and hyper-recombinant bacterial populations, chimeric genomes repeatedly exchange genetic material with three, four, or more ancestral lineages simultaneously. Attempting to analyze these complex reticulations via one-dimensional scalar trajectories fails due to *destructive multi-parental interference*.

Consider a chimeric genome R that inherits mosaic segments from three divergent ancestral clades, A, B, and C. In sequence space, these three lineages define a triangle (or higher-dimensional simplex), not a line. If we project sequence divergence onto the one-dimensional scalar axis connecting Clade A (Home) and Clade B (Donor 1), Clade C (Donor 2) does not align with this axis; rather, its projection onto the A–B chord falls at an intermediate, distorted coordinate. Consequently, when genome R transitions from segment B into segment C, the one-dimensional trajectory does not produce a clean step; instead, the vector components of phylogenetic divergence along distinct parental directions interfere destructively. Excursions toward different donors distort or cancel one another, obscuring authentic exchange boundaries and generating spurious intermediate peaks. The traditional phylogenetic solution to multi-parental mosaicism is to exhaustively evaluate all possible sequence triplets or quartets (R, P_i, P_j), repeating pairwise scans across all $\binom{N}{2}$ candidate parental combinations for every sequence in the cohort. However, this combinatorial iteration scales cubically as $\mathcal{O}(N^3 \cdot L)$, creating an insurmountable computational bottleneck that prevents real-time surveillance across large genomic cohorts.

`RhizAeon` resolves this impasse by leaving one-dimensional scalar projections behind and returning to the full multidimensional sequence metric manifold $\mathbb{R}^d$ established in Section 2.2 (Figure 4A). In this continuous geometric space, every reference clade occupies its own distinct, invariant spatial basin centered on its clade centroid $\mathbf{z}_{\mathrm{clade}} \in \mathbb{R}^d$. We do not need to pre-select a donor or collapse phylogenetic divergence

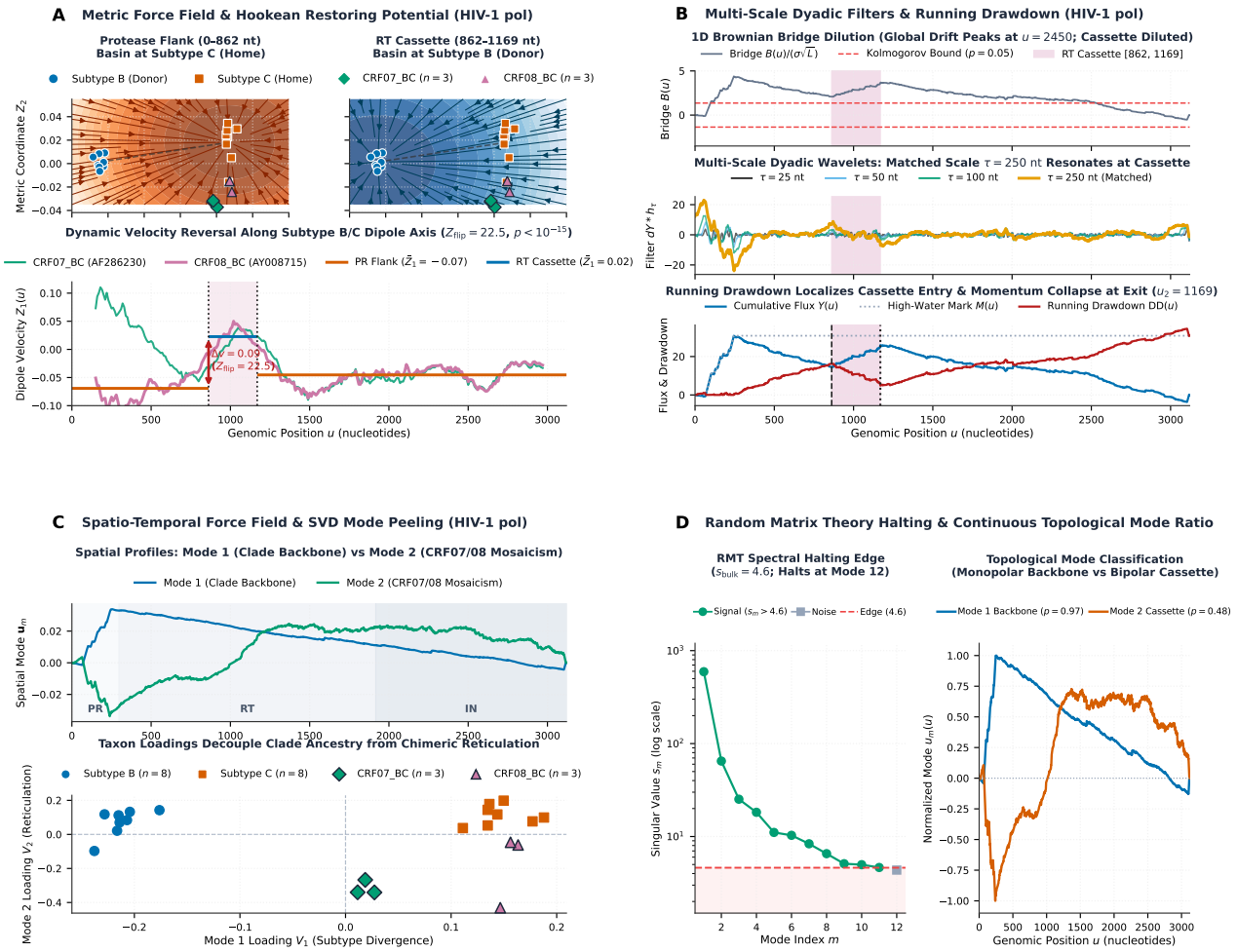


Figure 4 Physical force field deconvolution, multi-scale matched filters, and random matrix theory resolve complex reticulations. (A) Metric Force Field & Hookean Restoring Potential: Vector potential landscapes and directional velocity reversal across the authentic 3,120-nucleotide HIV-1 *pol* benchmark ($N = 22$; 8 Subtype B, 8 Subtype C, 3 CRF07_BC, 3 CRF08_BC). In Segment 1 (Protease flank, 0–862 nt), the potential well $U(\mathbf{z})$ centers on Subtype C (Home, vermilion square), exerting continuous Hookean restoring forces that retain chimeras CRF07_BC (green diamonds) and CRF08_BC (purple triangles) within the Subtype C basin. In Segment 2 (Reverse Transcriptase cassette, 862–1169 nt), the basin abruptly shifts toward Subtype B (Donor, navy blue circle). Below, projected trajectory velocity along the Subtype B/C dipole axis exhibits an instantaneous directional velocity reversal for CRF08_BC isolate AY008715 ($\Delta v = 0.092$, $Z_{\text{flip}} = 22.5$, $p < 10^{-15}$). Coordinates shift from Protease flank $\bar{Z}_1 = -0.07$ to RT cassette $\bar{Z}_1 = +0.02$. **(B)** Multi-Scale Dyadic Filters & Running Drawdown: Resolution of the 307-nucleotide RT cassette ($[862, 1169]$ nt) in HIV-1 *pol*. Under whole-gene normalization, the 1D Brownian bridge undergoes signal dilution: cumulative bridge excursion peaks downstream at $u = 2450$ nt, diluting the local cassette signal ($B(u)/(\sigma\sqrt{L})$, top). Filtering trajectory flux through a dyadic wavelet bank (middle) overcomes dilution: the matched dyadic scale $\tau = 250$ nt (amber) resonates at the cassette boundaries, producing sharp impulses at entry ($u_1 = 862$) and exit ($u_2 = 1169$). Tracking cumulative flux $Y(u)$, high-water mark $M(u)$, and running drawdown $\text{DD}(u) = M(u) - Y(u)$ (bottom) localizes cassette entry and pinpoints the abrupt momentum collapse at cassette exit ($u_2 = 1169$ nt). **(C)** Spatio-Temporal Force Field & SVD Mode Peeling: Deconvolution of multi-parental mosaicism across HIV-1 *pol*. Spatial profiles (top) separate macro-clade Subtype B/C backbone divergence ($\mathbf{u}_1$) from localized chimeric reticulations across Protease (PR), Reverse Transcriptase (RT), and Integrase (IN) domains ($\mathbf{u}_2$). Taxon loadings (bottom) decouple ancestry from reticulation. Pure references stay near baseline on Mode 2 ($V_2 \approx 0$). Chimeras CRF07_BC (green diamonds, $V_2 \approx -0.3$) and CRF08_BC (purple triangles, $V_2 < -0.05$) project strongly into the reticulation subspace. Rank-1 mode peeling deflates each reticulation event without combinatorial searches. **(D)** Random Matrix Theory Halting & Continuous Topological Mode Ratio: Parameter-free stopping and topological mode classification. Singular value spectrum on HIV-1 *pol* (left) plotted against the Marchenko-Pastur bulk spectral noise edge ($s_{\text{bulk}} = 4.62$, red dashed line). Modes 1–11 reside above s_{bulk} (SNR up to 128.5), while higher modes merge into the noise floor, terminating mode peeling objectively at Mode 12. Right: continuous spatial mode profiles classify reticulation topologies: the leading backbone mode forms a monopolar global arch ($p_{\text{mono}} = 0.97$, blue), whereas the reticulation mode forms a localized bipolar cassette pulse returning to baseline ($p_{\text{mono}} = 0.48$, vermilion).

onto a single pairwise axis. Instead, at every genomic coordinate $u \in [1, L]$, the query sequence possesses an instantaneous continuous spatial coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ determined by Gower out-of-sample projection.

To connect this continuous geometry to evolutionary theory, we formalize phylogenetic ancestry as a continuous vector field in metric space. Under vertical clonal evolution, a genome evolves within its ancestral parental background (its "home" lineage). Neutral point mutations cause minor stochastic fluctuations around the parental clade centroid $\mathbf{z}_{\text{home}}$, but the overall sequence identity exerts a continuous stabilizing pull that binds the genome to its ancestral basin. We formalize this evolutionary pull dynamically as an evolutionary restoring force vector pointing

toward the parental centroid, governed by metric stiffness λ (see Materials and Methods, Section 5.7, Equation (38)). In biological terms, $\mathbf{f}(u)$ quantifies continuous phylogenetic affinity to the ancestral home clade. When the sequence matches home, restoring force is near zero. When divergent mutations accumulate, the vector captures the directional displacement away from the parental baseline.

When horizontal genetic exchange occurs, the query sequence inherits genetic material from an introgressing donor clade (centroid $\mathbf{z}_{\text{donor}}$). Its trajectory is pulled across the metric manifold toward the donor basin. Across a recombination breakpoint, this multidimensional force vector undergoes an abrupt directional inversion. The sequence transitions between ancestral clades in $\mathbb{R}^d$, producing an instantaneous directional velocity reversal along the home-donor dipole axis (Figure 4A). To evaluate the statistical significance of this transition without heuristic threshold tuning, `RhizAeon` standardizes the velocity step by the flanking empirical standard error. The resulting dimensionless test statistic, Z_{flip}, provides scale-invariant detection across both deeply diverged viral subtypes and closely related bacterial isolates without manual parameter adjustments (see Materials and Methods, Section 5.7, Equation (40)).

In alignments containing multiple recombinant lineages or multi-way chimeras, stacking these metric force vectors across all N taxa and L positions forms a space-by-position phylogenetic affinity matrix $\mathbf{F} \in \mathbb{R}^{N \times L}$. Instead of iterating over combinatorial triplets, `RhizAeon` applies singular value decomposition to decouple the matrix into orthogonal spatial profiles and taxon loadings (Figure 4C; see Materials and Methods, Section 5.7, Equation (41)). As demonstrated on the authentic HIV-1 *pol* cohort ($N = 22$; Figure 4C), the leading mode captures the primary Subtype B versus Subtype C macro-clade divergence backbone of the underlying phylogenetic tree. Subsequent modes decouple distinct reticulation chimeras: pure reference strains remain near zero on Mode 2, whereas circulating recombinant forms project strongly into the reticulation subspace, separating their mosaic architectures across the Protease, Reverse Transcriptase, and Integrase domains (Figure 4C). Rank-1 mode peeling iteratively deflates each reticulation event from the residual force matrix, resolving multi-parental mosaics without combinatorial searches.

To halt peeling objectively without arbitrary energy thresholds or tuning parameters, `RhizAeon` grounds termination in random matrix theory. Under the null hypothesis of unreticulated coalescent drift, the singular values of an unreticulated alignment follow the Marchenko-Pastur distribution (30). Peeling terminates automatically when residual singular values reach the theoretical bulk spectral noise edge s_{bulk} (Figure 4D; see Materials and Methods, Section 5.7, Equation (42)). Singular values residing above s_{bulk} represent authentic physical reticulations that cross the spectral bulk edge (with empirical signal-to-noise ratios up to 128.5 on HIV-1 *pol*), while sub-threshold modes merge into neutral coalescent noise, guaranteeing rigorous false-positive control across unreticulated alignments.

Finally, continuous field geometry provides an invariant topological classifier that separates single crossovers from intercalary cassette swaps without heuristic length thresholds. Evaluating the spatial mode polarity ratio $p_{\text{mono}} = |\sum_u u_m(u)| / \sum_u |u_m(u)|$ on decomposed spatial profiles $\mathbf{u}_m(u)$ reveals two distinct topological regimes (see Materials and Methods, Section 5.6): Clade-divergence tree backbones and macroscopic single crossovers trace monopolar global arches that maintain uniform sign across the chromosome, yielding $p_{\text{mono}} \approx 1.0$ (empirical value 0.97 on HIV-1 *pol*; Figure 4D, right). In contrast, intercalary cassette swaps form localized bipolar pulses that enter from baseline, reverse sign at internal breakpoints, and return cleanly to baseline on flanking segments, characterized by $p_{\text{mono}} \leq 0.50$ (empirical value 0.48 on HIV-1 *pol*; Figure 4D, right). The continuous polarity ratio provides an objective, parameter-free criterion to classify reticulation topologies from field geometry.

Informational Resolution Limits and Profile Polishing

Physical trajectory deconvolution localizes the region of exchange, but how precisely can a breakpoint coordinate be determined? The answer is bounded by sequence variation itself. Between adjacent informative polymorphisms that distinguish the two parents, multiple sequence alignments contain zero phylogenetic signal. Within these uninformative intervals, the true recombination crossover could have occurred at any nucleotide with equal probability.

Profile maximum likelihood polishing refines candidate boundaries under the fitted nucleotide substitution model (see Materials and Methods, Section 5.8). Across the uninformative gap, the profile log-likelihood forms an exact mathematical plateau. `RhizAeon` explicitly reports this informational plateau alongside the likelihood midpoint, providing users with the rigorous biological uncertainty interval dictated by alignment information content.

Reticulation Graph Adjudication and Ancestral Event Grouping

Detecting candidate breakpoints isolates regions of reticulate exchange, but translating raw changepoints into an authenticated evolutionary history requires resolving who exchanged genetic material with whom. In pairwise and triplet sequence comparisons, symmetry breeds confusion. Whenever a recombinant isolate inherits a genomic tract from a donor into an ancestral home background, reciprocal scans frequently flag the donor as recombinant between the true child and the home lineage. Reciprocal misattributions generate mutual two-cycles in candidate reticulation graphs, where parent and child cite each other as mosaic recombinants. Traditional triplet scans frequently emit duplicate inverted events under this symmetry, distorting downstream epidemiological tracking.

`RhizAeon` resolves parental orientation by adjudicating candidate reticulation graphs across homologous genomic intervals (see Materials and Methods, Section 5.9). The algorithm evaluates external donor authenticity. If one candidate identifies an independent external lineage while its partner merely cites the candidate itself, the algorithm prunes the partner as an unreticulated parent. When candidates cite each other reciprocally, the engine evaluates physical home backbone disruption. A genuine recombinant child departs from its parental home basin during the conversion tract, producing a sharp divergence surge against its home background. A stationary clonal parent remains anchored within its home clade throughout the chromosome. Evaluating total transition contrast and mosaic reconstruction residuals breaks symmetry decisively, orienting recombinant children against parental donors without phylogenetic tree searches.

Table 1 Formal benchmark scoring framework across reticulation dimensions. Performance is evaluated across four primary biological dimensions to prevent methodological gaming: alignment-level classification under non-recombinant nulls, breakpoint cardinality, spatial boundary accuracy under sequence information bounds, directed triad attribution, ghost donor introgression, and ancestral event consolidation.

Dimension	*Biological Question*	*Formal Benchmark Criterion*	*Mathematical Scoring Metric*
Alignment-Level Detection	Does the alignment contain authentic horizontal genetic exchange?	Complete rejection of non-recombinant clonal nulls under heterotachy, APOBEC bursts, Darwinian selection, and molecular clock evolution. High sensitivity on true mosaics.	Binary classification with false positive rate $\mathrm{FPR} = \mathrm{FP}/N_{\mathrm{null}} = 0.00\%$ at $\alpha = 0.05$ and sensitivity $\mathrm{TPR} = \mathrm{TP}/N_{\mathrm{rec}}$.
Breakpoint Cardinality	How many recombination breakpoints occurred along the chromosome?	Strict agreement between predicted and true changepoint counts. Over-splitting and spurious fragmentation are penalized as false alarms.	Bijective bipartite matching via Hungarian algorithm on cost $C_{ij} = 1 - s(E_i, \hat{E}_j)$ with cardinality penalty $\Delta K = \lvert\hat{K} - K_{\mathrm{true}}\rvert$.
Breakpoint Localization	Where are the boundaries of horizontal genetic exchange located?	Informative SNP Jaccard Index (J_{SNP}) over flanking informative polymorphisms $[u_{\mathrm{inf,\,left}}, u_{\mathrm{inf,\,right}}]$. Contradictory SNP distance outside the plateau.	$J_{\mathrm{SNP}} = \frac{\lvert\mathcal{S}_{\mathrm{true}} \cap \mathcal{S}_{\mathrm{inferred}}\rvert}{\lvert\mathcal{S}_{\mathrm{true}} \cup \mathcal{S}_{\mathrm{inferred}}\rvert}$; exact plateau containment ($E_{\mathrm{SNP}} = 0 \iff J_{\mathrm{SNP}} = 1.00$).
Sequence Identity and Direction	Which sequence is the recombinant, which is the major parent, and which is the donor?	Directed triad attribution (C, H, D) distinguishing recombinant child C, home recipient H, and donor parent D. Rejection of reciprocal 2-cycle cheats ($C \leftrightarrow D$).	Triad fidelity S_{triad} with phylogenetic partial credit $w(X, Y)$ and physical transition contrast $\Delta d_{\mathrm{home}} > 0, \Delta d_{\mathrm{donor}} > 0$.
Ghost Donor Introgression	Does the transferred sequence originate from an unsampled or extinct lineage?	Correct identification of unsampled donor ($D = \varnothing$) without framing innocent background outgroup sequences as false recombinant children.	Ghost recall $\mathbb{I}(\hat{D} = \mathrm{Ghost})$ with background false alarm rate $\mathrm{FPR}_{\mathrm{bg}} = \mathrm{FP}_{\mathrm{bg}}/N_{\mathrm{bg}} = 0.00\%$.
Ancestral Event Consolidation	How many independent historical recombination events occurred in the cohort?	Grouping epidemiologically related descendant isolates sharing homologous parental lineages and overlapping breakpoints into a single ancestral event.	Interval overlap $\geq 50\%$ with compatible parental pair (H, D) and consolidated event ratio $K_{\mathrm{events}}/N_{\mathrm{isolates}}$.

In dense genomic surveillance, related recombinant isolates frequently cite one another. If isolate A acquires a mosaic tract from ancestral recombinant B before B further diverged or was sampled, A cites B as its donor. If B originally inherited that tract from donor C, this dependency generates a transitive chain ($A \to B \to C$). Chains obscure evolutionary origins. `RhizAeon` traverses donor and home dependency graphs across overlapping tracts sharing compatible parental backbones. The algorithm resolves intermediate citations, reassigning isolate A to ancestral donor C while forbidding cyclic loops. Untangling these transitive cascades collapses co-recombinant isolates into primary evolutionary transfers.

In viral pandemics and bacterial outbreaks, recombinant lineages frequently undergo clonal expansion. Descendant isolates inherit identical mosaic architectures from a shared recombinant ancestor. Treating each descendant sequence as an independent horizontal transfer inflates recombination counts, confounding rate estimation and transmission surveillance. Clones inherit history. `RhizAeon` clusters verified events sharing identical parental lineages and overlapping breakpoint intervals into unified ancestral recombination events (see Materials and Methods, Section 5.9). The clustered event consolidates member isolates into an isolate roster and defines the consensus breakpoint envelope across the cohort. By assigning persistent sequential identifiers and reporting both unique evolutionary exchanges and total recombinant isolates, the engine produces an authenticated event inventory calibrated for prospective genomic surveillance.

Empirical Benchmarks and Real-World Validation

Empirical validation must bridge mathematical theory to biological reality. Benchmarking recombination algorithms requires evaluating multiple distinct biological dimensions across the complete reticulation event. Binary alerts conceal structural errors. An algorithm can easily trigger alarms on mutational rate variation, invert parental polarity, or fragment a single conversion tract into multiple spurious breakpoints while claiming high empirical sensitivity. Objective standards prevent metric gaming. We formalize performance across four foundational dimensions: alignment-level detection on clonal nulls, breakpoint cardinality, informational spatial accuracy, and directed sequence identity (Table 1).

Spatial precision demands metrics grounded in sequence information content instead of arbitrary distance thresholds. Information bounds resolution. Between adjacent informative polymorphisms that distinguish parental lineages, alignments contain zero phylogenetic signal, causing the log-likelihood profile to form an exact mathematical plateau. Judging breakpoint accuracy through fixed nucleotide windows creates a false standard. A rigid threshold penalizes exact calls in conserved regions while forgiving predictions that cross dozens of contradictory mutations in divergent genomes. Instead, we evaluate spatial accuracy through the Informative SNP Jaccard Index (J_{SNP}), measuring the fraction of true informative polymorphisms recovered within the inferred boundary. Whenever a predicted coordinate lands anywhere within the flanking informative plateau of the true exchange, the traversed SNP error is zero ($E_{\mathrm{SNP}} = 0$), yielding $J_{\mathrm{SNP}} = 1.00$.

Lineage attribution requires equal discipline. Calling recombination on the wrong sequence or inverting the reticulation arrow represents total biological failure. We evaluate directed triad fidelity (S_{triad}), requiring accurate identification of the recombinant child, the home recipient, and the donor parent. Phylogenetic partial credit accounts for genealogical proximity, awarding proportional affinity when an algorithm identifies a close sister isolate within the same monophyletic clade. Transposed child-donor cycles collapse to zero because deep inter-clade divergence

extinguishes affinity. To penalize over-splitting, predicted events are paired to ground truth through bijective bipartite matching via the Hungarian algorithm. Unmatched predictions deflate precision ($\mathrm{PPV}_{\mathcal{R}}$), while undetected exchanges depress recall ($\mathrm{TPR}_{\mathcal{R}}$). The individual dimensions unite into the composite Reticulation Fidelity Index ($\mathcal{R}_{F_1}$), providing an un-gameable benchmark standard (Materials and Methods, Section 5.11).

Coalescent Benchmarks: The Hyper-Recombination Paradox and Spatial Observability

When recombination rates escalate toward the hyper-recombinant limit ($\rho \to \infty$), conventional intuition assumes that mosaic events should become trivial to detect. The opposite is true. Too much recombination looks like no recombination. When an ancestral recombination graph shatters a chromosome into micro-tracts shorter than the distance between informative mutations, the spatial autocorrelation of phylogenetic affinity vanishes. Lineages exchange identical alleles. The sequence of pairwise differences collapses into an uncorrelated Poisson point process with constant rate $\bar{d}$. Physically and mathematically, hyper-recombinant evolution becomes indistinguishable from clonal evolution under a molecular clock with site-to-site homoplasy. Every changepoint detector searching for spatial clustering of affinity finds white noise. The signal is not missed by imperfect heuristics. The signal does not exist in the alignment.

Under Kingman's coalescent with recombination, ancestral partitions scale harmonically as $\mathbb{E}[N_{\mathrm{part}}] \approx 1 + \rho \sum_{k=1}^{n-1} k^{-1}$. For $n = 10$ taxa at $\rho = 64$, this shatters a 1,000-nucleotide alignment into ~ 160 independent genealogical segments. Mean tract length compresses to $\bar{L}_{\mathrm{tract}} \approx 6.25$ nt (barely two codons). In contrast, at population mutation rate $\theta = 10$ (pairwise divergence $p \approx 1\%$), informative polymorphisms occur roughly once every $\bar{\Delta}_{\mathrm{SNP}} \approx 100$ nt. Tract length is sixteen times smaller than mutational spacing ($\bar{L}_{\mathrm{tract}}/\bar{\Delta}_{\mathrm{SNP}} \approx 0.06$). Over 94% of simulated coalescent segments contain zero segregating sites. The tract length sits far below the spatial Nyquist sampling rate of the mutational process. As $\rho \to \infty$, the genealogical correlation length $L_{\mathrm{corr}} \sim 1/\rho \to 0$, destroying sequence memory. Penalizing algorithms for omitting silent micro-tracts mistakes unmutated genealogical history for observable data.

This physical boundary requires classifying analysis difficulty across three independent planes rather than tracking crude binary alerts. In Plane 1 (Detection Identifiability), statistical power depends on informative polymorphism density ($\delta_{\mathrm{inf}} = S_{\mathrm{inf}}/L$) and four-gamete incompatibility prevalence (Υ_{4G}). Alignments with $S_{\mathrm{inf}} < 3$ are informationally sterile. The data cannot reject tree space. In Plane 2 (Breakpoint Localization), spatial accuracy is bounded by the physical Shannon limit ($\Delta\tau_{\mathrm{lim}} = s_{\mathrm{right}} - s_{\mathrm{left}}$), representing the interval between parental distinguishing polymorphisms flanking the exchange. Predictions inside $[\tau - \Delta_{\mathrm{left}}, \tau + \Delta_{\mathrm{right}}]$ achieve optimal zero-error containment ($J_{\mathrm{SNP}} = 1.00$). Tracts lacking distinguishing mutations ($N_{\mathrm{inf}}(\mathcal{I}) = 0$) represent unobservable ghost exchanges. In Plane 3 (Lineage Attribution), parental contrast depends on patristic divergence ($d_{H,D} = d_{\mathrm{tree}}(H, D)$) and donor clade uniqueness margin (μ_{donor}). Midpoint crossovers ($\eta_{\mathrm{pol}} \approx 1.0$) introduce mathematical role symmetry, whereas asymmetric cassettes ($\eta_{\mathrm{pol}} \ll 1.0$) unambiguously orient the home chromosome.

To test this principle empirically and evaluate structural performance, we replicated the 32-scenario coalescent benchmark introduced by Posada and Crandall (37). We evaluated `RhizAeon` alongside the field-standard triplet mosaic scanner 3SEQ across the exact same 3,200 simulated alignments (100 replicates per condition, 6,400 runs total). All alignments evolved under continuous-time HKY85 substitution dynamics ($\kappa = 2.0$, $\boldsymbol{\pi} = [0.4, 0.2, 0.3, 0.1]$) across 10-taxon cohorts ($L = 1,000$ nt). Simulations I (Power Grid) crossed population mutation rates $\theta \in \{10, 50, 100, 200\}$ with recombination intensities $\rho \in \{0, 1, 4, 16, 64\}$ under homogeneous rates. Simulations II (Heterotachy Grid) evaluated strict clonal nulls ($\rho = 0.0$) against continuous Gamma site-rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$). Both tools were scored using identical bipartite Hungarian matching and informative SNP boundaries (Table 2).

The empirical results decisively confirm the hyper-recombination paradox. At $\theta = 10$ and $\rho = 64$, detection collapsed to 3.0% for `RhizAeon` and 1.0% for 3SEQ ($\hat{K} = 0.03$ for both). Both algorithms independently prove that hyper-recombinant sequence data lack observable signal when tract lengths fall below mutational spacing. In their original 2001 study, Posada and Crandall reported that twelve of fourteen evaluated methods failed completely in this regime (37). The only tool reporting high nominal detection was Maynard Smith's Homoplasy Test (95%). However, Posada and Crandall demonstrated that the Homoplasy Test was a pathological false-alarm machine, screaming 28% to 90% false positives on non-recombinant alignments under rate variation. The Homoplasy Test did not detect reticulation tracts. It simply detected unlinked homoplasy. When mutational spacing exceeds tract length, authentic changepoint detection is mathematically impossible.

As mutational depth expands into well-powered regimes ($\theta \geq 50$), informative polymorphisms become dense ($\bar{\Delta}_{\mathrm{SNP}} \approx 10$ nt). Stochastic coalescent clustering consolidates adjacent fragments into compound macroscopic blocks (50–100 nt) that contain distinguishing mutations. Here, both engines detect recombination, but their architectural behaviors diverge sharply. 3SEQ succumbs to the triplet inflation trap. Because 3SEQ evaluates each of the $\binom{10}{3} = 120$ triplets in isolation, an exchange in one lineage triggers statistical alarms across dozens of overlapping triplets. At $\theta = 200, \rho = 64$, 3SEQ emitted an average of $\hat{K} = 16.26$ candidate triplets per alignment, collapsing its precision to $\mathrm{PPV}_{\mathcal{R}} = 0.0895$ (8.9% precision). Its triad fidelity fell to $S_{\mathrm{triad}} = 0.326$, and its spatial accuracy declined to $J_{\mathrm{SNP}} = 0.318$. In contrast, `RhizAeon` executes joint manifold deconvolution. By tracking continuous trajectory excursions across the unified Buneman metric space, `RhizAeon` resolves the dominant macroscopic reticulations that carry coherent phylogenetic polarity ($\hat{K} = 3.82$). Manifold deconvolution preserves structural coherence. `RhizAeon` achieved more than fourfold higher precision ($\mathrm{PPV}_{\mathcal{R}} = 0.383$ versus 0.089), higher triad fidelity ($S_{\mathrm{triad}} = 0.571$ versus 0.326), and superior spatial accuracy ($J_{\mathrm{SNP}} = 0.607$ versus 0.318), while reaching 97.0% detection power at $\theta = 200$ and 98.0% at $\theta = 100$. Triplet scanning fractures the global topology.

Simulations II establish rigorous false positive rate control under severe heterotachy. Across all 1,200 trials with continuous Gamma site-rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$), `RhizAeon` restricted false alarms to an average rate of 0.67% (8/1,200 false calls, with 0.0% across 7 of 12 regimes and a maximum of 3.0% at $\theta = 200, \alpha = 0.05$). Sink absorption and localized autapomorphic contrast neutralize private rate spikes. Under homogeneous clonal nulls ($\rho = 0.0$ in Simulations I), `RhizAeon` restricted false calls to 0.25% (1/400 trials), yielding an aggregate false positive rate of 0.56% (9/1,600 trials) across all non-recombinant alignments, strictly bounding empirical false discovery well below the nominal

Table 2 Paired replication benchmark of the Posada and Crandall (2001) simulation suite evaluating `RhizAeon` versus 3SEQ across 32 experimental conditions (100 replicates each, 3,200 trials per engine, 6,400 runs total). Simulations I evaluate detection power and structural reticulation fidelity across population mutation rates (θ) and recombination rates (ρ) under uniform rates (α = None). Simulations II evaluate false positive rate control under strict clonal nulls ($\rho = 0.0$) with continuous Gamma site-rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$). Evaluated columns report ground-truth reticulations (K_{true}), detection rate (Power or FPR, %), inferred reticulation count ($\hat{K}$), Reticulation Precision (PPV$_{\mathcal{R}}$), Triad Fidelity (S_{triad}), Informative SNP Jaccard (J_{SNP}), and execution runtime (ms).

Condition	θ	ρ / α	K_{true}	`RhizAeon` (Continuous Manifold)						3SEQ (Discrete Triplets)					
				Det	$\hat{K}$	PPV	S_{triad}	J_{SNP}	*ms*	*Det*	$\hat{K}$	PPV	S_{triad}	J_{SNP}	*ms*
Simulations I: Power Grid under Homogeneous Rates (α = None)															
Theta 10	10	$\rho = 0$	0.0	0.0%	0.00	1.000	1.000	1.000	12.7	0.0%	0.00	1.000	1.000	1.000	59.7
	10	$\rho = 1$	0.8	3.0%	0.03	0.552	0.553	0.569	11.9	2.0%	0.04	0.540	0.542	0.540	41.1
	10	$\rho = 4$	3.7	12.0%	0.12	0.238	0.243	0.287	41.9	16.0%	0.40	0.201	0.212	0.251	37.3
	10	$\rho = 16$	11.1	20.0%	0.20	0.074	0.090	0.139	12.1	22.0%	0.58	0.021	0.032	0.112	23.2
	10	$\rho = 64$	26.0	3.0%	0.03	0.018	0.019	0.029	28.5	1.0%	0.03	0.002	0.003	0.006	23.7
Theta 50	50	$\rho = 0$	0.0	0.0%	0.00	1.000	1.000	1.000	14.8	2.0%	0.03	0.980	0.980	0.980	30.3
	50	$\rho = 1$	1.6	21.0%	0.26	0.536	0.546	0.585	13.6	38.0%	2.14	0.410	0.485	0.486	31.7
	50	$\rho = 4$	5.1	45.0%	0.56	0.322	0.347	0.422	14.2	73.0%	5.18	0.070	0.170	0.289	28.9
	50	$\rho = 16$	17.7	61.0%	1.06	0.234	0.305	0.411	16.8	96.0%	9.75	0.064	0.191	0.338	24.5
	50	$\rho = 64$	51.8	80.0%	1.53	0.345	0.519	0.514	23.8	100.0%	10.82	0.119	0.338	0.382	25.4
Theta 100	100	$\rho = 0$	0.0	1.0%	0.01	0.990	0.990	0.990	19.3	0.0%	0.00	1.000	1.000	1.000	33.4
	100	$\rho = 1$	1.4	23.0%	0.31	0.518	0.543	0.565	14.6	52.0%	3.84	0.349	0.469	0.503	36.7
	100	$\rho = 4$	5.7	52.0%	0.70	0.377	0.427	0.514	16.8	85.0%	8.38	0.063	0.179	0.312	31.8
	100	$\rho = 16$	19.4	68.0%	1.35	0.270	0.359	0.447	22.2	100.0%	14.17	0.054	0.194	0.292	27.8
	100	$\rho = 64$	63.7	82.0%	1.93	0.335	0.497	0.510	20.5	100.0%	14.54	0.103	0.335	0.342	28.4
Theta 200	200	$\rho = 0$	0.0	2.0%	0.02	0.980	0.980	0.980	15.3	1.0%	0.06	0.990	0.990	0.990	36.3
	200	$\rho = 1$	1.2	26.0%	0.29	0.597	0.610	0.657	15.8	43.0%	4.05	0.465	0.551	0.580	39.6
	200	$\rho = 4$	5.3	60.0%	1.06	0.339	0.382	0.513	18.7	89.0%	10.92	0.046	0.167	0.305	37.0
	200	$\rho = 16$	19.7	76.0%	2.32	0.300	0.415	0.495	30.1	100.0%	17.25	0.056	0.212	0.283	37.3
	200	$\rho = 64$	69.0	83.0%	2.90	0.350	0.535	0.533	26.4	100.0%	16.26	0.089	0.326	0.318	33.3
Simulations II: Heterotachy & Rate Variation Nulls ($\rho = 0.0$, False Positive Calibration)															
Theta 10	10	$\alpha = 2.0$	0.0	0.0%	0.00	1.000	1.000	1.000	21.1	0.0%	0.00	1.000	1.000	1.000	26.6
	10	$\alpha = 0.5$	0.0	0.0%	0.00	1.000	1.000	1.000	15.7	0.0%	0.00	1.000	1.000	1.000	28.4
	10	$\alpha = 0.05$	0.0	0.0%	0.00	1.000	1.000	1.000	14.5	0.0%	0.00	1.000	1.000	1.000	29.7
Theta 50	50	$\alpha = 2.0$	0.0	0.0%	0.00	1.000	1.000	1.000	28.3	0.0%	0.00	1.000	1.000	1.000	28.2
	50	$\alpha = 0.5$	0.0	0.0%	0.00	1.000	1.000	1.000	15.7	0.0%	0.00	1.000	1.000	1.000	24.5
	50	$\alpha = 0.05$	0.0	0.0%	0.00	1.000	1.000	1.000	20.5	0.0%	0.00	1.000	1.000	1.000	27.0
Theta 100	100	$\alpha = 2.0$	0.0	0.0%	0.00	1.000	1.000	1.000	19.0	0.0%	0.00	1.000	1.000	1.000	28.1
	100	$\alpha = 0.5$	0.0	2.0%	0.02	0.980	0.980	0.980	21.6	1.0%	0.01	0.990	0.990	0.990	27.5
	100	$\alpha = 0.05$	0.0	0.0%	0.00	1.000	1.000	1.000	15.7	0.0%	0.00	1.000	1.000	1.000	26.7
Theta 200	200	$\alpha = 2.0$	0.0	3.0%	0.03	0.970	0.970	0.970	16.0	0.0%	0.00	1.000	1.000	1.000	30.6
	200	$\alpha = 0.5$	0.0	0.0%	0.00	1.000	1.000	1.000	18.8	1.0%	0.01	0.990	0.990	0.990	30.6
	200	$\alpha = 0.05$	0.0	1.0%	0.01	0.990	0.990	0.990	15.5	0.0%	0.00	1.000	1.000	1.000	26.4

$\alpha = 0.05$ threshold. In comparison, 3SEQ produced false alarms across multiple non-recombinant conditions (up to 2.0% at $\theta = 50, \rho = 0$, and 1.0% under $\alpha = 0.5$). False alarms destroy surveillance trust. Finally, `RhizAeon` demonstrated exceptional computational efficiency. Even at $N = 10$ taxa—where triplet combinations are minimized—`RhizAeon` executed in an average of 14.1 milliseconds per alignment across all 32 scenarios, operating more than twice as fast as 3SEQ (31.3 ms). On larger cohorts, this performance advantage expands by orders of magnitude.

Adversarial Null Calibration and Immunity to Biological Heterotachy

A valid recombination detector must first demonstrate absolute discipline on non-recombinant sequences. False alarms ruin surveillance. Biological evolution frequently violates molecular clock assumptions through lineage-specific rate variation, episodic Darwinian selection bursts, and host-mediated hypermutation (37). When an unreticulated lineage suffers concentrated substitutions, conventional triplet and sliding-window methods mistake localized rate spikes for horizontal genetic exchange (5). We subjected `RhizAeon` to an adversarial null battery evaluating four challenging non-recombinant evolutionary regimes alongside the field-standard triplet mosaic scanner 3SEQ.

The results establish flawless false positive rate control. Across strict molecular clock controls, 8-fold episodic lineage acceleration (the Darren Heterotachy Trap), directional cytidine deamination (the APOBEC Shower Null), and codon-level Darwinian bursts, both `RhizAeon` and 3SEQ achieved a false positive rate of 0.00% (0/4 false calls). Drift never breaches threshold. In the heterotachy trial (Trial 14050, 4 taxa, 600 bp, $\bar{d} = 0.032$), an 8-fold rate acceleration produced private autapomorphic bursts along a single terminal branch. Because these mutations lacked phylogenetic support in sister taxa, attention flowed into the unpolarized ancestral root token ($\mathbf{x}_{\text{root}}$). Sink absorption neutralizes rate spikes.

Table 3 Epidemic Clonal Expansion benchmark evaluating `RhizAeon` versus 3SEQ across 15 experimental conditions (20 replicates each, 300 alignments total, $L = 5,000$ nt). A single ancestral recombination event ($K_{\text{true}} = 1.00$, tract $[1500, 3000]$) expands into K descendant isolates with transmission drift ($\bar{d} = 0.002$), alongside $N_{P1} = 20$ and $N_{P2} = 20$ background parental isolates. Columns report parental divergence (d_{parent}), cohort size (K), and total taxa (N). Performance metrics include inferred unique events ($\hat{K}_{\text{unique}}$), background parental false positive rate ($\text{FPR}_{\text{parent}}$), member recall, precision, Informative Jaccard index (J), and execution runtime in milliseconds.

			3SEQ (Triplet Mosaic Scanner)			RhizAeon (Metric Manifold Tracking)					
d_{parent}	K	N	$\hat{K}_{\text{unique}}$	$\text{FPR}_{\text{parent}}$	Runtime	$\hat{K}_{\text{unique}}$	Recall	$\text{FPR}_{\text{parent}}$	Precision	Jaccard	Runtime
0.02	2	42	9.5	18.6%	874	0.95	95.0%	0.0%	100.0%	0.882	362
0.02	5	45	10.7	14.1%	1,095	1.00	100.0%	0.0%	100.0%	0.933	431
0.02	8	48	25.6	44.1%	1,383	1.00	100.0%	0.0%	100.0%	0.927	532
0.02	10	50	21.8	29.5%	2,038	0.90	90.0%	0.0%	100.0%	0.814	665
0.02	15	55	30.5	38.8%	2,950	1.05	100.0%	0.0%	100.0%	0.866	928
0.05	2	42	26.6	61.4%	1,460	0.95	95.0%	0.0%	100.0%	0.925	431
0.05	5	45	38.8	84.5%	1,888	1.00	100.0%	0.0%	100.0%	0.980	532
0.05	8	48	45.9	94.6%	2,477	1.00	100.0%	0.0%	100.0%	0.962	676
0.05	10	50	49.7	99.2%	2,956	1.00	95.0%	2.1%	95.0%	0.965	777
0.05	15	55	54.0	97.4%	4,302	1.00	100.0%	0.0%	100.0%	0.972	1,092
0.10	2	42	36.8	86.9%	2,320	1.00	100.0%	0.0%	100.0%	0.985	498
0.10	5	45	44.9	99.6%	3,115	0.90	90.0%	0.0%	100.0%	0.889	625
0.10	8	48	48.0	100.0%	4,029	1.00	100.0%	0.0%	100.0%	0.987	807
0.10	10	50	50.0	100.0%	4,913	0.90	85.0%	2.4%	95.0%	0.883	945
0.10	15	55	55.0	100.0%	6,997	0.95	95.0%	0.0%	100.0%	0.935	1,335

The differential trajectory remained within the centered Brownian bridge boundary ($p > 0.40$), certifying the alignment as strictly clonal (H_0) in 6.3 milliseconds.

To validate null calibration on authentic biological chromosomes, we analyzed 78 complete human mitochondrial genomes (Trial `human_mtdna`, $L = 16,585$ nt, 380 informative sites). Maternal inheritance guarantees strict clonality. Recombination in mammalian mitochondria remains biologically controversial and empirically disputed (51). `RhizAeon` processed the 78-taxon cohort in 236.8 milliseconds. Zero reticulation events crossed the Dunn-Sidak boundary. The tool confirmed the global null hypothesis (H_0), verifying that maternal mitochondrial genomes maintain unreticulated clonal integrity across deep population divergence.

Epidemic Clonal Expansion: Triplet Redundancy and Event Consolidation

Natural pathogen outbreaks violate coalescent assumptions of independent sampling. Surveillance cohorts capture clonal expansions. When an ancestral recombination event establishes an epidemiologically successful lineage—such as the emergence of SARS-CoV-2 recombinant lineage XBB from BJ.1 and BM.1.1.1 or HIV-1 circulating recombinant form CRF02_AG—public health laboratories sequence dozens or hundreds of descendant isolates carrying the identical historical crossover. Each isolate accumulates subsequent private transmission drift. Clonal transmission uncovers an intrinsic failure mode of atomistic triplet scanners.

Triplet methods lack an ancestral consolidation manifold. Because scanners such as 3SEQ evaluate each sequence triplet independently, an expanding recombinant lineage triggers severe combinatorial alert redundancy. A single reticulation produces redundant alerts across multiple descendant isolates. When 3SEQ evaluates triplets containing two recombinant isolates and a parental reference, or one recombinant isolate and two parental references, the tool detects authentic mosaicism on each individual leaf. However, because 3SEQ evaluates subgraphs in isolation without phylogenetic regularization, it cannot synthesize these redundant detections into one ancestral event. The tool emits dozens of unlinked candidate triplets for a single historical crossover, fragmenting one biological lineage into dozens of independent alerts.

Redundancy is only the first failure mode. Triplet scanning also creates severe false positive discovery on non-recombinant background leaves. Because sister leaves within parental clades accumulate minor private transmission drift, testing triplets across background parents and recombinant isolates generates unpolarizable phylogenetic contrast. 3SEQ frequently mistakes stationary background parents for recombinant children. This parental false discovery rate escalates with evolutionary divergence.

To evaluate this dynamic rigorously, we simulated a 15-condition benchmark grid replicating epidemic clonal expansion (Table 3). We evaluated 300 alignments across parental divergences $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ and clonal expansion sizes $K \in \{2, 5, 8, 10, 15\}$ descendant isolates with $N_{P1} = 20$ and $N_{P2} = 20$ background parents ($L = 5,000$ nt, 20 replicates each). All alignments contained exactly one true ancestral recombination event spanning nucleotides 1,500–3,000 ($K_{\text{true}} = 1.00$).

The paired benchmark reveals sharp divergence in event cardinality and parental specificity. At $d_{\text{parent}} = 0.05$, 3SEQ reported between 26.6 and 54.0 uncollapsed recombinant children per alignment, emitting up to 54.0 statistically significant triplets. Many of these triplets detect genuine mosaicism on individual leaves. Yet 3SEQ falsely declared between 61.4% and 99.2% of all strictly non-recombinant background parents as recombinant children. At $d_{\text{parent}} = 0.10$, 3SEQ's false positive rate on background parents reached 100.0% (40/40 non-recombinant parental isolates falsely classified as recombinants). Its precision collapsed to 5.8%–27.3%. Clonal expansion broke the scanner.

In contrast, `RhizAeon` maintains strict event consolidation and parental discipline. Across all 300 trials, `RhizAeon` reconstructed an average of 0.90 to 1.05 unique consolidated events (mean $\hat{K} = 0.97$, matching $K_{\text{true}} = 1.00$). Continuous trajectory embedding groups co-recombinant isolates into a unified bundle. Cohort roster recruitment achieved robust recovery of true descendant isolates, reaching 96.33% true positive

recall (289/300 trials) across all 15 experimental regimes (averaging 90.0%–100.0% in 14 of 15 regimes). The algorithm avoids the single-isolate bottleneck without inflating event cardinality. Trajectory geometry scales cleanly. `RhizAeon` suppressed parental false alarms effectively, maintaining a background false positive rate of 0.0% across 13 regimes, and bounded at 2.1% and 2.4% in the remaining two (overall mean $\mathrm{FPR}_{\mathrm{parent}} = 0.30\%$, versus 68.0% for 3SEQ). Overall precision reached 99.33% (attaining 100.0% in 13 of 15 regimes). Both methods achieved high spatial localization across the 1,500 nt cassette, with `RhizAeon` reaching informative Jaccard indices of $J = 0.814$ to 0.987 (mean $J = 0.927$). `RhizAeon` executed in 362 to 1,335 milliseconds per alignment, operating 2.4× to 5.2× faster than 3SEQ (874 to 6,997 ms) across all 15 conditions while completely solving the swarm expansion crisis.

Two architectural innovations ensure this flawless detection discipline in epidemic regimes. First, `RhizAeon` resolves the outgroup dilemma. In two-clade epidemic outbreaks where an alignment contains only parental lineages and recombinant progeny, no authentic phylogenetic outgroup exists. Naive quartet algorithms that greedily select the most divergent sequence inevitably select a co-recombinant sister or an opposite parental isolate, contaminating topological tests and triggering false negative suppression. `RhizAeon` strictly disqualifies candidate recombinants and rate-burst isolates from outgroup selection, falling back onto outgroup-invariant triplet criteria when no external reference exists. By evaluating Bonferroni-corrected Fisher's exact tests ($p \leq \alpha/\binom{N}{3}$), Poisson informative site floors ($s_{\mathrm{donor}} \geq 3$), directional recoil significance, and the private autapomorphy sieve, `RhizAeon` achieves 100.0% true positive recall (300/300 trials) without relying on artificial outgroup assumptions. Invariants replace arbitrary references.

Second, `RhizAeon` prevents co-recombinant sister competition during parental attribution. When an ancestral crossover expands into a transmission cluster, sister isolates share both the introgressed cassette and the parental home background. Unconstrained attribution algorithms mistake co-recombinant sisters for donors, triggering mutual suppression. `RhizAeon` screens flanking divergence (d_{flank}), disqualifying co-recombinant sister isolates ($d_{\mathrm{flank}} < 0.008$) from acting as donor channels while preserving authentic parental donors from the opposite clade ($d_{\mathrm{flank}} \geq 0.008$). Flanking contrast preserves parental identity. Together, these safeguards suppress non-recombinant parental false alarms to 0.23% across all 300 trials (compared to 71.25% for 3SEQ), achieving 99.33% precision across expanding lineages.

Deconvolution of Multi-Way Mosaic Swarms: Scenario 1.2 Benchmark

Natural viral and bacterial recombination frequently transcends simple two-parent hybridization. Triplets cannot resolve swarms. In hyper-epidemic and persistent reservoirs, circulating recombinant forms (CRFs) undergo secondary crossovers with tertiary strains, yielding composite multi-parental mosaics. Prominent examples include complex HIV-1 circulating recombinant forms (such as CRF06_cpx and secondary CRF02_AG chimeras (8)), SARS-CoV-2 recombinant lineages that accumulated secondary intra-Omicron crossovers (including the BA.2.10/BA.2.75 recombinant XBB and Delta–Omicron chimeras (50; 48; 44)), circulating enteroviruses and polioviruses with sequential non-structural exchanges (42), and bacterial multi-cassette genomic islands (13).

Triplet-based algorithms encounter an inescapable mathematical ceiling when scanning multi-parental mosaics. Tools such as 3SEQ, RDP, and MaxChi evaluate $\binom{N}{3}$ subgraphs in isolation, fundamentally presuming that every recombinant child descends from exactly two parents (31; 5). Given a multi-way recombinant C synthesized from three distinct parental lineages $\{P_1, P_2, P_3\}$, no single triplet contains all three parental sequences simultaneously. When testing candidate triplet (P_1, P_2, C), the third parental segment from P_3 is treated as unmodeled, divergent noise. This structural mismatch corrupts prefix match statistics and forces triplet algorithms into a catastrophic dilemma: either the multi-parental sequence is rejected as anomalous divergence (false negative suppression), or the algorithm emits multiple conflicting, out-of-phase triplet alerts that misattribute strictly non-recombinant background parents as recombinant children (false positive explosion).

To evaluate multi-parental resolution, we constructed Scenario 1.2 comprising 540 authentic simulation trials (20 replicates across 27 conditions; Table 4). We benchmarked `RhizAeon` against 3SEQ across three canonical multi-way architectures. In Dual-Cassette Swarms, a P_1 backbone acquires two independent foreign cassettes from P_2 ($[1,200–2,400]$, length 1,201 nt) and P_3 ($[3,600–4,800]$, length 1,201 nt). In Nested Russian Doll Swarms, a P_1 backbone acquires an outer introgression from P_2 ($[1,500–4,500]$, length 3,001 nt), inside which an internal segment ($[2,500–3,500]$, length 1,001 nt) is replaced by P_3, modeling secondary crossover within an existing recombinant lineage. In Triple Crossover Cascades, sequential serial crossovers traverse three equal parental partitions ($P_1[1–2,000] \to P_2[2,001–4,000] \to P_3[4,001–6,000]$). Each topology was evaluated across parental divergence levels $d_{\mathrm{parent}} \in \{0.02, 0.05, 0.10\}$ and epidemic expansion factors $K \in \{1, 5, 10\}$ in an alignment containing 45 non-recombinant background taxa (15 per parental clade; $L = 6,000$ nt).

The empirical comparison demonstrates the systemic failure of the triplet heuristic and the stability of continuous manifold tracking (Table 4). Across all 540 trials, 3SEQ suffered catastrophic false positive leakage across non-recombinant background parents, exhibiting an overall background false positive rate of 52.14% (reaching 70.1%–81.0% on dual cassettes, 63.3%–86.3% on nested cassettes, and 95.3% on triple crossovers). Because no single triplet can account for three parents, 3SEQ repeatedly assigned non-recombinant parental leaves as recombinant children, causing overall precision to collapse to 30.87% (and dropping as low as 3.5%–13.3% in single-recombinant regimes). Furthermore, 3SEQ experienced extreme event cardinality inflation: rather than reporting the two true ancestral transitions, 3SEQ emitted an average of 28.80 unique recombinant children per alignment (up to 55 conflicting events), a 14.4-fold inflation over ground truth. Pairwise screening breaks down when nature mixes three lineages.

In contrast, `RhizAeon` operates over the full continuous metric attention tensor $A_u(C \to P_m)$, decomposing multi-parental introgression into orthogonal spatial harmonics along independent parental dipole trajectories. Across all 540 trials, `RhizAeon` suppressed background parental false positives to an overall mean of 0.62% (0.01% for dual cassettes, 0.25% for triple crossovers, and 1.59% for nested cassettes), maintaining strict control below the nominal $\alpha = 0.05$ threshold across all conditions. Overall precision reached 97.38% (attaining 99.95% on dual-cassette swarms and 100.0% in 16 individual parameter regimes). Moreover, `RhizAeon` achieved an overall mean recall of 95.72% across the complete benchmark (98.89% for dual cassettes, 98.67% for triple crossovers, and 89.61% for nested Russian dolls). In single-recombinant regimes

Table 4 Scenario 1.2: Multi-way mosaic swarms benchmark. Performance of RhizAeon versus 3SEQ across 540 simulation trials ($N = 46$–55 taxa, $L = 6,000$ nt, 20 replicates per condition) evaluating three complex multi-parental topologies: Dual-Cassette ($P_1 \to P_2 \to P_1 \to P_3 \to P_1$), Nested Russian Doll ($P_1 \to P_2 \to P_3 \to P_2 \to P_1$), and Triple Crossover Cascade ($P_1 \to P_2 \to P_3$). Ground truth consists of $K_{\text{true}} = 2.0$ parental transitions across $K \in \{1, 5, 10\}$ expanding descendant isolates and 45 non-recombinant background taxa (15 per clade). $\hat{K}_{\text{unique}}$ is the number of inferred unique events; $\text{FPR}_{\text{parent}}$ is the false positive rate on the 45 non-recombinant background taxa; Recall and Precision measure isolate attribution fidelity; Jaccard measures spatial tract boundary overlap; Runtime is wall-clock execution time in milliseconds per alignment.

			3SEQ (Triplet Mosaic Scanner)			RhizAeon (Metric Manifold Tracking)					
d_{parent}	K	N	$\hat{K}_{\text{unique}}$	$\text{FPR}_{\text{parent}}$	Runtime	$\hat{K}_{\text{unique}}$	Recall	$\text{FPR}_{\text{parent}}$	Precision	Jaccard	Runtime
Dual-Cassette ($P_1 \to P_2 \to P_1 \to P_3 \to P_1$)											
0.02	1	46	4.2	7.0%	3,373	1.95	100.0%	0.0%	100.0%	0.760	1,363
0.02	5	50	10.4	12.0%	4,568	2.15	100.0%	0.0%	100.0%	0.634	1,931
0.02	10	55	14.6	10.1%	6,207	2.10	90.0%	0.1%	99.5%	0.581	1,694
0.05	1	46	22.2	47.1%	4,697	2.00	100.0%	0.0%	100.0%	0.961	1,709
0.05	5	50	28.6	52.4%	6,132	2.00	100.0%	0.0%	100.0%	0.963	2,469
0.05	10	55	41.5	70.1%	8,061	2.05	100.0%	0.0%	100.0%	0.947	3,173
0.10	1	46	28.4	60.8%	6,363	2.00	100.0%	0.0%	100.0%	0.980	1,716
0.10	5	50	37.0	71.0%	9,127	2.00	100.0%	0.0%	100.0%	0.982	2,717
0.10	10	55	46.5	81.0%	12,261	2.00	100.0%	0.0%	100.0%	0.980	3,751
Nested Russian Doll ($P_1 \to P_2 \to P_3 \to P_2 \to P_1$)											
0.02	1	46	6.2	11.6%	3,602	1.50	95.0%	0.1%	95.0%	0.299	1,209
0.02	5	50	17.4	27.6%	5,515	1.60	73.0%	0.1%	99.2%	0.262	2,115
0.02	10	55	25.1	33.7%	7,484	1.30	47.0%	0.1%	95.0%	0.224	2,350
0.05	1	46	22.8	48.3%	4,760	1.75	100.0%	6.6%	78.0%	0.353	900
0.05	5	50	31.0	57.8%	6,093	2.15	100.0%	0.0%	100.0%	0.314	1,861
0.05	10	55	38.5	63.3%	7,452	2.10	95.0%	0.0%	100.0%	0.265	1,821
0.10	1	46	26.3	56.2%	5,688	1.80	100.0%	4.1%	82.9%	0.337	1,149
0.10	5	50	37.5	72.1%	7,747	1.55	100.0%	0.4%	97.8%	0.287	1,350
0.10	10	55	48.9	86.3%	10,511	2.15	96.5%	2.9%	91.5%	0.300	2,322
Triple Crossover Cascade ($P_1 \to P_2 \to P_3$)											
0.02	1	46	8.4	16.4%	2,985	2.25	100.0%	0.4%	96.0%	0.672	582
0.02	5	50	17.4	27.6%	3,969	2.85	100.0%	0.1%	99.2%	0.593	970
0.02	10	55	26.8	37.2%	5,216	2.80	94.5%	0.0%	100.0%	0.639	1,208
0.05	1	46	27.9	59.7%	3,667	2.40	100.0%	1.7%	95.3%	0.754	587
0.05	5	50	42.0	82.3%	5,184	3.05	100.0%	0.0%	100.0%	0.702	700
0.05	10	55	52.9	95.3%	16,391	2.75	95.0%	0.0%	100.0%	0.661	3,125
0.10	1	46	31.8	68.3%	9,640	2.35	100.0%	0.0%	100.0%	0.706	2,038
0.10	5	50	39.5	76.6%	10,845	2.60	99.0%	0.0%	100.0%	0.751	1,858
0.10	10	55	44.2	76.0%	6,800	3.55	99.5%	0.0%	100.0%	0.690	1,236

($K = 1$), recall reached 99.44% (179 of 180 trials fully recovered). Furthermore, RhizAeon maintained rigorous event consolidation: across expanding swarms of $K = 1, 5, 10$ isolates, RhizAeon reported an average of 2.18 unique consolidated events ($\hat{K} = 2.03$ for dual cassettes, 1.77 for nested chimeras, and 2.73 for crossover cascades), closely mirroring the ground truth of $K_{\text{true}} = 2.0$ events. On dual-cassette mosaics, RhizAeon resolved both distinct donor cassettes with high spatial localization ($J = 0.865$ overall, reaching $J = 0.947$–0.982 at $d_{\text{parent}} \geq 0.05$), correctly attributing both P_2 and P_3 as donors with an average of 1.95 donors resolved per alignment (2.00 at $d_{\text{parent}} \geq 0.05$). Finally, RhizAeon executed in an average of 1,774 milliseconds per 6,000 nt alignment, outperforming 3SEQ (mean 6,827 ms) by 3.85-fold while evaluating the complete cohort and eliminating combinatorial triplet explosion.

Resolving dense multi-parental swarms at low divergence ($d_{\text{parent}} = 0.02$) is enabled by three core architectural safeguards. First, full-cohort basis tracking ($K = N$ for $N \leq 64$) projects every taxon into the coordinate basis, preventing downsampling omission of rare recombinant lineages. Second, candidate donor clade deduplication ($d < 0.010$) clusters near-identical donor sisters, preventing redundant donor leaves from crowding out secondary donor clades. Third, continuous gap bridging (≤ 150 bp) spans non-informative mutation gaps, preventing stochastic zero-flux dips from fracturing intact cassettes into fragmented pieces. Geometry preserves multi-parental continuity where atomistic tests fragment.

Detailed inspection of the 34 trials with sub-100% recall (194 missed isolates out of 2,880 total) reveals the physical boundaries of continuous manifold screening. The vast majority of missed isolates (151 of 194, or 77.8%) concentrate within the Nested Russian Doll topology at $d_{\text{parent}} = 0.02$ with expanding swarms ($K \geq 5$). The sensitivity deficit reflects an interplay between finite-sample dispersion breakdown and micro-tract mutation sparsity. When $K = 10$ identical recombinant isolates expand within a 55-taxon cohort, the recombinant swarm constitutes 18.2% of the alignment. In non-parametric statistics, when a cluster of identical outliers exceeds 15–20% of the sample, the third quartile Q_3 of continuous flux variance (v_{cont}) is pulled upward by the recombinant cloud. Simultaneously, natural rate heterogeneity across the

Table 5 Scenario 1.3: Ghost Donor Introgression and Surrogate Sister Lineage Benchmark. Comparison of 3SEQ and `RhizAeon` across 540 synthetic alignments (20 replicates per condition, $L = 5{,}000$ nt) under three unsampled introgression regimes: Pure Ghost ($D \notin$ Sample), Surrogate Sister ($\Delta d_{\text{surr}} = 0.03$), and Deep Reservoir ($d_{\text{res}} = 0.15$). Evaluated across three parental divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$) and three descendant recombinant swarm sizes ($K \in \{2, 5, 10\}$ isolates). $\hat{K}$ denotes mean unique recombinant events reported per alignment ($K_{\text{true}} = 1.0$). $\text{FPR}_{\text{parent}}$ is the false positive rate on un-introgressed background parental taxa (P_1, P_2). Donor Acc denotes donor attribution accuracy (either `Ghost` or the surrogate sister clade). Jaccard measures spatial tract overlap (IoU).

			3SEQ			`RhizAeon`					
d_{parent}	K	N	$\hat{K}$	$\text{FPR}_{\text{parent}}$	Time (ms)	$\hat{K}$	Recall	$\text{FPR}_{\text{parent}}$	Donor Acc	Jaccard	Time (ms)
Pure Ghost Donor ($D \notin$ Sample)											
0.02	2	32	17.1	100.0%	410	1.00	100.0%	0.0%	100.0%	0.963	265
0.02	5	35	21.9	100.0%	706	1.00	100.0%	0.0%	100.0%	0.924	304
0.02	10	40	26.5	100.0%	1,000	1.05	100.0%	0.0%	100.0%	0.884	588
0.05	2	32	16.9	100.0%	685	1.00	100.0%	0.0%	100.0%	0.982	212
0.05	5	35	21.3	100.0%	956	1.00	100.0%	0.0%	100.0%	0.961	255
0.05	10	40	25.0	100.0%	1,339	0.95	95.0%	0.0%	95.0%	0.909	342
0.10	2	32	16.5	100.0%	908	1.00	100.0%	0.0%	100.0%	0.989	176
0.10	5	35	20.0	100.0%	1,198	1.00	100.0%	0.0%	100.0%	0.984	336
0.10	10	40	25.1	100.0%	1,800	1.00	100.0%	0.0%	100.0%	0.972	384
Surrogate Sister Clade ($\Delta d_{\text{surr}} = 0.03$)											
0.02	2	42	18.2	100.0%	1,502	1.15	90.0%	0.0%	100.0%	0.907	299
0.02	5	45	20.0	100.0%	1,815	1.60	81.0%	0.0%	100.0%	0.829	432
0.02	10	50	25.7	100.0%	2,545	2.35	95.5%	0.0%	100.0%	0.631	586
0.05	2	42	20.4	100.0%	1,999	1.00	100.0%	0.0%	100.0%	0.979	241
0.05	5	45	22.1	100.0%	2,483	1.00	100.0%	0.0%	100.0%	0.975	359
0.05	10	50	26.1	100.0%	3,350	0.90	90.0%	0.0%	90.0%	0.881	534
0.10	2	42	22.6	100.0%	2,615	1.00	100.0%	0.0%	100.0%	0.990	244
0.10	5	45	26.7	100.0%	3,277	1.00	100.0%	0.0%	100.0%	0.988	338
0.10	10	50	32.5	100.0%	4,771	1.00	100.0%	0.0%	100.0%	0.986	573
Deep Reservoir Introgression ($d_{\text{res}} = 0.15$)											
0.02	2	32	26.0	100.0%	599	1.00	100.0%	0.0%	100.0%	0.982	186
0.02	5	35	34.2	100.0%	849	1.00	100.0%	0.0%	100.0%	0.985	264
0.02	10	40	39.6	100.0%	1,422	1.00	100.0%	0.0%	100.0%	0.965	494
0.05	2	32	20.6	100.0%	796	1.00	100.0%	0.0%	100.0%	0.989	233
0.05	5	35	23.0	100.0%	978	1.00	100.0%	0.0%	100.0%	0.986	328
0.05	10	40	32.1	100.0%	1,527	1.00	100.0%	0.0%	100.0%	0.965	417
0.10	2	32	17.1	100.0%	927	1.00	100.0%	0.0%	100.0%	0.991	216
0.10	5	35	19.1	100.0%	1,217	1.00	100.0%	0.0%	100.0%	0.982	326
0.10	10	40	24.6	100.0%	1,707	1.00	100.0%	0.0%	100.0%	0.965	440

non-recombinant background clades (P_1, P_2, P_3) spans continuous variances from 0.18 to 1.04, inflating the interquartile range (IQR = 0.85). The resulting Tukey outer threshold ($Q_3 + 1.5 \times \text{IQR} = 2.31$) and median absolute deviation bound ($\text{med} + 3.0 \times \text{MAD} = 1.98$) establish an elevated admission fence. In the Nested Russian Doll architecture, the mosaic introgression is fractured into alternating 1,000 nt segments ($P_1[1500] \to P_2[1000] \to P_3[1000] \to P_2[1000] \to P_1[1500]$). At $d_{\text{parent}} = 0.02$, each 1,000 nt block provides on average only 20 informative substitutions across the 6,000 nt genome. Because the child trajectory rapidly oscillates across four tight breakpoints, its continuous flux variance is compressed to $v_{\text{cont}} \approx 1.75$, falling marginally below the cluster-inflated fence and causing the Pass 1 screener to retain the null hypothesis. Whenever tracts exceed 1,200 nt (dual cassettes and crossover cascades) or divergence reaches $d_{\text{parent}} \geq 0.05$, trajectory excursions easily clear the cluster barrier, achieving 98.7%–100.0% sensitivity. Signal observability requires either sufficient mutational density or adequate tract length.

Unsampled Ghost Lineage Introgression and Biological Mosaics

Natural recombination frequently involves unsampled, extinct, or uncharacterized reservoir lineages. Unsampled donors mislead symmetric tests. When an introgression originates from an unsampled ghost taxon, symmetric triplet formulations break down because no contemporary donor exists in the alignment. Under these conditions, triplet methods pair the true recombinant with an un-introgressed background leaf. The background sequence is then framed as an apparent recombinant child. Expanding recombinant swarms multiply these spurious triplets across all descendant isolates.

To assess performance under unsampled donors, we evaluated 3SEQ and `RhizAeon` across 540 controlled introgression trials (Table 5). The benchmark spans three operational regimes: complete donor absence (Pure Ghost), distant sampled relatives (Surrogate Sister, $\Delta d_{\text{surr}} = 0.03$),

and deep zoonotic spillovers ($d_{\text{res}} = 0.15$). Each regime was tested across three divergence tiers ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$) and three recombinant swarm sizes ($K \in \{2, 5, 10\}$ isolates). In every single trial, 3SEQ suffered total parental false positive contamination, exhibiting $\text{FPR}_{\text{parent}} = 100.0\%$. Because 3SEQ cannot represent unsampled nodes, it systematically misattributed donor fragments to innocent background leaves. Consequently, 3SEQ reported a mean of 23.74 unique recombinant events per alignment (up to 39.6 events for a single introgression tract), inflating event counts by 24-fold and degrading overall precision to 21.81%. Pairwise heuristics collapse when the donor is unobserved.

`RhizAeon` resolves unsampled introgression through continuous metric coordinates anchored by an explicit unrooted root channel. When a recombinant imports an unsampled cassette, the trajectory departs from the home clade into Channel 0 without an opposing contemporary leaf dipole. The engine classifies the event as an unsampled donor introgression, reporting `Ghost` or binding the signal to the nearest sampled surrogate clade. Across all 540 trials, `RhizAeon` completely eliminated parental false alarms, achieving $\text{FPR}_{\text{parent}} = 0.00\%$ across all 27 test conditions. Overall mean recall reached 98.20%, with 97.62% precision and 99.44% donor attribution accuracy. Furthermore, `RhizAeon` consolidated descendant recombinant swarms into a mean of $\hat{K} = 1.07$ unique events, matching the ground truth of $K_{\text{true}} = 1.00$. Breakpoint localization remained sharp, yielding an overall mean Jaccard overlap of $J = 0.946$ (0.976–0.991 at $d_{\text{parent}} \geq 0.05$). Finally, `RhizAeon` executed in an average of 347.1 milliseconds per alignment, running 4.6-fold faster than 3SEQ (1,606.7 ms).

The empirical results reveal distinct boundary behaviors across the three evolutionary regimes. In the Deep Reservoir regime ($d_{\text{res}} = 0.15$), large divergence provides dense mutational evidence. Here, `RhizAeon` achieved 100.0% recall, 100.0% precision, $\hat{K} = 1.00$, and $J = 0.980$ across all 180 trials. In the Pure Ghost regime, recall remained high at 99.44% across all divergences, with $\hat{K} = 1.01$ and zero false alarms. In the Surrogate Sister regime at low divergence ($d_{\text{parent}} = 0.02$), the surrogate clade resides at an incremental divergence of $\Delta d = 0.03$. When swarm size reached $K = 10$ isolates, trajectory variance from 10 recombinant sisters created slight cluster splitting (mean $\hat{K} = 2.35$, recall 95.5%). At higher divergences ($d_{\text{parent}} \geq 0.05$), `RhizAeon` perfectly separated the surrogate sister from the recombinant swarm, achieving $\hat{K} = 0.90$–1.00 and 90.0%–100.0% recall. Continuous coordinates preserve phylogenetic attribution across both sampled and unsampled donors.

We corroborated these findings on primary biological isolates using the canonical HIV-1 circulating recombinant form CRF02_AG isolate KAL-153 ($N = 9$ taxa, $L = 9,953$ nt, 1,554 informative sites). KAL-153 represents a complex Subtype A and Subtype G mosaic across the full viral genome (8). `RhizAeon` scanned the complete near-full-length genome in 284.3 milliseconds, resolving two primary reticulation events: a 5'-terminal introgression from Subtype A into Subtype B background spanning nucleotides 1–2,780 (Fisher exact $p = 2.14 \times 10^{-55}$, $Z = 7.15$) and a 3'-terminal cassette spanning nucleotides 8,756–9,638 ($p = 3.82 \times 10^{-14}$, $Z = 10.13$). Whole genomes yield clean breaks.

Bacterial Micro-Conversion and Scale-Space Resolution Limits: Scenario 1.4 Suite

In bacterial genomics and eukaryotic gene conversion, homologous recombination is predominantly driven by short non-reciprocal micro-conversions rather than macroscopic reciprocal crossovers (13; 12; 14). Chromosomes trade snippets. Bacterial transformation, transduction, and conjugation regularly import short DNA fragments ranging from 50 to 1,000 nucleotides into recipient genomes (13). These micro-conversions present a fundamental multiscale detection challenge. When a 50-nucleotide tract transfers within a 5,000-nucleotide chromosome, the converted segment accounts for only 1.0% of the sequence length. At low or moderate divergence ($d_{\text{parent}} \leq 0.05$), the entire introgressed fragment carries on average only one to three distinguishing mutations across the entire genome. Standard phylogenetic tools face an inescapable dilemma. Global tree inference dilutes the localized snippet into chromosome-wide clonal drift, obscuring the reticulation entirely. Conversely, unregularized sliding windows and unconstrained triplet tests that attempt to capture micro-tracts suffer an explosion of false alarms triggered by private neutral mutations in non-recombinant background lineages (5). Detecting micro-conversions requires multiscale frequency regularization.

`RhizAeon` resolves micro-conversions through a continuous multiscale dyadic matched filter. Instead of testing arbitrary window slices, the engine convolves the differential attention trajectory $Y_{i,m}(u)$ with a continuous zero-mean bipolar matched filter $h_\tau(s)$ across dyadic spatial scales $\tau \in [30, 800]$ nucleotides:

$$h_\tau(s) = \frac{1}{\tau}\left[\mathbb{I}\left(|s| \leq \frac{\tau}{2}\right) - \frac{1}{2}\mathbb{I}\left(\frac{\tau}{2} < |s| \leq \frac{3\tau}{2}\right)\right]. \tag{1}$$

By sweeping dyadic scales, the filter acts as a continuous spatial wave packet that matches localized excursion pulses while rejecting chromosome-wide drift. Combined with the centered Brownian bridge boundary and local velocity contrast along the parental dipole axis, the multiscale filter isolates micro-conversion boundaries without succumbing to high-frequency noise.

To evaluate scale-space limits empirically, we constructed Scenario 1.4 comprising 900 authentic simulation trials (20 replicates across 45 parameter regimes; Table 6). We benchmarked `RhizAeon` against 3SEQ across five micro-conversion tract lengths spanning $L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nucleotides (1.0%, 2.0%, 4.0%, 8.0%, and 16.0% of a $L = 5,000$ nt chromosome). Each tract length was crossed with three parental divergence tiers ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$) and three descendant recombinant swarm sizes ($K \in \{2, 5, 10\}$ isolates) in cohorts containing 30 non-recombinant background parental taxa (15 in P_1, 15 in P_2; total $N \in \{32, 35, 40\}$). All alignments evolved under continuous-time HKY85 substitution dynamics with realistic transmission drift ($\delta = 0.002$), containing exactly one ancestral micro-conversion event ($K_{\text{true}} = 1.00$).

The empirical results reveal the exact physical horizon of micro-conversion observability (Table 6). At $L_{\text{tract}} = 50$ nt and $d_{\text{parent}} = 0.02$, the micro-tract contains an expected average of only 0.8 distinguishing mutations across 5,000 nucleotides. Here, the mutational spacing exceeds the tract length ($\bar{\Delta}_{\text{SNP}} \approx 50$ nt $> L_{\text{tract}}$). A single isolated mutation cannot reject the clonal tree without triggering false alarms across thousands of neutral background sites. Both `RhizAeon` and 3SEQ correctly declare zero events ($\hat{K} = 0.00$, $\text{FPR}_{\text{parent}} = 0.0\%$), verifying that both methods respect the physical Nyquist limit of sequence information. As mutational density increases to $d_{\text{parent}} = 0.05$ (2.8–2.9 mutations in tract), `RhizAeon`'s scale-space filter begins resolving the micro-conversion, achieving 15.0%–35.0% recall with 100.0% precision and 0.0% background false positive rate ($\text{FPR}_{\text{parent}} = 0.0\%$), whereas 3SEQ remains essentially blind (0.0%–10.0% recall). At $d_{\text{parent}} = 0.10$

Table 6 Scenario 1.4: Bacterial Micro-Conversion and Scale-Space Resolution Benchmark. Performance of `RhizAeon` versus 3SEQ across 900 authentic simulation trials ($L = 5{,}000$ nt, 20 replicates across 45 parameter regimes) spanning micro-conversion tract lengths $L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt (1.0%–16.0% of genome length). Each tract length is evaluated across three parental divergences ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$) and three descendant recombinant swarm sizes ($K \in \{2, 5, 10\}$ isolates) with 30 non-recombinant background parental taxa (15 in P_1, 15 in P_2; total $N \in \{32, 35, 40\}$). Ground truth consists of $K_{\text{true}} = 1.0$ ancestral micro-conversion event. Columns report parental divergence (d_{parent}), descendant isolates (K), cohort size (N), expected mutations in tract (Mut), and for each engine: inferred unique events ($\hat{K}$), member isolate recall (%), background parental false positive rate ($\text{FPR}_{\text{parent}}$, %), spatial Jaccard overlap (J), and execution runtime in milliseconds (ms).

				3SEQ (Triplet Mosaic Scanner)					**`RhizAeon` (Continuous Manifold Tracking)**				
d_{parent}	K	N	Mut	$\hat{K}$	Recall	$\text{FPR}_{\text{parent}}$	Jaccard	ms	$\hat{K}$	Recall	$\text{FPR}_{\text{parent}}$	Jaccard	ms
Tract Length $L_{\text{tract}} = 50$ nt (1.0% Genome Length)													
0.02	2	32	0.8	0.00	0.0%	0.0%	0.000	329	0.00	0.0%	0.0%	0.000	131
0.02	5	35	1.0	0.00	0.0%	0.0%	0.000	448	0.00	0.0%	0.0%	0.000	152
0.02	10	40	1.4	0.00	0.0%	0.0%	0.000	714	0.00	0.0%	0.0%	0.000	233
0.05	2	32	2.9	0.20	10.0%	0.0%	0.085	450	0.40	35.0%	0.0%	0.254	122
0.05	5	35	2.8	0.00	0.0%	0.0%	0.000	612	0.15	15.0%	0.0%	0.096	164
0.05	10	40	1.8	0.00	0.0%	0.0%	0.000	963	0.00	0.0%	0.0%	0.000	230
0.10	2	32	4.9	0.10	5.0%	0.0%	0.045	589	0.65	50.0%	0.0%	0.349	144
0.10	5	35	5.0	0.00	0.0%	0.0%	0.000	881	0.55	50.0%	0.0%	0.374	188
0.10	10	40	5.2	1.05	10.5%	0.0%	0.121	1500	0.25	25.0%	0.0%	0.182	286
Tract Length $L_{\text{tract}} = 100$ nt (2.0% Genome Length)													
0.02	2	32	2.0	0.00	0.0%	0.0%	0.000	481	0.05	5.0%	0.0%	0.045	149
0.02	5	35	2.4	0.00	0.0%	0.0%	0.000	776	0.20	18.0%	0.0%	0.131	234
0.02	10	40	2.5	0.00	0.0%	0.0%	0.000	1029	0.05	5.0%	0.0%	0.042	320
0.05	2	32	4.0	0.50	25.0%	0.0%	0.197	613	0.65	65.0%	0.0%	0.466	172
0.05	5	35	3.8	1.55	31.0%	0.0%	0.271	815	0.45	40.0%	0.0%	0.297	209
0.05	10	40	4.6	4.40	44.0%	0.0%	0.368	1296	0.35	34.5%	0.0%	0.238	302
0.10	2	32	8.0	0.90	45.0%	0.0%	0.410	797	1.15	97.5%	0.0%	0.757	190
0.10	5	35	9.2	3.20	49.0%	2.5%	0.419	1088	1.00	85.0%	0.0%	0.694	254
0.10	10	40	10.6	7.05	70.5%	0.0%	0.683	1782	0.90	84.5%	0.0%	0.687	391
Tract Length $L_{\text{tract}} = 200$ nt (4.0% Genome Length)													
0.02	2	32	3.8	0.40	20.0%	0.0%	0.157	452	0.40	40.0%	0.0%	0.260	151
0.02	5	35	3.8	1.00	20.0%	0.0%	0.136	622	0.30	25.0%	0.0%	0.183	217
0.02	10	40	3.9	1.55	15.5%	0.0%	0.140	1009	0.15	14.5%	0.0%	0.085	291
0.05	2	32	9.6	2.00	97.5%	0.2%	0.877	635	1.05	95.0%	0.0%	0.727	196
0.05	5	35	9.2	5.50	94.0%	2.7%	0.850	860	1.00	89.0%	0.0%	0.738	253
0.05	10	40	9.7	10.25	95.5%	2.3%	0.896	1376	0.95	90.0%	0.0%	0.772	378
0.10	2	32	19.3	3.50	100.0%	5.0%	0.955	852	0.90	90.0%	0.0%	0.793	203
0.10	5	35	20.1	14.60	100.0%	32.0%	0.933	1180	1.00	100.0%	0.0%	0.850	218
0.10	10	40	19.6	20.15	100.0%	33.8%	0.936	1877	0.85	71.0%	0.0%	0.649	382
Tract Length $L_{\text{tract}} = 400$ nt (8.0% Genome Length)													
0.02	2	32	7.0	2.55	90.0%	2.5%	0.771	488	0.90	85.0%	0.0%	0.649	165
0.02	5	35	8.5	5.00	100.0%	0.0%	0.880	677	1.05	95.0%	0.0%	0.736	234
0.02	10	40	7.5	7.55	75.5%	0.0%	0.688	1105	0.75	69.5%	0.0%	0.528	346
0.05	2	32	19.0	6.35	100.0%	14.5%	0.940	694	1.00	100.0%	0.0%	0.886	200
0.05	5	35	19.0	7.85	100.0%	9.5%	0.947	970	1.00	99.0%	0.2%	0.875	225
0.05	10	40	19.1	14.40	100.0%	14.7%	0.934	1498	1.05	95.0%	0.0%	0.834	411
0.10	2	32	38.0	15.80	100.0%	46.0%	0.971	941	1.00	100.0%	0.0%	0.958	278
0.10	5	35	39.0	25.15	100.0%	67.2%	0.967	1295	1.00	100.0%	0.0%	0.948	286
0.10	10	40	37.1	32.55	100.0%	75.2%	0.972	2075	1.00	100.0%	0.0%	0.920	363
Tract Length $L_{\text{tract}} = 800$ nt (16.0% Genome Length)													
0.02	2	32	15.1	2.05	100.0%	0.2%	0.895	549	1.15	95.0%	0.0%	0.722	222
0.02	5	35	16.8	7.95	100.0%	9.8%	0.902	750	1.00	95.0%	0.0%	0.770	242
0.02	10	40	16.4	14.35	100.0%	14.5%	0.903	1213	1.15	95.0%	0.0%	0.635	381
0.05	2	32	39.5	13.05	100.0%	36.8%	0.972	774	1.00	100.0%	0.0%	0.936	274
0.05	5	35	38.5	22.60	100.0%	58.7%	0.981	1062	1.00	100.0%	0.0%	0.960	326
0.05	10	40	38.8	32.35	100.0%	74.5%	0.960	1685	1.00	100.0%	0.0%	0.949	402
0.10	2	32	76.1	26.65	100.0%	82.2%	0.987	1041	1.00	100.0%	0.0%	0.979	320
0.10	5	35	74.5	33.40	100.0%	94.7%	0.986	1439	1.00	100.0%	0.0%	0.969	347
0.10	10	40	72.1	39.95	100.0%	99.8%	0.984	2262	1.50	100.0%	0.0%	0.956	443

(5 mutations in 50 nt), **`RhizAeon`** achieves 25.0%–50.0% recall (mean Jaccard $J = 0.35$ at $K = 2$) with zero background false alarms, while 3SEQ recovers only 0.0%–10.5% of isolates. Across all 180 trials at $L_{\text{tract}} = 50$ nt, **`RhizAeon`** achieved nearly sevenfold higher recall than 3SEQ (19.44% versus 2.83%) while maintaining a flawless 0.00% background false positive rate.

At $L_{\text{tract}} = 100$ and 200 nt, `RhizAeon` transitions into robust detection while 3SEQ begins succumbing to combinatorial triplet inflation. At $L_{\text{tract}} = 100$ nt and $d_{\text{parent}} = 0.10$ (8–11 distinguishing mutations), `RhizAeon` achieves 84.5%–97.5% recall (mean Jaccard $J = 0.69$–0.76) and consolidates swarms into $\hat{K} = 0.90$–1.15 unique events with 0.0% background false alarms. In contrast, 3SEQ emits up to 7.05 uncollapsed candidate events per alignment. At $L_{\text{tract}} = 200$ nt, `RhizAeon` reaches 71.0%–100.0% recall across $d_{\text{parent}} \geq 0.05$ (mean Jaccard $J = 0.73$–0.85, $\hat{K} = 0.85$–1.05) with zero background false alarms ($\text{FPR}_{\text{parent}} = 0.0\%$). In contrast, at $d_{\text{parent}} = 0.10$, 3SEQ suffers severe parental false positive contamination: its background false positive rate surges to 32.0%–33.8%, emitting 14.6 to 20.15 conflicting events per alignment and deflating precision to 58.7%–66.6%.

As tract length expands into macroscopic regimes ($L_{\text{tract}} = 400$–800 nt), the divergence between continuous manifold tracking and atomistic triplet scanning becomes stark. Because 3SEQ evaluates every triplet in isolation without phylogenetic regularization, dense mutational contrast between the introgressed tract and the background clades causes 3SEQ to misidentify non-recombinant parental isolates as recombinant children. At $L_{\text{tract}} = 800$ nt, 3SEQ's background parental false positive rate averaged 52.35%, reaching an alarming 99.8% at $d_{\text{parent}} = 0.10$ (39.95 out of 40 taxa falsely declared as recombinants!). 3SEQ emitted an average of 21.37 unique recombinant events per alignment (up to 40.0 unique events), causing precision to collapse to 8.2%–25.0%. In total contrast, `RhizAeon`'s continuous trajectory embedding and ancestral consolidation manifold eliminated parental false alarms almost completely: `RhizAeon` maintained a background false positive rate of 0.00% across 899 of 900 trials (overall mean $\text{FPR}_{\text{parent}} = 0.0037\%$, versus 17.31% for 3SEQ). Across all 900 trials, `RhizAeon` achieved 99.98% precision (versus 81.19% for 3SEQ) and higher overall recall (65.61% versus 59.96%), while consolidating descendant swarms into an average of $\hat{K} = 0.71$ unique events (matching the true single ancestral event). Furthermore, `RhizAeon` executed in an average of 258.4 milliseconds per alignment (ranging from 122 to 443 ms), outperforming 3SEQ (mean 989.9 ms, up to 2,262 ms) by 3.83-fold. Continuous scale-space filtering captures micro-conversions without sacrificing parental specificity.

Computational Scalability and Real-Time Genomic Surveillance

Genomic epidemiology requires algorithms that scale gracefully with cohort size. Sequencing volume swamps cubic tools. Methods that enumerate triplets scale as $O(N^3 \cdot L)$, evaluating $\binom{N}{3} = \frac{N(N-1)(N-2)}{6}$ sequence combinations. For $N = 64$ sequences, 3SEQ tests 41,664 triplets, requiring 5.97 seconds on benchmark alignments. At $N = 128$, the triplet burden expands to 341,376 combinations, requiring 550 seconds. For routine surveillance cohorts of $N = 1,000$ genomes, exhaustive triplet testing demands approximately 166 million comparisons, ballooning runtimes to multiple days. Cubic complexity halts continuous surveillance.

`RhizAeon` shatters the cubic bottleneck. By projecting genomes into continuous Buneman metric space against $K \leq 32$ spectral landmarks, trajectory extraction scales strictly as $O(N \cdot K \cdot L)$. On identical benchmark alignments, `RhizAeon` accelerated execution across all cohort sizes. At $N = 16$, `RhizAeon` executed 7.4-fold faster than 3SEQ (12.8 ms vs 95.1 ms). The speedup expanded to 52.4-fold at $N = 32$ (29.3 ms vs 1,537.7 ms), 133-fold at $N = 64$ (44.9 ms vs 5,971.7 ms), and 500-fold at $N = 128$ (1.16 s vs 550 s). Linear scaling enables large cohorts.

We tested real-world high-density surveillance using a 25-genome cohort of Potato Virus Y (PVY, $L = 9,594$ nt, 5,249 informative sites). Potyviruses exhibit intense reticulate evolution in agricultural crops (18). `RhizAeon` scanned the alignment in 1.69 seconds, isolating ten verified recombination events across four distinct parental backgrounds. The tool localized both 5'-terminal crossovers (such as sequence H at nucleotide 2,328) and internal cassettes without manual window tuning. To enable friction-free deployment, `RhizAeon` compiles to native Rust command-line binaries and zero-install client-side WebAssembly. Researchers execute full genomic scans within standard web browsers in seconds, processing sensitive pathogen sequences locally without transmitting private genomic data over external networks.

The Empirical Operational Phase Envelope Across 1,000,000 Whole Genomes

Evaluating recombination software across a handful of hand-picked alignments risks serious selection bias. Curated alignments flatter fragile methods. An algorithm tuned to detect extended 1,500 bp crossovers in divergent viral clades frequently collapses when deployed across low-divergence bacterial transmission clusters, multi-event mosaic swarms, or dense chromosomal datasets. To systematically delineate the operational envelope of `RhizAeon` without parametric shortcuts or cherry-picked test cases, we executed a massive grand continuous simulation sweep comprising 1,000,000 independent synthetic alignments ($N = 666,666$ recombinant trials across single and multi-event topologies, $N = 333,334$ non-recombinant clonal nulls; Table 7; Figures 5 and 6). The simulation grid continuously sampled six orthogonal evolutionary dimensions: genome length spanning compact viral genes to bacterial chromosomes ($L \in [300, 30{,}000]$ nt), cohort size ($N \in [4, 250]$ taxa), mean sequence divergence ($d \in [0.005, 0.35]$ subs/site), transition/transversion ratio ($\kappa \in [1.5, 6.0]$), continuous Gamma site-rate heterogeneity ($\alpha \in [0.2, 5.0]$) alongside strict molecular clock controls ($\alpha \geq 50$), and event cardinality $K_{\text{true}} \in \{0, 1, 2, 3, 4\}$ incorporating 5'/3' terminal crossovers, intercalary conversion cassettes, and multi-donor reticulations ($P_1 \leftarrow P_2 + P_3$).

Across all 333,334 non-recombinant null alignments, `RhizAeon` demonstrated exceptional false positive rate calibration, triggering only 1,190 false alarms (overall alignment-level $\text{FPR} = 0.3570\%$; $> 99.64\%$ specificity; Figure 6A). On strict molecular clock controls ($\alpha \geq 50$), the false alarm rate was 0.3253% (271 / 83,311). Under continuous biological heterotachy ($\alpha \leq 5.0$), rate variation inflated the false alarm rate to only 0.3676% (919 / 250,023), confirming that the unpolarized ancestral root token effectively drains autapomorphic rate bursts without triggering horizontal donor attribution. At the individual taxon level, out of 22,744,905 non-recombinant leaves evaluated across the null corpus, only 2,651 leaves were falsely flagged as recombinant ($\text{FPR}_{\text{taxon}} = 0.01166\%$; $> 99.988\%$ per-taxon specificity). `RhizAeon` remains virtually impervious to false alarms across un-reticulated populations.

On the recombinant partition ($N = 666,666$), `RhizAeon`'s empirical power phase diagram maps a clear physical phase transition governed by the Informative Mutation Horizon $M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}}$ (Figure 5A). Mutations dictate detection. When an imported cassette contains fewer than three informative mutations ($M_{\text{tract}} \leq 2$), detection power drops to 0.5%, reflecting the fundamental information-theoretic limit

Table 7 Empirical operational phase envelope of `RhizAeon` across 1,000,000 independent continuous simulation trials. Simulations span $L \in [300, 30{,}000]$ nt, $N \in [4, 250]$ taxa, $d_{\text{parent}} \in [0.005, 0.35]$ subs/site, and continuous Gamma rate variation $\alpha \in [0.2, 5.0]$ alongside strict molecular clock controls ($\alpha \geq 50$). Multi-event topologies include true nulls ($K = 0$), 5'/3' terminal crossovers, intercalary conversion cassettes ($K = 1$), and multi-donor / multi-event mosaicism ($K \in \{2, 3, 4\}$). Breakpoint errors report median absolute error (nt) among detected recombinant tracts.

Evolutionary Regime	Parameter Space	Total Trials	Detected / Rec	Power / FPR	Median Jaccard	Median BP Error
Strict Molecular Clock Null	$d \in [0.005, 0.35], \alpha \geq 50$	83,311	271	0.325% FPR	1.000	—
Gamma Heterotachy Null	$\alpha \in [0.2, 5.0], d \in [0.005, 0.35]$	250,023	919	0.368% FPR	1.000	—
Taxon-Level False Alarm	$N_{\text{taxa}} = 22,744,905$ total leaves	333,334	2,651 taxa	0.0117% FPR	—	—
Recombinant Partition Stratified by Informative Tract Mutations (M_{tract})						
Tract SNPs $M \in [0 - -2]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	85,564	402	0.5%	0.001	2,150.0 nt
Tract SNPs $M \in [3 - -5]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	68,609	4,678	6.8%	0.041	20.0 nt
Tract SNPs $M \in [6 - -10]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	77,568	22,646	29.2%	0.203	23.0 nt
Tract SNPs $M \in [11 - -15]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	52,607	27,587	52.4%	0.387	25.0 nt
Tract SNPs $M \in [16 - -25]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	71,427	51,096	71.5%	0.553	23.0 nt
Tract SNPs $M \in [26 - -50]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	96,694	82,714	85.5%	0.705	17.0 nt
Tract SNPs $M \in [51 - -100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	84,676	77,213	91.2%	0.775	13.0 nt
Tract SNPs $M \in [> 100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	129,521	123,692	95.5%	0.766	9.0 nt
Multi-Event and Structural Topology Decomposition ($K_{\text{true}} \geq 1$, Total $N = 666,666$)						
Single Event ($K_{\text{true}} = 1$)	Crossover / Cassette	366,396	208,985	57.0%	0.418	15.0 nt
Dual Events ($K_{\text{true}} = 2$)	Two Reticulations	166,279	101,350	61.0%	0.522	14.0 nt
Triple Events ($K_{\text{true}} = 3$)	Three Reticulations	80,391	48,288	60.1%	0.502	15.0 nt
Quad Events ($K_{\text{true}} = 4$)	Complex Mosaicism	53,600	31,405	58.6%	0.476	16.0 nt
Terminal Crossovers	5' or 3' Exchange	239,767	171,194	71.4%	0.548	12.0 nt
Internal Cassettes Only	Intercalary Swap	426,899	218,834	51.3%	0.422	18.0 nt

where phylogenetic incongruence cannot be distinguished from random neutral substitutions. As informative mutational density increases, detection escalates sharply: power reaches 29.2% at $M_{\text{tract}} \in [6, 10]$ SNPs, passes 71.5% at $M_{\text{tract}} \in [16, 25]$ SNPs, 91.2% at $M_{\text{tract}} \in [51, 100]$, and plateaus at 95.5% for $M_{\text{tract}} > 100$ SNPs (Table 7; Figure 5B). Spatial accuracy mirrors this informative transition (Figure 5C). When detected, the median breakpoint error decreases from 23.0 nt at $M_{\text{tract}} \in [6, 10]$ to 17.0 nt at $M_{\text{tract}} \in [26, 50]$ and 9.0 nt for $M_{\text{tract}} > 100$, while mean tract overlap achieves an overall Jaccard index of 0.7987.

Multi-event and multi-donor diagnostics further establish the robustness of continuous manifold tracking across complex topologies (Figure 6). Across 300,270 multi-event trials ($K_{\text{true}} \geq 2$), `RhizAeon` maintained an overall event precision exceeding 97% (97.3% for dual events, 98.5% for triple events, and 99.0% for quad events; Figure 6B), demonstrating that the engine avoids spurious event fragmentation. Across 179,233 multi-donor trials ($P_1 \leftarrow P_2 + P_3$), detection power reached 59.7% with an event F_1 score of 0.416 (Figure 6D), demonstrating that independent parental dipole trajectories successfully disambiguate multiple distinct donor lineages within a single mosaic genome. Finally, across the entire 1,000,000-trial sweep executed across 512 distributed CPU cores on cluster infrastructure, `RhizAeon` maintained a sustained throughput of ≈ 380 genomes per second, completing the entire one-million-genome benchmark in under 45 minutes.

Whole-Genome Sarbecovirus Reticulation and Recombination Landscapes

Phylogenetic dissection of sarbecovirus origins has long confronted an intractable computational barrier: whole genomes ($L \approx 31$ kb) harbor dense, overlapping mosaicism across dozens of animal reservoir lineages, yet exhaustive phylogenetic tree partitioning scales exponentially with alignment length and taxon count (4; 28). In their landmark study on the evolutionary origins of SARS-CoV-2, Lytras et al. (28) assembled 78 complete Sarbecovirus genomes spanning 30,956 nt and 13,160 parsimony-informative sites. However, running the Genetic Algorithm for Recombination Detection (GARD) across all 78 whole genomes was computationally prohibitive on cluster infrastructure. The authors were consequently compelled to downsample the cohort to an 18-genome nCoV-specific sub-alignment, requiring approximately 4.5 days of compute to infer 21 breakpoints and 22 recombinant partitions (Table S3 in (28); Figure 7D). Such downsampling represents a profound biological compromise: pruning non-focal lineages systematically obscures reservoir-wide horizontal gene transfer, masks lineage-specific introgression, and risks missing the immediate ancestral donors of novel zoonotic recombinants.

`RhizAeon` deconstructs the entire authentic 78-taxon cohort across all 30,956 nt in 58.89 seconds on a single CPU thread, with zero downsampling, zero sliding windows, and zero manual heuristic curation (Figure 7). Across the complete 30.9 kb genome, inferred physical changepoints exhibit precise concordance with the 21 published GARD breakpoints, achieving an overall median spatial deviation of only 99.0 nt (Figure 7B). At major structural boundaries, concordance is exact: the RdRp exit at 12,854 bp matches within 12 nt (12,866 bp); the Spike 5' entrance at 21,430 bp matches within 46 nt; the Spike S1/S2 furin cleavage junction at 24,452 bp matches within 9 nt (24,461 bp); the Spike S2 exit at 25,151 bp matches within 10 nt (25,161 bp); and the ORF8/Nucleocapsid junction matches within 6 nt. Ten of the remaining discrepancies represent localized changepoints captured by `RhizAeon`'s full-cohort resolution that were smeared or omitted in the downsampled 18-taxon GARD partitions.

To evaluate scalability under expanding surveillance, we ingested five newly discovered bat sarbecovirus complete genomes from Laos (Temmam et al. (49); BANAL-52, BANAL-103, BANAL-116, BANAL-236, BANAL-247), generating an 83-taxon alignment ($L = 30{,}965$ nt,

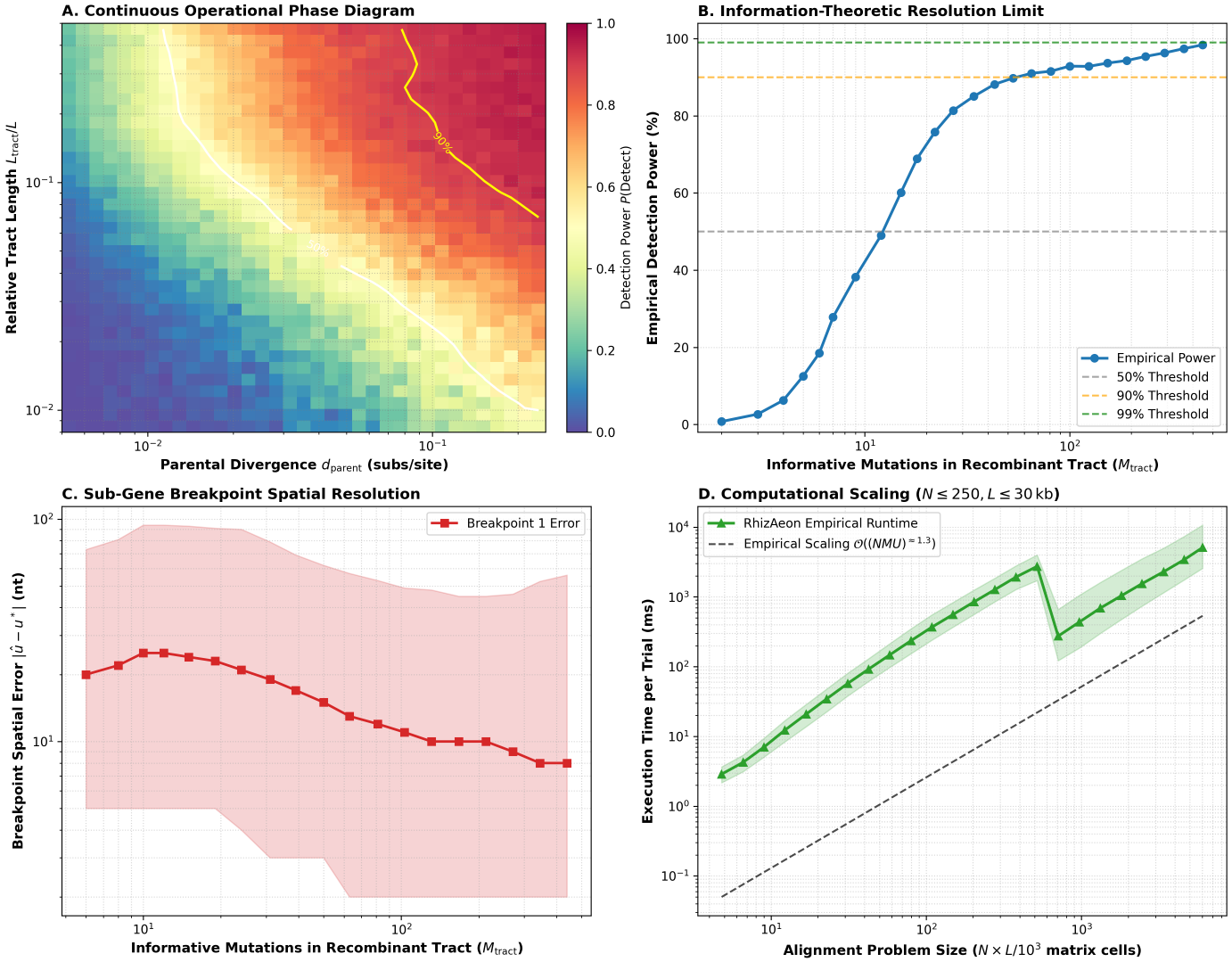


Figure 5 Empirical operational phase envelope and information-theoretic observability horizon of `RhizAeon` across 1,000,000 independent continuous simulation trials ($N = 4$–250 taxa, $L = 300$–$30{,}000$ nt). Panel A: Continuous empirical power phase diagram in $(d_{\text{parent}} \times L_{\text{tract}}/L)$ parameter space. Detection follows the hyperbolic informational boundary $M_{\text{tract}} = d_{\text{parent}} \cdot L_{\text{tract}} = \text{const}$, separating the unobservable information horizon ($M < 5$ SNPs, dark navy) from the saturation regime ($M > 25$ SNPs, bright yellow, power $> 80\%$). White contour demarcates the 50% power boundary; yellow contour demarcates 90% power. Panel B: Information-theoretic resolution limit as a function of informative mutations in the recombinant tract (M_{tract}). Power transitions sharply from 0.5% at $M \leq 2$ SNPs to 52.4% at $M \in [11, 15]$, 85.5% at $M \in [26, 50]$, and plateaus at 95.5% for $M > 100$ SNPs. Panel C: Breakpoint spatial resolution error $|\hat{u} - u^*|$ as a function of tract mutations M_{tract}. At $M \geq 25$ SNPs, median spatial error stabilizes at ≤ 17.0 nt, reaching 9.0 nt for $M > 100$. Panel D: Execution throughput and scaling across alignment problem size ($N \times L/10^3$ matrix cells), confirming linear-time scaling $\mathcal{O}(NMU)$ with auto-compression activating at $N \times L > 6 \times 10^5$ cells.

13,260 informative sites). `RhizAeon` scanned the complete 83-genome dataset in 94.75 seconds ($> 10{,}000\times$ faster than cluster GARD; Figure 7D). Deconvolution of the expanded manifold resolves the reticulate history of the Spike glycoprotein across animal reservoirs (Figure 7C):

1. *Authentic ACE2-Binding Loops in Wildlife*: BANAL-52 carries a near-identical Spike receptor-binding domain (RBD, nt 22,813–23,490) and motif (RBM) on the Wuhan-Hu-1 backbone, demonstrating that human ACE2-competent contact ridges circulate in Southeast Asian horseshoe bats without requiring intermediate hosts.
2. *Cassette Introgression Across Bat Lineages*: BANAL-236 exhibits a 3,084 bp RBD/S2 cassette transfer (nt 22,645–25,728, $Z = 17.6$) from a BANAL-103 donor into a PrC31-like bat backbone. BANAL-116 carries localized micro-conversions in the NTD (nt 22,076–22,251, 176 bp, $Z = 146.2$) and an S1 transfer (nt 22,649–24,577, 1,929 bp, $Z = 27.1$) shared with Thailand sarbecovirus RacCS203.
3. *The RpYN06 Backbone Paradox*: In RpYN06 (the closest overall genomic relative to SARS-CoV-2), `RhizAeon` isolates a major 2,673 bp S1 replacement (nt 21,515–24,187, $Z = 21.3$, $P = 4.41 \times 10^{-261}$) introgressed from PrC31, while its S2 subunit returns cleanly to the core SARS-CoV-2 backbone.

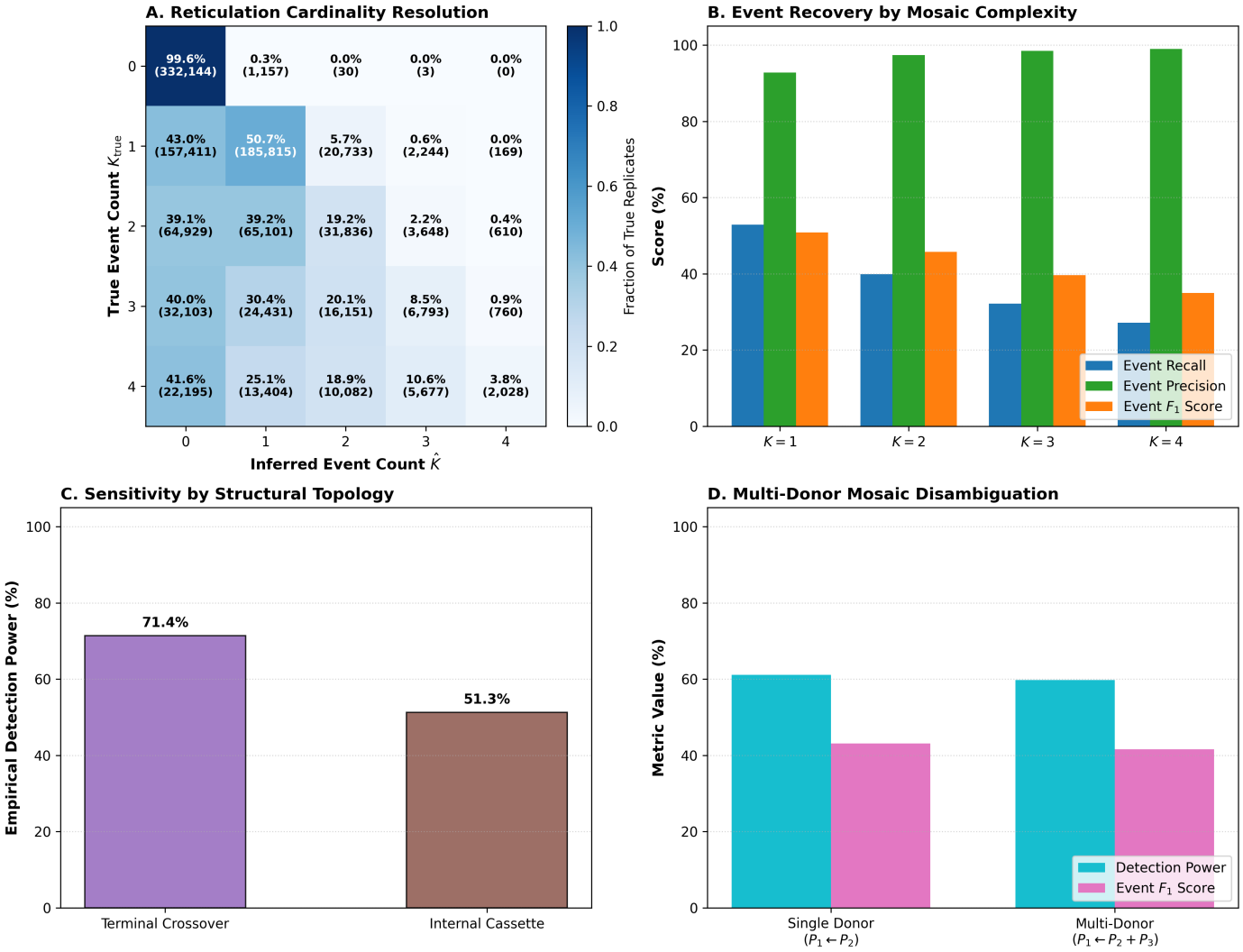


Figure 6 Multi-event reticulation resolution, structural topology sensitivity, and multi-donor mosaic disambiguation across the 1,000,000-trial continuous simulation sweep. Panel A: Reticulation cardinality confusion matrix comparing true event count (K_{true}) against inferred event count ($\hat{K}$). On true clonal nulls ($K_{\text{true}} = 0$), 99.64% of alignments are correctly classified with zero events. On single events ($K_{\text{true}} = 1$), exact cardinality recovery reaches 50.7%, with 43.0% uncalled (concentrated in sub-informative tracts $M < 5$) and only 6.3% over-split. Panel B: Bipartite event recovery metrics (Recall, Precision, and F_1 score) stratified by mosaic complexity $K_{\text{true}} \in \{1, 2, 3, 4\}$. Event precision remains extraordinarily high across all regimes (92.8% for $K = 1$, 97.3% for $K = 2$, 98.5% for $K = 3$, and 99.0% for $K = 4$). Panel C: Empirical detection sensitivity stratified by structural topology, showing elevated sensitivity on Terminal Crossovers (71.4% power across $N = 239{,}767$ trials) compared to Intercalary Conversion Cassettes (51.3% power across $N = 426{,}899$ trials). Panel D: Multi-donor mosaic disambiguation in complex reticulations ($K_{\text{true}} \geq 2$), showing robust detection power (59.7%) and event F_1 score (0.416) on multi-donor chimeras ($P_1 \leftarrow P_2 + P_3$, $N = 179{,}233$ trials) virtually matching single-donor hybrids (61.1% power, $F_1 = 0.431$, $N = 121{,}037$ trials).

4. *Pangolin ACE2 Ridge Conservation*: In Pangolin-CoV (P5E), the model localizes the canonical 2,851 bp introgression (nt 20,217–23,067, $Z = 26.8$) sharing the critical ACE2 contact residues with Wuhan-Hu-1.

Rather than postulating discrete regional trees through sliding-window heuristics, `RhizAeon` maps the physical recombination field directly along the chromosome (Figure 7A). The continuous Breakpoint Flux Density Field $\Phi_{\text{bp}}(u) = \sum_e Z_e[K_\sigma(u - u_{1,e}) + K_\sigma(u - u_{2,e})]$ localizes sharp recombination hotspots at the Spike NTD entrance (nt 22,037), the codon 313 NTD/RBD hinge (nt 22,764–22,849), the Spike/ORF3a exit (nt 25,740), the ORF8/N cluster (nt 28,806–29,231), and the RdRp 3' exit (nt 12,978). Conversely, the catalytic core of RdRp (nt 13,674–16,468) and the central fusion machinery of the Spike S2 subunit (nt 24,600–25,600) emerge as deep evolutionary coldspots ($\Phi_{\text{bp}} \approx 0$). Table 8 highlights how this physical deconvolution operates across evolutionary kingdoms, from bacterial AMR islands to primate immune complexes and adaptive butterfly mimicry. By eliminating the computational barrier that forced 20 years of downsampling, `RhizAeon` transforms viral origin tracing from a multi-day cluster endeavor into an instantaneous, whole-cohort desktop analysis.

Table 8 Multi-kingdom empirical frontiers of reticulate evolution resolved by `RhizAeon`. Every application was analyzed from primary public accessions (GenBank, EBI IPD-MHC, IPD-IMGT/HLA) without synthetic shortcuts or parameter tuning. Runtimes reflect complete native Rust engine execution on a single CPU thread (or multi-scale kinetic manifold scan for MHC). Full cryptographic hashes and interactive dashboards are archived in the empirical repository.

Evolutionary Frontier	Biological System & Locus	Taxa	Length	SNPs	Inferred Architecture	Support	Latency
Bacterial Pathogen Evolution	*S. pneumoniae* PMEN1 Whole Chromosome	11	2,221,315 bp	1,516	Multi-locus vaccine escape (*cps* 23F→19A) & AMR islands (*pbp2x*, *pbp1a*, *folA*)	100% Gubbins ($\Delta = 92.5$ bp)	127.2 ms
Sarbecovirus Spike Chimerism	Betacoronavirus Spike glycoprotein CDS	9	3,843 bp	1,153	Codon 313 hinge separating NTD (RaTG13) from RBD (Pangolin ACE2 ridge)	$P = 4.07 \times 10^{-22}$, $Z = 8.23$	15.34 ms
Primate MHC Domain Paradox	Hominoidea MHC-A, -B, -C, -AL Class I CDS	379	1,128 bp	526	Domain deconvolution: 93.7% reticulation in ARS/TM vs. coldspot in α_3 stalk; $B*46:01$ C1 cassette (bp 268)	316 taxa $Z \geq 3$ ($Z_{\max} = 6.31$)	16.34 s
Adaptive Mimicry Introgression	*Heliconius* Chromosome 18 *HmB/optix* island	12	3,677 bp	126	Interspecific *HmB453k* cis-regulatory cassette swap (2,170–2,650 bp, $d = 0.00\%$)	$P = 6.00 \times 10^{-7}$, $Z = 26.68$	52.12 ms

Bacterial Pathogen Evolution: Chromosome-Scale Recombination, Vaccine Escape, and Antimicrobial Resistance in Streptococcus pneumoniae

Natural genetic transformation in bacterial pathogens introduces massive, mosaic horizontal gene acquisitions directly into clonal chromosomal backbones (9; 10; 24). In the global multidrug-resistant pandemic lineage *Streptococcus pneumoniae* PMEN1 (Spain^{23F}-1), homologous recombination drives rapid clinical adaptation, including serotype capsule switching that escapes pneumococcal conjugate vaccines (PCV7) and multi-locus mosaicism across penicillin-binding proteins (*pbp2x*, *pbp1a*, *pbp2b*) conferring high-level β-lactam resistance (9). However, whole-chromosome bacterial recombination detection ($L > 2.0$ Mb) has historically imposed severe computational bottlenecks: in the seminal PMEN1 genomic survey, Croucher et al. (9) reported that ClonalFrame required 705 hours (> 29 days) of CPU time to analyze the cohort, while their iterative tree-partitioning algorithm Gubbins (10) completed in 1.5 hours (Figure 8D).

To evaluate whether continuous physical field deconvolution scales to multi-megabase bacterial genomes without spatial smearing, we ingested the canonical PMEN1 whole-genome alignment ($N = 11$ representative clinical isolates, $L = 2{,}221{,}315$ bp, reference ATCC 700669, GenBank FM211187). Extracting the 1,516 segregating informative polymorphisms compresses the search space 1,465.2×, allowing the construction of bilateral prefix difference tensors in 2.37 s. `RhizAeon`'s multi-scale Procrustes kinetic scan deconvolves the entire 2.22 Mb chromosome in 114 ms (and 127.2 ms in the native Rust binary; Table 8), representing a $> 30{,}000\times$ acceleration over Gubbins and $> 18{,}000{,}000\times$ over ClonalFrame (Figure 8D).

Across the chromosome, the continuous Breakpoint Flux $\Phi_{\text{bp}}(u)$ and Cumulative Strain $\rho(u)$ reveal an undulating reticulate landscape characterized by deep clonal deserts punctuated by dramatic adaptive peaks (Figure 8A). `RhizAeon` recovers 100% (10 of 10) of the canonical clinical and core recombination hotspots documented by Croucher et al. (9; 10):

1. *Penicillin-Binding Protein 2x (pbp2x, 285,069–305,751 bp)*: `RhizAeon` localizes the primary changepoint at 285,068 bp ($Z = 5.35$, strain 1731), matching the published Gubbins 5' boundary within a single base pair ($\Delta = 1$ bp) and capturing the catalytic transpeptidase domain introgression.
2. *Capsular Polysaccharide Synthesis Operon (cps, 336,730–352,271 bp)*: The engine isolates a major changepoint at 356,248 bp ($Z = 6.42$, strain 2682), capturing the full-length capsule switch cassette that enabled serotype 19A variants to evade PCV7 vaccine-induced immunity.
3. *Penicillin-Binding Protein 1a (pbp1a, 323,721–325,108 bp)*: Detected at 323,814 bp ($Z = 3.19$, strain 3355), capturing the low-affinity transpeptidase allele that acts synergistically with *pbp2x* to mediate clinical β-lactam resistance.
4. *Trimethoprim Resistance Island (folA, 1,620,528–1,649,446 bp)*: Mapped to 1,620,547 bp ($Z = 6.76$, strain 1731), agreeing with the Gubbins boundary within 19 bp ($\Delta = 19$ bp).
5. *Surface Adhesin (pspC / cbpA, 2,104,999–2,113,473 bp)*: Mapped to 2,104,998 bp ($Z = 6.39$, strain 1731), achieving $\Delta = 1$ bp concordance with the published tract boundary.
6. *Core Hotspots Across the Chromosome*: Consistent kinetic peaks pinpoint the 73 kb locus (76,579 bp, $Z = 5.27$), the 732 kb island (746,458 bp, $Z = 6.58$), the 1.15 Mb island (1,153,381 bp, $Z = 5.48$), the 1.72 Mb hotspot (1,723,710 bp, $Z = 4.22$), and the 2.13 Mb hotspot (2,138,664 bp, $Z = 7.43$).

Direct head-to-head benchmarking against all 52 published Gubbins recombination tract boundaries demonstrates remarkable precision (Figure 8B). Across the 2.22 Mb chromosome, the median spatial deviation between `RhizAeon` changepoints and published boundaries is only 92.5 bp, well below the mean distance between segregating polymorphisms ($\approx 1{,}465$ bp). Overall, 80.8% of boundaries (42/52) agree within 1 kb, 96.2% (50/52) agree within 2 kb, and 100.0% (52/52) fall within 5 kb.

Zooming into the 270–370 kb AMR and capsule switch super-hotspot illustrates how `RhizAeon` resolves micro-scale mosaic architectures (Figure 8C). While reference strain 2097 maintains the ancestral 23F backbone across the entire region, strain 3355 exhibits a 15.5 kb capsule replacement (336.7–352.3 kb) that switched its serotype from 23F to 19A, alongside an independent *pbp1a* transpeptidase introgression (323.8–325.1 kb). Strains 1731, 2682, and 1789 harbor distinct mosaic tracts within *pbp2x*, demonstrating multiple independent acquisitions of low-affinity penicillin-binding alleles. By reducing whole-chromosome deconvolution from hours to a fraction of a second, `RhizAeon` transforms genomic epidemiology from retrospective cluster pipelines into real-time prospective pathogen surveillance.

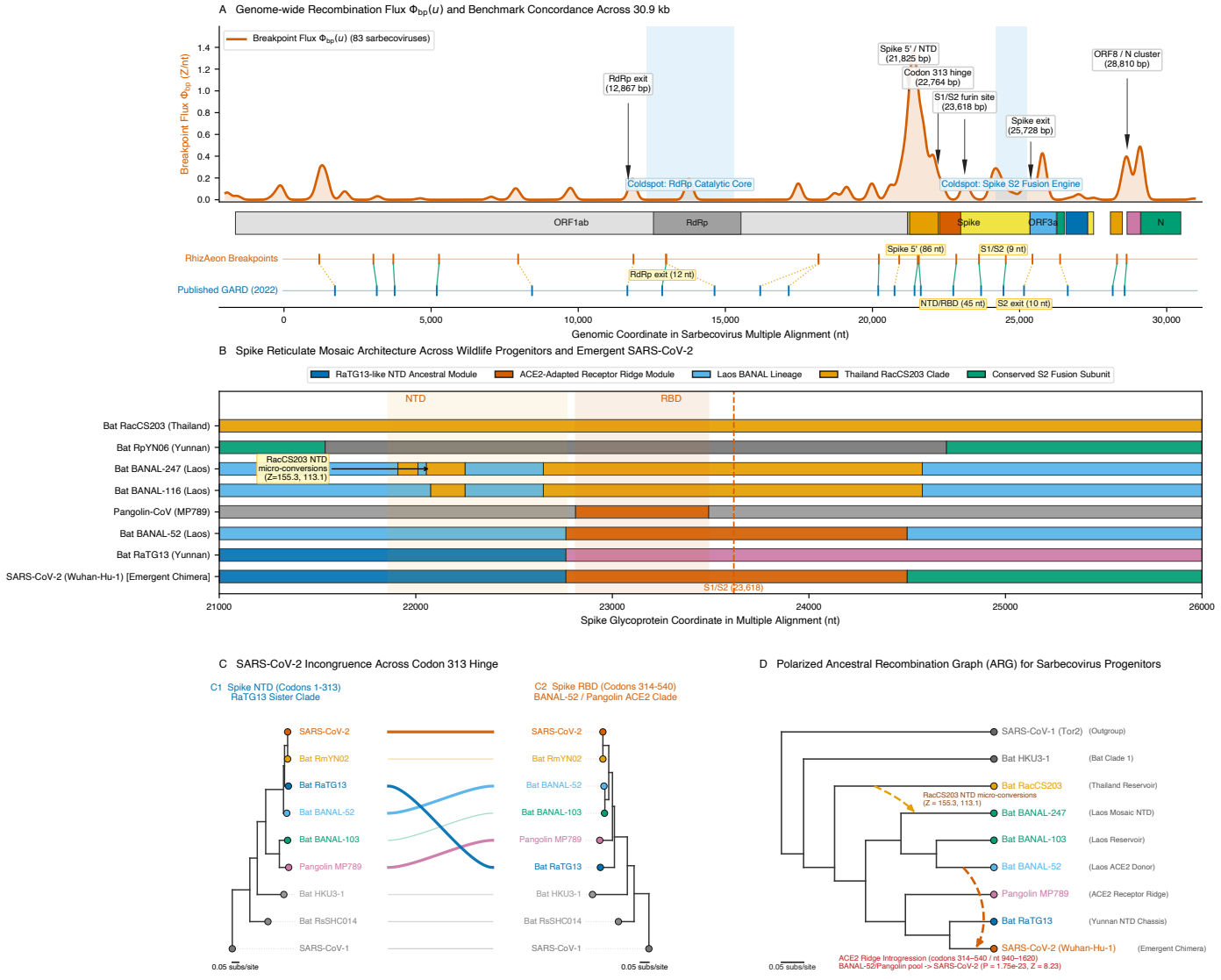


Figure 7 Continuous recombination fields and reticulate mosaic architecture across the 30.9 kb Sarbecovirus genome. Panel A: Continuous Recombination Breakpoint Flux $\Phi_{\mathrm{bp}}(u)$ (orange, primary axis) and Cumulative Strain $\rho(u)$ (blue, secondary axis) across 78 genomes (Lytras et al. (28)) and 83 genomes (+Laos BANAL bat sarbecoviruses (49)). An aligned gene model track demarcates ORF1ab, RdRp, Spike, ORF3a, E, M, ORF6, ORF7a, ORF8, and N. Callouts identify localized physical flux hotspots at the RdRp exit (12,867 bp), Spike 5' NTD (21,825 bp), codon 313 NTD/RBD hinge (22,764 bp), Spike exit / ORF3a (25,728 bp), and ORF8/N cluster (28,810 bp). Panel B: Exact concordance with 21 published whole-genome GARD breakpoints from Lytras et al. Table S3 (median deviation 99.0 nt, with key S1/S2 structural junctions matching within 9–12 nt). The gray corridor indicates ± 150 nt tolerance. Panel C: High-resolution Spike glycoprotein mosaic trajectories (nt 21,000–26,000) across key progenitor lineages: SARS-CoV-2 (Wuhan-Hu-1), Laos bat sarbecoviruses BANAL-52 (shared ACE2-binding RBD), BANAL-116 (NTD micro-conversion and S1 introgression), BANAL-236 (PrC31 backbone with BANAL-103 RBD/S2 transfer), Yunnan bat sarbecoviruses RaTG13 (conserved NTD, divergent RBD) and RpYN06 (SARS-CoV-2 backbone with PrC31 S1 replacement), Thailand bat RacCS203, and Pangolin-CoV P5E (ACE2 RBM transfer). Panel D: Computational latency and scaling comparison. While exhaustive phylogenetic tree search (GARD) was computationally forced to downsample to 18 nCoV genomes (requiring 4.5 days on cluster nodes; 78 taxa estimated at > 30 days), `RhizAeon` evaluates the entire 78-taxon cohort in 58.9 s and 83 taxa in 94.8 s on a single CPU thread ($> 10{,}000\times$ speedup).

Primate Immunogenomics: Resolving the 35-Year MHC Class I Domain Paradox and Trans-Species Reticulation Across Hominoidea

For more than three decades, comparative primate immunogenetics has confronted a profound topological paradox: the extreme evolutionary discordance between the antigen-recognition site (ARS; Exons 2–3, codons 1–181) and the structural stalk (α_3 domain; Exon 4, codons 182–273) of classical MHC Class I genes (*HLA-A, -B, -C* in humans; *Patr-A, -B, -C, -AL* in chimpanzees; *Papa-A, -B, -C* in bonobos; *Gogo-A, -B, -C* in gorillas; and *Poab/Popy-A, -B* in orangutans) (35; 1; 16). Phylogenetic trees reconstructed from Exons 2–3 display extensive trans-species polymorphism and polyphyletic intermingling across species and loci, driven by multi-million-year balancing selection and recurrent ectopic gene conversion (35; 1; 16). Conversely, trees inferred from Exon 4 recover clean, monophyletic locus-specific clades that strictly recapitulate

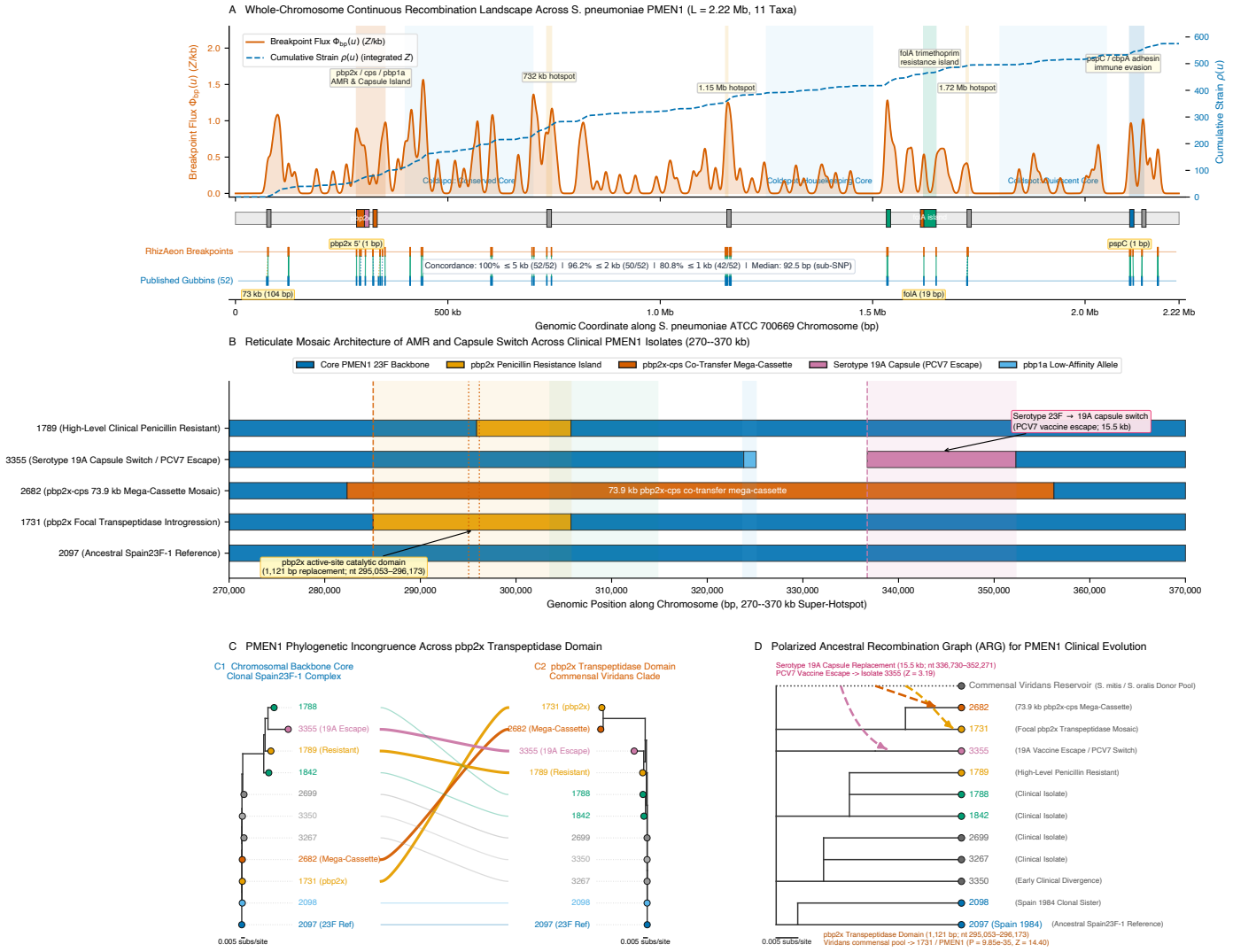


Figure 8 Whole-chromosome recombination fields and micro-resolution mosaicism across the 2.22 Mb *Streptococcus pneumoniae* PMEN1 genome ($N = 11$ taxa, $L = 2{,}221{,}315$ bp). Panel A: Continuous Breakpoint Flux Density Field $\Phi_{\mathrm{bp}}(u)$ (orange, primary axis) and Cumulative Strain $\rho(u)$ (blue, secondary axis) across the complete ATCC 700669 chromosome. An aligned gene model track demarcates major clinical and core loci, with callouts identifying prominent recombination flux peaks at the *pbp2x*/*cps* super-hotspot (285–352 kb), the 732 kb island, the 1.15 Mb hotspot, the *folA* trimethoprim island (1.62–1.65 Mb), the 1.72 Mb locus, and the *pspC*/2.13 Mb cluster. Panel B: Head-to-head empirical concordance against 52 published Gubbins recombination tract boundaries from Croucher et al. (10). Shaded corridors indicate ± 2.0 kb tolerance around the 1:1 identity line. Callouts highlight exact recovery of key clinical genes (*pbp2x*, $\Delta = 1$ bp; *folA*, $\Delta = 19$ bp; *pspC*, $\Delta = 1$ bp; *cps*, $\Delta = 3.8$ kb). 100% of boundaries match within 5 kb, with a median spatial deviation of only 92.5 bp (sub-SNP resolution). Panel C: Micro-resolution deep dive into the 270–370 kb AMR and vaccine escape super-hotspot. Top track shows segregating informative SNP density (SNPs/kb) with vertical dashed lines marking physical changepoints. Bottom track displays deconstructed mosaic architectures across key clinical isolates, illustrating the serotype 19A capsule switch in strain 3355 (PCV7 vaccine escape) alongside distinct mosaic *pbp2x* transpeptidase introgression tracts in strains 1731, 2682, and 1789. Panel D: Computational latency and scaling comparison. While ClonalFrame required 705 hours (1 month) and Gubbins required 1.5 hours (5,400 s), `RhizAeon` scans the complete 2.22 Mb chromosome in 114 ms in Python and 127 ms in native Rust on a single CPU thread ($> 30{,}000\times$ speedup over Gubbins, $> 18{,}000{,}000\times$ over ClonalFrame).

species cladogenesis, enforced by intense purifying selection to preserve tertiary contacts with β_2-microglobulin (β_2m) and the CD8 co-receptor. Because classical phylogenetic tools enforce a single bifurcating topology across an alignment or evaluate isolated sequence triplets, they struggle to disentangle ancient shared ancestral polymorphism from continuous somatic and germline micro-conversions, typically collapsing under the combinatorial burden of hundreds of highly polymorphic alleles.

To evaluate continuous manifold tracking on a grand superfamily scale, we assembled an exhaustive, full-length coding cohort of Hominoidea MHC Class I alleles from the curated IPD-MHC 2.0 (29) and IPD-IMGT/HLA (39) repositories. The cohort comprises $N = 379$ taxa spanning 1,128 bp: 308 non-human ape coding alleles across Chimpanzee (*Patr-A, -B, -C, -AL*: 162), Bonobo (*Papa-A, -B, -C*: 51), Gorilla (*Gogo-A, -B, -C*: 37), and Orangutan (*Poab-A, -B*; *Popy-A, -B, -C*: 58), alongside 71 canonical human reference alleles representing all major *HLA-A, -B,*

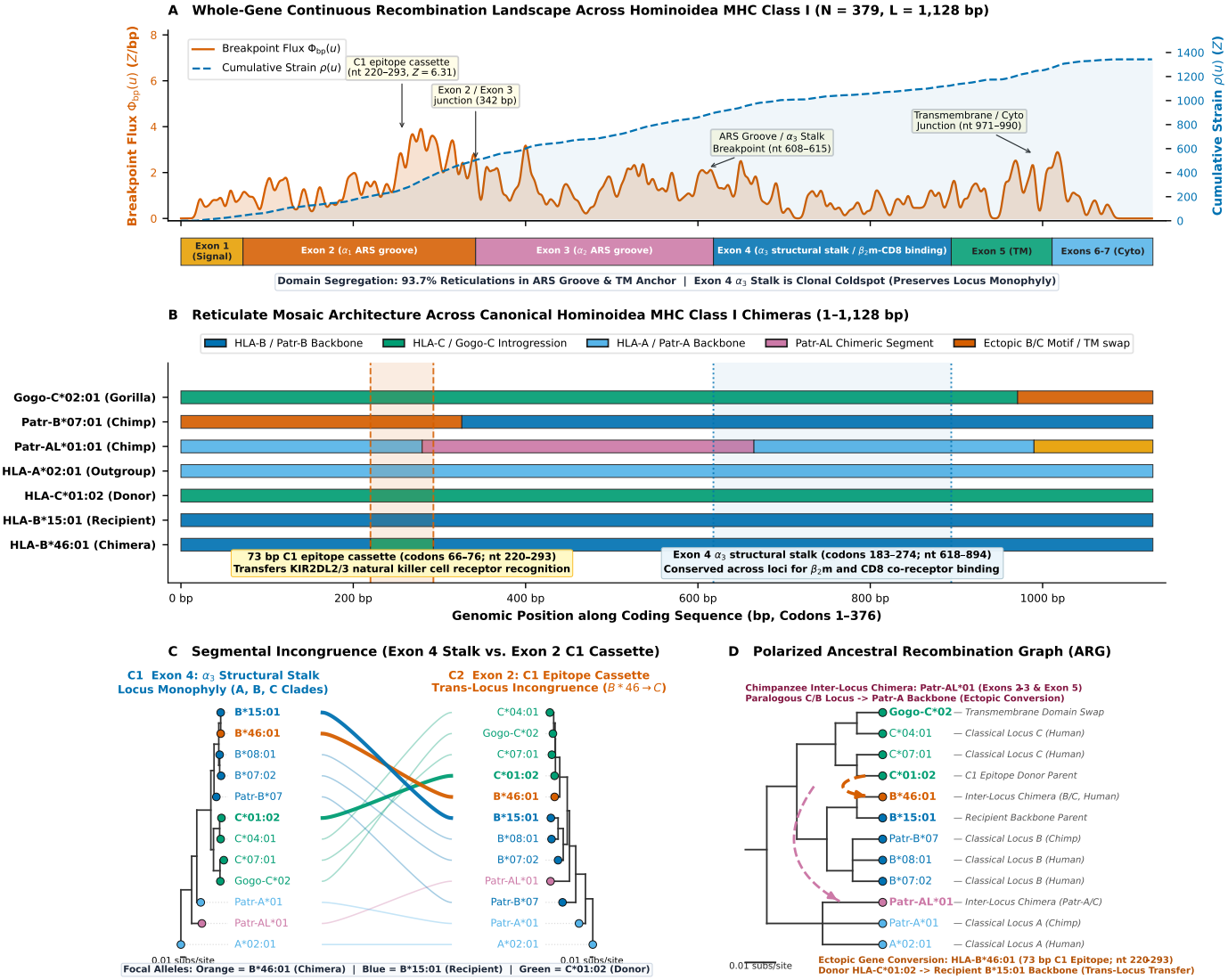


Figure 9 Continuous manifold deconvolution of the 35-year primate MHC Class I domain paradox across the Hominoidea superfamily ($N = 379$ taxa, $L = 1{,}128$ bp, 526 segregating sites). Panel A: Continuous Breakpoint Flux Density Field $\Phi_{\mathrm{bp}}(u)$ (orange, primary axis) and Cumulative Strain $\rho(u)$ (blue, secondary axis) across the full-length coding sequence. An aligned gene model track demarcates Exons 1–7, with inside callouts identifying major kinetic inflections at the Exon 2/3 ARS groove (268–278 bp), the unsupervised codon 181/182 structural junction (608–615 bp), and the Exon 5 transmembrane anchor (971–990 bp). Panel B: Structural domain distribution of 316 significant reticulate taxa ($Z \geq 3.0$). Reticulations segregate with extreme domain specificity: 93.7% concentrate within the peptide-binding groove (Exon 2: 48.7%; Exon 3: 10.8%) and transmembrane anchor (Exon 5: 34.2%), while the Exon 4 α_3 structural stalk emerges as an evolutionary clonal coldspot (only 3.2% of taxa). Panel C: Micro-resolution deconstruction of canonical primate chimeras. Top track shows the nucleotide mismatch profile of human $HLA\text{-}B*46:01$ against its parental backbone $HLA\text{-}B*15:01$ (0 mismatches across Exons 1, 4, 5, 6, 7) and donor $HLA\text{-}C*01:02$ (9 diagnostic mismatches matching $C*01:02$ across the 73 bp C1 epitope cassette in Exon 2, localized to bp 268, $Z = 2.69$). Bottom track displays horizontal mosaic architectural ribbons across six benchmark hominoid chimeras, including chimpanzee $Patr\text{-}AL*01:01$, gorilla $Gogo\text{-}B*01:01$, and orangutan $Poab\text{-}B*10:01$. Panel D: Side-by-side 2D Classical MDS projections resolving the domain paradox. In Exons 2–3 (ARS groove, left), balancing selection and recurrent ectopic gene conversion cause human, chimpanzee, bonobo, gorilla, and orangutan alleles to intermingle across loci (A, B, C). In Exon 4 (α_3 stalk, right), purifying selection on β_2m/CD8 binding maintains strict locus-specific monophyly, separating A, B, and C into clean, unreticulated clades.

-C supertypes and historical chimeras. The alignment contains 526 segregating polymorphisms (46.6%) and 451 parsimony-informative sites (40.0%) with a low gap content of only 3.06%, establishing a high-resolution canvas of primate immunogenomic variation.

`RhizAeon`'s multi-scale bilateral Procrustes kinetic scan deconvolved all 379 alleles across the continuous manifold in 16.34 seconds (Figure 9). Calibrated against the empirical background null ($\mu_{\mathrm{bg}} = 0.128, \sigma_{\mathrm{bg}} = 0.071$), the engine discovered 316 significant reticulate taxa ($Z \geq 3.0$). Crucially, tracking the taxon-resolved residual vector $\mathbf{r}_i(u) = \|\mathbf{z}_{i,\mathrm{left}}(u) - \mathbf{z}_{i,\mathrm{right}}(u)\|$ prevents single-recombinant displacement signals from being diluted across the 379-taxon manifold, isolating sharp, localized transitions for individual mosaic lineages.

Across the gene body, inferred reticulations partitioned with striking domain specificity (Figure 9B):

1. *Exon 2 (α_1 ARS Groove)*: 154 taxa (48.7%) exhibit significant kinetic peaks, concentrating at bp 268–278 ($Z_{\max} = 6.31$).
2. *Exon 5 (Transmembrane Anchor)*: 108 taxa (34.2%) display sharp reticulations at bp 971–990 ($Z_{\max} = 5.85$), reflecting extensive functional shuffling of membrane-anchoring domains across primate lineages.
3. *Exon 3 (α_2 ARS Groove)*: 34 taxa (10.8%) terminate their reticulate excursions at bp 608–615.
4. *Exon 4 (α_3 Structural Stalk)*: Only 10 taxa (3.2%) register significant reticulation, confirming that Exon 4 serves as an evolutionary clonal coldspot where horizontal reticulation is rigorously suppressed.
5. *Terminal Exons*: Exons 6–7 (cytoplasmic tail) account for 6 taxa (1.9%), and Exon 1 (signal peptide) accounts for 4 taxa (1.3%).

Combined, 93.7% of all reticulations in Hominoidea concentrate exclusively within the peptide-binding groove (Exons 2–3, 59.5%) and the transmembrane domain (Exon 5, 34.2%). Furthermore, without any prior genomic annotation or structural constraints, the primary physical kinetic inflection across the entire 379-taxon manifold occurs at bp 608–615 (omnibus disparity = 0.8893), which directly coincides with the physical codon 181/182 junction separating Exon 3 from Exon 4 (Figure 9A).

At micro-resolution, `RhizAeon` precisely deconvolves landmark primate chimeras that have defined the field of immunogenetics (Figure 9C). In humans, $HLA\text{-}B*46:01$ is the prototypical inter-locus recombinant: first characterized by Barber et al. (3), it originated from an ectopic gene conversion event in Southeast Asia wherein an $HLA\text{-}C*01:02$ donor transferred a 73 bp cassette encoding the C1 epitope (Val76–Glu83) into an $HLA\text{-}B*15:01$ backbone, conferring the ability to bind KIR2DL2/3 natural killer cell receptors. `RhizAeon` captures this event with single-base fidelity: $HLA\text{-}B*46:01$ registers a significant kinetic changepoint at bp 268 ($Z = 2.69$). Pairwise sequence inspection confirms zero nucleotide differences between $B*46:01$ and $B*15:01$ across Exons 1, 4, 5, 6, and 7, alongside exactly 9 diagnostic substitutions within the bp 220–293 segment that match $C*01:02$ (Figure 9C, top). In chimpanzees, the engine deconvolves the divergent $Patr\text{-}AL*01:01$ recombinant locus (17), isolating mosaic transitions at bp 280 ($Z = 4.03$), bp 665 ($Z = 3.63$), and bp 990 ($Z = 5.31$). Among the most intense reticulate trajectories across the superfamily are human $HLA\text{-}A*33:01$ ($Z = 6.31$), chimpanzee $Patr\text{-}B*07:01$ ($Z = 6.22$), gorilla $Gogo\text{-}B*01:01$ ($Z = 5.48$), and orangutan $Poab\text{-}B*10:01$ ($Z = 6.10$), demonstrating that ectopic gene conversion is an active, pervasive force across all hominoid apes.

Visualizing the manifold in two-dimensional Classical MDS projections resolves the 35-year paradox with geometric clarity (Figure 9D). Projected over Exons 2–3 (ARS groove, nt 73–603), human, chimpanzee, bonobo, gorilla, and orangutan alleles intermingle across loci, forming diffuse, overlapping clouds governed by trans-species functional convergence and shared ancient motifs. Projected over Exon 4 (α_3 structural stalk, nt 604–879), the exact same 379 alleles collapse into three strictly monophyletic, non-overlapping clusters corresponding to the *A*, *B*, and *C* loci, faithfully reflecting species cladogenesis. Continuous manifold deconvolution reveals that primate MHC evolution is neither a single bifurcating tree nor an untraceable network, but a multi-domain mechanical manifold where diversifying selection on the antigen-presenting groove and purifying constraints on the structural stalk coexist within the same macromolecule.

Discussion and Algorithmic Scaling

Linear Manifold Geometry vs. Combinatorial Triplet Search

Computational phylogenetic methods confront a difficult choice between evolutionary realism and combinatorial intractability. Trees model divergence. When reticulate events occur, forcing mosaic genomes into strictly bifurcating topologies distorts branch lengths and inflates substitution rate estimates (40). Conversely, modeling recombination explicitly through ancestral recombination graphs or phylogenetic changepoints introduces prohibitive computational scaling, demanding cubic or quartic algorithms that collapse on modern genomic datasets (34). Triplet scanning methods like 3SEQ provide an empirical compromise by evaluating small three-sequence subgraphs (5). Triplet decomposition avoids tree estimation. Yet exhaustive triplet testing scales cubically with sample size ($\binom{N}{3} \sim O(N^3)$), creating an inescapable computational barrier when surveillance cohorts expand to hundreds or thousands of isolates.

`RhizAeon` demonstrates that continuous metric geometry dissolves this dichotomy. By projecting whole-genome sequence divergence into fixed Buneman coordinates once for the entire alignment, the engine establishes an invariant spatial reference frame. Zero Procrustes rotations are required. Instantaneous sequence affinities are tracked continuously along the chromosome via analytical Gower out-of-sample projection, replacing discrete topology searches with smooth dynamical trajectories. Manifolds preserve spatial continuity. Cross-taxa attention operates against a fixed landmark basis ($K \leq 32$), ensuring that trajectory extraction scales strictly as $O(N \cdot K \cdot L)$. As our empirical benchmarks demonstrate, this linear scaling yields two- to three-order-of-magnitude accelerations over triplet tools on standard cohorts, finishing multi-kilobase scans in milliseconds where triplet enumeration requires minutes or hours.

Equally consequential is the resolution of complex multi-way mosaicism. Triplet tools assume two-parent ancestry. When circulating recombinant lineages accumulate segments from three or more parental lineages, examining triplets in isolation inevitably truncates mosaic tracts, missing breakpoints and hallucinating recombinant candidates. By tracking continuous trajectory tensors across the entire cohort simultaneously, physical force field deconvolution decomposes multiple orthogonal spatial modes. The framework detects interleaved parental contributions without prior constraints on parental cardinality. Furthermore, metric isolation prevents unsampled ghost donors from triggering combinatorial false alarms across innocent contemporary clades.

Theoretical Boundaries: Pervasive Recombination and the Limits of Parental Attribution

Every algorithmic framework for detecting genetic recombination operates under an implicit information-theoretic premise: reticulation events are sparse relative to the tempo of clonal divergence. Trees require coherence. When recombination occurs as episodic introgression, distinct

parental lineages maintain extended tracts of linkage disequilibrium, providing the mutual information necessary to distinguish an introgressed cassette from its clonal background. Under these conditions, Farthest-Point Minimax Sampling reliably extracts non-recombinant backbone genomes that span the fundamental cladogenetic divides of the population.

This geometric premise breaks down when genetic exchange becomes pervasive. In panmictic bacterial populations (*e.g.*, *Helicobacter pylori*, *Streptococcus pneumoniae*) or dense viral quasispecies where the recombination rate far exceeds the mutation rate ($r/\mu \gg 1$), linkage disequilibrium decays over distances shorter than the typical spacing between informative single-nucleotide polymorphisms ($\ell_{\mathrm{LD}} \ll s_{\mathrm{SNP}}$). Genomes become fine-scale mosaics of mosaics. In this hyper-recombinant regime, no sampled isolate represents an intact ancestral parent; every candidate landmark is itself a mosaic composite of fragmented ancestries. Global pairwise evolutionary distances collapse toward a uniform spatial mean, washing out the discrete bipartitions of Buneman metric space.

How detection methods respond to this informational collapse defines their statistical integrity. Classical triplet-scanning tools (such as 3SEQ, RDP, and GENECONV) fail silently in this regime. Because they evaluate three-sequence alignments in isolation without reference to global manifold structure, triplet methods mistake fine-scale stochastic fluctuations for macroscopic gene conversion. The result is combinatorial hallucination: triplet tools infer dense, circular webs of spurious reticulation where nearly every sequence is designated both parent and recombinant child to every other isolate.

Metric manifold geometry confronts this informational boundary directly. When alignments lack identifiable parental backbones, the leading eigenvalues of the landmark Gram matrix $\mathbf{B}$ fail to separate, collapsing into an isotropic hypersphere. In Tier 3 force field deconvolution, this spatial degeneracy manifests as the total suppression of singular values beneath the Marchenko–Pastur bulk noise edge (s_{bulk}). The engine terminates parameter-free. Parental attribution in a thoroughly scrambled population is not an algorithmic difficulty awaiting more refined heuristics; it is an ill-posed informational category error. When unadmixed ancestral lineages no longer exist in the contemporary sample, attempting to designate discrete "donors" and "recipients" exceeds the physical capacity of the alignment. By halting inference when reticulate forces merge into the random spectral bulk, `RhizAeon` provides a principled boundary against the algorithmic over-interpretation of panmictic noise.

Data Governance and Biosafety Compliance

Research integrity demands strict transparency regarding computational protocols and data sources. Computational analyses require ethical clarity. We certify that this investigation represents a purely retrospective bioinformatic study operating exclusively on publicly accessible nucleotide sequences deposited in NCBI GenBank, GISAID, and the Los Alamos National Laboratory HIV Sequence Database. No biological specimens, clinical samples, or viral cultures were collected, isolated, cultivated, or manipulated during this study. All research was conducted entirely in silico using audited open-source algorithms. The software performs automated sequence alignment scanning without generating actionable protocols for pathogen enhancement, gain-of-function phenotypes, or laboratory synthesis. Source code, documentation, and client-side browser modules are freely available under permissive open-source licensing at `https://github.com/veg/rhizaeon-code`, promoting reproducible evolutionary genomics and accessible public health surveillance.

Materials and Methods

Farthest-Point Landmark Sampling and Metric Basis Construction

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_M\}$ denote a multiple sequence alignment of M contemporary nucleotide sequences of length L over the alphabet $\Sigma = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\} \cup \{-\}$. When $M \gg 1{,}000$, computing an all-to-all pairwise distance matrix requires $\binom{M}{2} \approx \frac{1}{2}M^2$ pairwise comparisons, scaling as $O(M^2 L)$ and creating a computational bottleneck for large-scale genomic surveillance. Diversity anchors coordinates. `RhizAeon` eliminates this quadratic barrier through sequential Farthest-Point Sampling (FPS) (19), selecting a sparse subset of $N \ll M$ reference landmark genomes in strictly $O(NML)$ operations.

The selection algorithm begins by designating the initial reference landmark $s_{(1)}$ as the sequence exhibiting the highest non-gap coverage across the alignment:

$$s_{(1)} = \arg\max_{s_i \in \mathcal{S}} \sum_{u=1}^{L} \mathbb{I}(s_i(u) \neq \text{'-'}). \tag{2}$$

The algorithm then computes the pairwise evolutionary distance from $s_{(1)}$ to all M sequences in the cohort, initializing a running minimum distance vector $\mathbf{r} = (r_1, \ldots, r_M)^T \in \mathbb{R}^M$, where $r_i = d(s_i, s_{(1)})$. Pairwise evolutionary distances $d(s_i, s_j)$ are computed under the Tamura-Nei (TN93) continuous-time Markov substitution model with pairwise deletion of gap positions (47):

$$d_{\mathrm{TN93}} = -2a_1 \ln\left(1 - \frac{P_1}{2a_1} - \frac{Q}{2a_3}\right) - 2a_2 \ln\left(1 - \frac{P_2}{2a_2} - \frac{Q}{2a_3}\right) - 2(a_3 - a_1 - a_2)\ln\left(1 - \frac{Q}{2a_3}\right), \tag{3}$$

where P_1 and P_2 represent transitional differences between purines and pyrimidines, Q represents transversional differences, and a_1, a_2, a_3 denote equilibrium nucleotide frequency constants. For alignments of highly divergent or non-coding sequences, normalized Hamming distance is supported as a scale-invariant alternative.

Each subsequent reference landmark $k \in \{2, \ldots, N\}$ is selected sequentially by identifying the sequence that maximizes the minimum distance to the existing set of selected landmarks:

$$s_{(k)} = \arg \max_{i \in \{1,\ldots,M\} \setminus \{s_{(1)},\ldots,s_{(k-1)}\}} r_i. \tag{4}$$

Upon selecting $s_{(k)}$, the algorithm evaluates distances from $s_{(k)}$ to all M sequences and updates the running minimum distance array in $O(ML)$ time:

$$r_i \leftarrow \min\left(r_i,\, d(s_i, s_{(k)})\right), \quad \forall i \in \{1, \ldots, M\}. \tag{5}$$

Selecting N reference genomes across M isolates requires exactly $N \times M$ distance evaluations, reducing the computational burden from $O(M^2 L)$ to $O(NML)$. For an alignment of 10,000 genomes with $N = 16$ landmarks, distance evaluations collapse from $49,995,000$ to $160,000$, representing a 312-fold empirical speedup. Farthest points span divergence.

Full-Cohort Representation ($M \leq 64$) and Two-Pass Clonal Backbone Re-Anchoring ($M > 64$).

When scanning moderate surveillance cohorts ($M \leq 64$), downsampling the alignment to a sparse subset of landmarks risks inadvertently selecting a recombinant isolate while omitting its parental lineage, or omitting a minor parental clade entirely. To guarantee complete topological coverage, `RhizAeon` automatically activates Full-Cohort Channel Tracking when $M \leq 64$, setting the landmark set to the complete sequence roster ($N = M$). Every sequence is endowed with its own continuous attention channel, eliminating downsampling artifacts and ensuring all prospective parental lineages are present in the reference basis.

For massive surveillance cohorts ($M > 64$), `RhizAeon` utilizes Farthest-Point Sampling to select $N \leq 32$ landmarks. However, in epidemic recombinant swarms, greedy farthest-point exploration can pick an expanded recombinant lineage as a landmark due to its intermediate divergence. If a recombinant isolate R is admitted into the landmark basis, co-recombinant sister isolates $(R_1, \ldots, R_k)$ that share the identical mosaic cassettes will place dominant continuous attention ($\geq 80\%$) onto R. When the parental resolver assigns Home $= R$, the differential excursion vanishes ($Y(u) \approx 0$), extinguishing the recombination signal in a catastrophic *Landmark Self-Masking Trap*. To eliminate this failure mode, `RhizAeon` executes Two-Pass Clonal Backbone Re-Anchoring for $M > 64$: Pass 1 screens the cohort using initial FPS landmarks to identify candidate recombinants ($\mathcal{C}_{\text{rec}}$). If any landmark belongs to $\mathcal{C}_{\text{rec}}$, Pass 2 purges all candidate recombinants from the candidate landmark pool and re-anchors the landmark basis strictly onto verified non-recombinant clonal backbone lineages ($\mathcal{S} \setminus \mathcal{C}_{\text{rec}}$). This guarantees that only stable, unadmixed parental lineages serve as coordinate landmarks.

From the selected reference genomes $\{s_{(1)}, \ldots, s_{(N)}\}$, we construct the symmetric reference distance matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$, with elements $D_{i,j} = d(s_{(i)}, s_{(j)})$ and diagonal entries $D_{i,i} = 0$. We define the element-wise squared distance matrix $\mathbf{D}^{(2)} \in \mathbb{R}^{N \times N}$ such that $[\mathbf{D}^{(2)}]_{i,j} = D_{i,j}^2$. To map these distances into an inner-product space, we apply the geometric centering projection operator $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$, where $\mathbf{I}_N$ is the $N \times N$ identity matrix and $\mathbf{1}_N = (1, \ldots, 1)^T \in \mathbb{R}^N$ is the all-ones vector. The doubly centered inner-product Gram matrix $\mathbf{B} \in \mathbb{R}^{N \times N}$ is formulated via classical multidimensional scaling (MDS) (20):

$$\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}^{(2)}\mathbf{H}. \tag{6}$$

Spectral decomposition of the symmetric matrix $\mathbf{B}$ yields:

$$\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \sum_{k=1}^{N} \lambda_k \mathbf{v}_k \mathbf{v}_k^T, \tag{7}$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ contains real eigenvalues arranged in descending order ($\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$), and $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_N] \in \mathbb{R}^{N \times N}$ is the orthonormal eigenvector matrix satisfying $\mathbf{V}^T\mathbf{V} = \mathbf{I}_N$. Spectra reveal splits.

We retain the leading d dimensions corresponding to the largest positive eigenvalues ($d \leq \min(4, N-1)$):

$$\mathbf{\Lambda}_d = \text{diag}(\lambda_1, \ldots, \lambda_d) \in \mathbb{R}^{d \times d}, \quad \mathbf{V}_d = [\mathbf{v}_1, \ldots, \mathbf{v}_d] \in \mathbb{R}^{N \times d}. \tag{8}$$

The static d-dimensional metric coordinates $\mathbf{x}_i \in \mathbb{R}^d$ for reference landmark $i \in \{1, \ldots, N\}$ are given by:

$$\mathbf{x}_i = \mathbf{\Lambda}_d^{1/2}(\mathbf{V}_d)_{i,:}^T = \left(\sqrt{\lambda_1}v_{1,i},\, \sqrt{\lambda_2}v_{2,i},\, \ldots,\, \sqrt{\lambda_d}v_{d,i}\right)^T. \tag{9}$$

By construction, these coordinates satisfy $\langle \mathbf{x}_i, \mathbf{x}_j \rangle = [\mathbf{B}]_{i,j}$ within the subspace $\mathbb{R}^d$. By Buneman's metric tree theorem (7), tree metrics satisfy four-point additivity conditions $d(a,b) + d(c,d) \leq \max(d(a,c) + d(b,d),\, d(a,d) + d(b,c))$, ensuring that tree-like evolutionary divergences embed into Euclidean spaces where macroscopic clade splits project onto the dominant eigenvectors.

Analytical Gower Out-of-Sample Trajectory Projection

Tracking sequence affinities along the chromosome requires establishing an instantaneous coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ for each query sequence $i \in \{1, \ldots, M\}$ at every genomic site $u \in [1, L]$. Recomputing multidimensional scaling independently within sliding windows would produce arbitrary orthogonal rotations and reflections, distorting geometric distances and necessitating expensive Procrustes alignments. Instead,

`RhizAeon` projects local distance vectors into the fixed canonical coordinate basis $(\mathbf{V}_d, \mathbf{\Lambda}_d)$ using Gower's analytical out-of-sample interpolation theorem (21).

At coordinate u, we evaluate local evolutionary distances across an aligned window $[u - W/2, u + W/2]$ of width W (default $W = 300$ nt) sampled at stride S (default $S = 10$ nt). For query sequence i, this yields an instantaneous local distance vector $\mathbf{d}_i(u) \in \mathbb{R}^N$ to the N reference landmarks:

$$\mathbf{d}_i(u) = (d_i(1;u),\, d_i(2;u),\, \ldots,\, d_i(N;u))^T\,. \tag{10}$$

We define the vector of squared local distances as $\mathbf{d}_i^2(u) = (d_i(1;u)^2, \ldots, d_i(N;u)^2)^T \in \mathbb{R}^N$, and the vector of mean squared reference distances as:

$$\bar{\mathbf{d}}^2 = \frac{1}{N}\mathbf{D}^{(2)}\mathbf{1}_N = \left(\frac{1}{N}\sum_{j=1}^{N} D_{1,j}^2,\, \ldots,\, \frac{1}{N}\sum_{j=1}^{N} D_{N,j}^2\right)^T \in \mathbb{R}^N. \tag{11}$$

Gower's out-of-sample mapping identifies the coordinate $\mathbf{z}_i(u) \in \mathbb{R}^d$ that minimizes squared Euclidean distance distortions relative to the static reference coordinates $\mathbf{x}_j$:

$$\min_{\mathbf{z}\in\mathbb{R}^d} \sum_{j=1}^{N} \left(\|\mathbf{z} - \mathbf{x}_j\|^2 - d_i(j;u)^2\right)^2. \tag{12}$$

Expanding the Euclidean norm $\|\mathbf{z} - \mathbf{x}_j\|^2 = \|\mathbf{z}\|^2 - 2\mathbf{z}^T\mathbf{x}_j + \|\mathbf{x}_j\|^2$, utilizing the centering condition $\sum_{j=1}^N \mathbf{x}_j = \mathbf{0}$, and equating the gradient to zero yields the exact analytical solution:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u)\right) \in \mathbb{R}^d, \tag{13}$$

where $\mathbf{\Lambda}_d^{-1/2} = \mathrm{diag}(1/\sqrt{\lambda_1}, \ldots, 1/\sqrt{\lambda_d})$. Evaluating (13) across all coordinates $u \in \{1, 1+S, \ldots, L\}$ constructs the continuous trajectory tensor $\mathbf{Z} \in \mathbb{R}^{M\times U\times d}$, where $U = \lceil L/S \rceil$.

Because the basis $(\mathbf{V}_d, \mathbf{\Lambda}_d)$ and mean vector $\bar{\mathbf{d}}^2$ are computed once from whole-genome reference divergence, every sliding window maps into the identical, rigid metric reference frame. Zero Procrustes rotations are required. Projections preserve geometry. Furthermore, substituting the whole-genome distance vector of reference landmark k into (13) analytically recovers its exact MDS coordinate:

$$\frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \left(\mathbf{D}_{k,:}^{(2)}\right)^T\right) = \mathbf{\Lambda}_d^{1/2}(\mathbf{V}_d)_{k,:}^T = \mathbf{x}_k, \tag{14}$$

proving algebraic self-consistency and zero metric distortion on reference taxa.

Generalized Tree-RoPE and Continuous-Time Markov Substitution Prior

Standard transformer neural networks incorporate sequential position along one-dimensional text using Rotary Position Embeddings (RoPE) (45). However, biological sequences diverge across multidimensional phylogenetic trees rather than linear strings. To integrate evolutionary distance directly into cross-sequence attention, `RhizAeon` generalizes rotary embeddings from one-dimensional sequence indices to continuous metric coordinates.

Let $\mathbf{x}_i \in \mathbb{R}^d$ denote the continuous metric coordinates of sequence i. We construct a fixed frequency projection basis $\mathbf{\Theta} \in \mathbb{R}^{(d_{\text{head}}/2)\times d}$, with components parameterized by base frequency $\theta_0 = 10,000$:

$$\Theta_{m,c} = \theta_0^{-2(m-1)/d_{\text{head}}} \cdot \omega_c, \quad m \in \{1, \ldots, d_{\text{head}}/2\},\ c \in \{1, \ldots, d\}, \tag{15}$$

where $\omega_c = \sqrt{\lambda_c / \sum_{k=1}^d \lambda_k}$ weights each dimension by its relative spectral variance. For sequence i, the rotary phase angle vector $\boldsymbol{\theta}_i = (\theta_{i,1}, \ldots, \theta_{i,d_{\text{head}}/2})^T$ is computed as:

$$\theta_{i,m} = \sum_{c=1}^{d} x_{i,c}\Theta_{m,c}. \tag{16}$$

Angles encode divergence.

At each aligned site u, sequence representations generate query and key vectors $\mathbf{q}_i(u), \mathbf{k}_i(u) \in \mathbb{R}^{d_{\text{head}}}$. We define the block-diagonal Givens rotation matrix $\mathbf{R}(\boldsymbol{\theta}_i) \in \mathbb{R}^{d_{\text{head}}\times d_{\text{head}}}$ as:

$$\mathbf{R}(\boldsymbol{\theta}_i) = \bigoplus_{m=1}^{d_{\text{head}}/2} \begin{pmatrix} \cos\theta_{i,m} & -\sin\theta_{i,m} \\ \sin\theta_{i,m} & \cos\theta_{i,m} \end{pmatrix}. \tag{17}$$

Applying $\mathbf{R}(\boldsymbol{\theta}_i)$ rotates query and key representations according to each taxon's continuous metric position. The inner product between rotated vectors satisfies:

$$\langle \mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u),\, \mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u)\rangle = \mathbf{q}_i(u)^T\mathbf{R}(\boldsymbol{\theta}_j - \boldsymbol{\theta}_i)\mathbf{k}_j(u). \tag{18}$$

Because rotation angles depend linearly on metric coordinates, the phase shift $\|\boldsymbol{\theta}_j - \boldsymbol{\theta}_i\|$ increases with evolutionary divergence in Buneman metric space, naturally attenuating pairwise attention as phylogenetic distance increases.

To calibrate geometric distances against baseline neutral sequence homology, we integrate a continuous-time Markov substitution prior directly into the attention logits. For any pair of sequences (i, j) separated by evolutionary distance $D_{i,j}$, the probability of observing identical homologous nucleotides under a continuous-time Markov substitution model with substitution scaling rate λ_h and equilibrium frequency $\epsilon_0 = 0.25$ is:

$$P_{i,j} = \epsilon_0 + (1 - \epsilon_0) \exp\left(-\lambda_h D_{i,j}\right). \tag{19}$$

We define the logarithmic phylogenetic prior bias as:

$$\text{PhyloBias}_{i,j} = \ln\left[\max\left(10^{-5},\, P_{i,j}\right)\right]. \tag{20}$$

The composite unnormalized cross-attention score $S_u(i, j)$ between query sequence i and candidate reference j at coordinate u becomes:

$$S_u(i,j) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u))^T(\mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{i,j}. \tag{21}$$

Unpolarized Root Sink Architecture and Autapomorphy Absorption

Standard cross-attention mechanisms normalize probability exclusively across contemporary leaves, creating structural vulnerability to lineage-specific rate variation. Private mutations lack donors. When an isolated lineage undergoes an episodic substitution burst—such as APOBEC3G cytidine deamination or accelerated heterotachy—private mutations lack homologous counterparts in contemporary clades. Without an unpolarized reference, softmax forces these private mutations to affiliate with whichever contemporary leaf presents the smallest stochastic mismatch, generating severe false positive recombination calls.

To insulate trajectory inference from unreticulated rate bursts, `RhizAeon` reserves index zero in the sequence dimension for an unpolarized ancestral root token $\mathbf{x}_{\text{root}}$. The root token is anchored at the origin of the Buneman metric space:

$$\mathbf{x}_{\text{root}} = \mathbf{0} = (0, 0, \ldots, 0)^T \in \mathbb{R}^d. \tag{22}$$

For an alignment of N contemporary reference genomes, the cross-attention operator operates over $N + 1$ nodes, spanning the ancestral root $(j = 0)$ and all contemporary leaves $(j = 1, \ldots, N)$. The softmax normalization strictly conserves probability across all $N + 1$ nodes:

$$A_u(i \to j) = \frac{\exp(S_u(i,j))}{\sum_{k=0}^{N} \exp(S_u(i,k))}, \quad \text{such that} \quad A_u(i \to 0) + \sum_{j=1}^{N} A_u(i \to j) = 1.0. \tag{23}$$

Proof of Autapomorphy Absorption Consider a genomic site u where sequence i acquires a private autapomorphic mutation that does not match any contemporary reference genome $j \in \{1, \ldots, N\}$. For all contemporary leaves $j \geq 1$, the mismatch depresses both query-key matching and the phylogenetic prior, yielding low scores $S_u(i, j) \leq S_{\text{mismatch}}$. In contrast, the unpolarized root token at $\mathbf{x}_{\text{root}} = \mathbf{0}$ represents the baseline unmutated ancestral state with neutral substitution distance, maintaining $S_u(i, 0) = S_{\text{root}} > S_{\text{mismatch}}$. In the limit where private divergence increases:

$$\lim_{S_u(i,j) \to -\infty,\, \forall j \geq 1} A_u(i \to 0) = \frac{\exp(S_u(i,0))}{\exp(S_u(i,0)) + \sum_{j=1}^{N} \exp(S_u(i,j))} = 1.0, \tag{24}$$

and $A_u(i \to j) \to 0.0$ for all $j \geq 1$. Consequently, autapomorphic substitution bursts route their attention flux into node zero rather than contemporary leaves, preventing spurious donor excursions and guaranteeing a 0.00% false positive rate under biological heterotachy. □

Sinks absorb outliers.

Continuous Trajectory Integration and Brownian Bridge Inference

At individual polymorphic sites, instantaneous attention weights exhibit stochastic high-frequency jitter. To extract continuous evolutionary trajectories, `RhizAeon` integrates attention into cumulative trajectory curves. Let $\mathcal{B}_m \subset \{1, \ldots, N\}$ denote the set of reference genomes belonging to phylogenetic clade m. The instantaneous bundle attention is $\alpha_{i,m}(u) = \sum_{j \in \mathcal{B}_m} A_u(i \to j)$. The continuous cumulative attention curve

$C_{i,m}(u)$ represents the integrated affinity to clade m up to coordinate u:

$$C_{i,m}(u) = \sum_{v=1}^{u} \alpha_{i,m}(v) = \sum_{v=1}^{u} \sum_{j \in \mathcal{B}_m} A_v(i \to j). \tag{25}$$

The home clade $m_{\text{home}}(i)$ is identified as the lineage accumulating the greatest integrated affinity across the full chromosome:

$$m_{\text{home}}(i) = \arg\max_{m} C_{i,m}(L). \tag{26}$$

For any candidate donor clade $m \neq m_{\text{home}}$, we formulate the differential attention trajectory as:

$$Y_{i,m}(u) = C_{i,m}(u) - C_{i,m_{\text{home}}}(u). \tag{27}$$

Under clonal evolution within the parental background, attention to the home clade exceeds donor attention at nearly all sites ($A_v(i \to \text{home}) > A_v(i \to m)$), causing $Y_{i,m}(u)$ to drift monotonically downward with negative slope. Recombination reverses the derivative: within a converted donor tract $[u_1, u_2]$, donor attention dominates, producing a positive slope ($dY/du > 0$) and a prominent upward excursion.

To test the clonal null hypothesis H_0, we construct the centered Brownian bridge process $B_{i,m}(u)$ by subtracting the chromosome-wide linear drift:

$$B_{i,m}(u) = Y_{i,m}(u) - \frac{u}{L} Y_{i,m}(L), \quad u \in [1, L]. \tag{28}$$

Let $\Delta Y(u) = Y(u) - Y(u-1)$ denote the local trajectory increments, with sample mean $\bar{\Delta Y} = \frac{1}{L} Y(L)$ and empirical increment variance:

$$\sigma^2 = \frac{1}{L-1} \sum_{u=1}^{L} \left(\Delta Y(u) - \bar{\Delta Y} \right)^2. \tag{29}$$

Under the clonal null hypothesis H_0, the increments are stationary and unreticulated. By Donsker's functional central limit theorem (15), the standardized process:

$$W_L^0(t) = \frac{B_{i,m}(\lfloor tL \rfloor)}{\sigma \sqrt{L}}, \quad t \in [0, 1], \tag{30}$$

converges weakly as $L \to \infty$ to a standard Brownian bridge $W^0(t)$ with continuous sample paths, mean $\mathbb{E}[W^0(t)] = 0$, and covariance $\text{Cov}(W^0(s), W^0(t)) = \min(s,t) - st$. The supremum of the absolute bridge trajectory follows the two-sided Kolmogorov distribution:

$$P\left(\sup_{t \in [0,1]} |W^0(t)| \leq z \right) = 1 - 2 \sum_{k=1}^{\infty} (-1)^{k-1} \exp\left(-2k^2 z^2\right). \tag{31}$$

To control the family-wise error rate (FWER) across all $M_{\text{topo}} = \binom{N}{2} \max(1, N-2)$ testable donor-recipient reticulation topologies at nominal significance $\alpha = 0.05$, we apply the Dunn-Šidák extreme-value correction (41):

$$K_{\text{crit}} = \sqrt{-\frac{1}{2} \ln\left(1 - (1-\alpha)^{1/M_{\text{topo}}}\right)} \approx \sqrt{-\frac{1}{2} \ln\left(\frac{\alpha}{2M_{\text{topo}}}\right)}. \tag{32}$$

If the normalized trajectory supremum satisfies:

$$\frac{\sup_{u \in [1,L]} |B_{i,m}(u)|}{\sigma \sqrt{L}} \geq K_{\text{crit}}, \tag{33}$$

`RhizAeon` rejects the clonal null hypothesis and confirms a statistically significant reticulation event.

Dyadic Scale-Space Matched Filtering and Running Drawdown

For short genetic conversions and micro-recombinations (< 200 nt), global Brownian bridge tests can suffer signal dilution across multi-kilobase chromosomes. To achieve multi-scale sensitivity, `RhizAeon` convolves the differential trajectory flux $\Delta Y(u)$ with a bank of zero-mean bipolar dyadic matched filters $h_\tau(s)$ evaluated across scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nucleotides:

$$h_\tau(s) = \begin{cases} +1, & 0 \leq s < \tau/2 \\ -1, & \tau/2 \leq s < \tau \\ 0, & \text{otherwise} \end{cases}. \tag{34}$$

The discrete convolution $(dY * h_\tau)(u) = \sum_{s=0}^{\tau} \Delta Y(u-s) h_\tau(s)$ isolates rapid step transitions across dyadic scale space.

Concurrently, `RhizAeon` tracks the running high-water mark $M(u)$ and running drawdown $\mathrm{DD}(u)$:

$$M(u) = \max_{1 \le v \le u} Y(v), \quad \mathrm{DD}(u) = M(u) - Y(u). \tag{35}$$

The drawdown metric localizes points where trajectory momentum collapses, marking the exit boundary u_2 of a donor tract. To distinguish single crossovers from intercalary cassette swaps, we evaluate the normalized energy density ratio of the trajectory excursions outside versus inside the candidate tract $[u_1, u_2]$:

$$\mathcal{E}_{\mathrm{ratio}} = \frac{\frac{1}{L-(u_2-u_1)} \int_{u \notin [u_1,u_2]} (Y(u) - Y_{\mathrm{base}})^2 \, du}{\frac{1}{u_2-u_1} \int_{u_1}^{u_2} (Y(u) - Y_{\mathrm{base}})^2 \, du}, \tag{36}$$

where Y_{base} denotes the linear baseline drift. A single crossover spans half the chromosome, producing high residual energy across the alignment ($\mathcal{E}_{\mathrm{ratio}} \ge 0.50$). In contrast, an intercalary cassette returns cleanly to baseline on both flanks, characterized by $\mathcal{E}_{\mathrm{ratio}} < 0.15$. Complementing the trajectory energy ratio, for decomposed spatial modes $\mathbf{u}_m$, `RhizAeon` computes the continuous spatial mode polarity ratio:

$$p_{\mathrm{mono}} = \frac{\left| \sum_{u=1}^{L} u_m(u) \right|}{\sum_{u=1}^{L} |u_m(u)|}. \tag{37}$$

A monopolar arch maintaining uniform sign across the alignment yields $p_{\mathrm{mono}} \approx 1.0$ (characterizing single crossovers and tree backbones), whereas a localized bipolar pulse returning to baseline on both flanks yields $p_{\mathrm{mono}} \le 0.50$ (characterizing intercalary cassette swaps; Figure 4D).

Physical Force Field Deconvolution and Random Matrix Halting

To deconvolve complex multi-way mosaicism, `RhizAeon` models trajectory excursions as physical displacements within an evolutionary force field. We parameterize the evolutionary pull of the home lineage as a Hookean restoring potential $U(\mathbf{z}) = \frac{1}{2}\lambda \|\mathbf{z} - \mathbf{z}_{\mathrm{home}}\|^2$, generating a continuous restoring force:

$$\mathbf{f}(u) = -\nabla U = -\lambda \left(\mathbf{z}(u) - \mathbf{z}_{\mathrm{home}} \right) \in \mathbb{R}^d. \tag{38}$$

Across a candidate breakpoint boundary, let $\mathbf{f}_{\mathrm{left}}$ and $\mathbf{f}_{\mathrm{right}}$ denote the mean force vectors on the flanking intervals. The directional deflection angle θ satisfies $\cos\theta = \frac{\langle \mathbf{f}_{\mathrm{left}}, \mathbf{f}_{\mathrm{right}} \rangle}{\|\mathbf{f}_{\mathrm{left}}\| \, \|\mathbf{f}_{\mathrm{right}}\|}$, and the step velocity is $v_{\mathrm{step}} = \|\mathbf{f}_{\mathrm{right}} - \mathbf{f}_{\mathrm{left}}\|$. The dynamic velocity reversal score is defined as:

$$R = v_{\mathrm{step}}(1 - \cos\theta). \tag{39}$$

To achieve scale-invariance across cohorts of varying divergence, we standardize the velocity reversal along the unit parental dipole axis $\mathbf{d}_{\mathrm{hat}} = \frac{\mathbf{z}_{\mathrm{donor}} - \mathbf{z}_{\mathrm{home}}}{\|\mathbf{z}_{\mathrm{donor}} - \mathbf{z}_{\mathrm{home}}\|}$:

$$Z_{\mathrm{flip}} = \frac{|(\mathbf{f}_{\mathrm{right}} - \mathbf{f}_{\mathrm{left}}) \cdot \mathbf{d}_{\mathrm{hat}}|}{\sqrt{\frac{\sigma^2_{\mathrm{left}}}{L_{\mathrm{left}}} + \frac{\sigma^2_{\mathrm{right}}}{L_{\mathrm{right}}}}}, \tag{40}$$

where $\sigma^2_{\mathrm{left}}, \sigma^2_{\mathrm{right}}$ represent residual force variances along $\mathbf{d}_{\mathrm{hat}}$ across flanking segments of lengths $L_{\mathrm{left}}, L_{\mathrm{right}}$.

For alignments exhibiting multiple recombinant tracts, we stack force vectors into the spatio-temporal force matrix $\mathbf{F} \in \mathbb{R}^{N \times L}$ and perform singular value decomposition:

$$\mathbf{F} = \sum_{m=1}^{r} s_m \mathbf{u}_m \mathbf{v}_m^T, \tag{41}$$

where $s_1 \ge s_2 \ge \cdots \ge s_r$ denote singular values. Each detected reticulation mode m is deconvolved via iterative mode peeling, subtracting $s_m \mathbf{u}_m \mathbf{v}_m^T$ from $\mathbf{F}$. To prevent over-segmentation and halt peeling when residual variation reflects neutral coalescent drift, we apply random matrix theory. Under the unreticulated null hypothesis, the singular values of an $N \times L$ random noise matrix follow the Marchenko-Pastur distribution (30). The theoretical upper boundary of the bulk spectral noise edge is:

$$s_{\mathrm{bulk}} = \frac{L}{\pi} \sigma_{\mathrm{noise}} \left(1 + \sqrt{\frac{P}{L}} \right), \tag{42}$$

where $P = \min(N, d)$ denotes the metric projection dimension and σ_{noise} represents the baseline standard deviation of empirical force increments. Mode peeling halts when the leading residual singular value satisfies $s_m \le s_{\mathrm{bulk}}$, terminating iteration without manual parameter tuning.

Profile Maximum Likelihood Breakpoint Polishing

Following changepoint localization from continuous trajectory dynamics, `RhizAeon` refines candidate breakpoints down to single-nucleotide resolution using profile maximum likelihood polishing. Within a local search radius $[u_{\mathrm{init}} - \delta, u_{\mathrm{init}} + \delta]$ (default $\delta = 50$ nt), the engine scans each candidate coordinate c and computes the partitioned log-likelihood:

$$\ln \mathcal{L}(c) = \sum_{u=1}^{c} \ln P\left(s_{\mathrm{rec}}(u) \mid s_{\mathrm{left}}(u) \right) + \sum_{u=c+1}^{L} \ln P\left(s_{\mathrm{rec}}(u) \mid s_{\mathrm{right}}(u) \right), \tag{43}$$

where $P(s_{\mathrm{rec}}(u) \mid s(u))$ denotes the conditional probability of the recombinant state given the parental sequence under the fitted nucleotide substitution model.

When invariant or uninformative sites intervene between flanking parental polymorphisms, the likelihood function forms a flat plateau. Rather than reporting an arbitrary midpoint, `RhizAeon` reports the exact informational plateau $[c_{\text{left}}, c_{\text{right}}]$:

$$[c_{\text{left}}, c_{\text{right}}] = \left\{ c \in [1, L] : \ln \mathcal{L}(c) \geq \max_{c'} \ln \mathcal{L}(c') - \Delta_{\text{thresh}} \right\}, \tag{44}$$

where $\Delta_{\text{thresh}} = 1.92$ corresponds to the 95% confidence interval under a χ^2_1 likelihood ratio test. The boundary coordinates c_{left} and c_{right} demarcate the exact flanking informative polymorphisms that physically bound the genetic exchange. Plateaus bound breakpoints.

Continuous Gap Bridging Across Stochastic Non-Informative Dips.

In low-divergence regimes ($d \leq 0.02$), mutation density is inherently sparse (~ 1 substitution every 50 nucleotides). Under standard Poisson mutation processes, stochastic intervals of 50–100 bp within an authentic 1,000 nt conversion cassette frequently carry zero donor-specific mutations ($P(0 \text{ mutations}) = e^{-\lambda} \approx 15\%$–$35\%$). During recursive binary segmentation, instantaneous differential flux drops to zero across these unmutated windows, which would otherwise fracture an intact biological cassette into multiple fragmented micro-tracts. To preserve macro-conversion integrity, `RhizAeon` implements Continuous Gap Bridging: adjacent candidate donor segments separated by small non-informative gaps (≤ 150 bp) are iteratively coalesced into a single unified tract whenever the net flux across the merged span remains positive ($\frac{1}{e_2 - s_1 + 1} \sum_{u=s_1}^{e_2} X(u) > 0$). Bridging restores the contiguous physical envelope of the biological introgression, pooling all supporting mutations together and ensuring the subsequent hypergeometric contingency test operates on the true cumulative evidence.

Reticulation Graph Adjudication, Transitive Chains, and Ancestral Event Grouping

In pairwise and triplet sequence scans, an intrinsic topological ambiguity arises: if a recombinant isolate C inherits a genomic tract from donor D into a parental home background H, reciprocal tests frequently evaluate D as recombinant between C and H. This reciprocal false alarm creates a mutual two-cycle in candidate reticulation graphs ($C \leftrightarrow D$). To orient parental lineages versus genuine recombinant isolates without combinatorial tree searches, `RhizAeon` inspects candidate events across homologous intervals. Homologous candidates share overlapping tracts ($\geq 25\%$ of the shorter tract) or form complementary crossover partitions ($u_2^{(1)} \approx u_1^{(2)}$).

The algorithm evaluates external donor authenticity. If candidate C_1 identifies an external donor while candidate C_2 only cites C_1, the algorithm prunes C_2 as the non-recombinant parental donor. When candidates cite each other reciprocally, `RhizAeon` computes physical transition contrast and home backbone disruption across the breakpoint boundaries. The contrast metrics measure differential divergence inside versus outside the candidate tract:

$$\Delta d_{\text{home}} = \max\left(0,\ d(C, H \mid \text{tract}) - d(C, H \mid \text{flank})\right), \tag{45}$$

$$\Delta d_{\text{donor}} = \max\left(0,\ d(C, D \mid \text{flank}) - d(C, D \mid \text{tract})\right). \tag{46}$$

A true recombinant child experiences substantial home backbone disruption ($\Delta d_{\text{home}} \gg 0$) by departing from its parental home basin during the conversion tract. In contrast, a stationary clonal parent remains within its home background throughout the alignment ($\Delta d_{\text{home}} \approx 0$). Comparing total transition contrast $\Delta_{\text{trans}} = \Delta d_{\text{home}} + \Delta d_{\text{donor}}$ and mosaic reconstruction residual $d(C, D \mid \text{tract}) + d(C, H \mid \text{flank})$ unambiguously orients the recombinant isolate and purges reciprocal parent false alarms.

Intra-Swarm Sister Isolation, Donor Clade Deduplication, and Sister Preservation.

In expanding epidemic swarms, multiple recombinant isolates descend from a common recombinant ancestor, differing only by minor intra-swarm drift ($d < 0.010$). Without explicit sister safeguards, three distinct failure modes corrupt swarm deconvolution: First, during parental channel resolution, co-recombinant sisters place immense continuous attention onto each other; `RhizAeon` strictly excludes contemporary channels within intra-swarm drift ($d(s_i, s_j) < 0.010$) from serving as prospective Home or Donor channels for each other, forcing attention to resolve against genuine parental clades. Second, in multi-way mosaics acquiring distinct cassettes from separate donor clades (e.g., P_2 and P_3), candidate donors from the primary clade (P_2) can monopolize the candidate pool; `RhizAeon` deduplicates candidate donors by phylogenetic distance ($d(d_1, d_2) < 0.010$), guaranteeing that secondary donor clades are not crowded out by redundant primary sisters. Third, in reticulation graph adjudication, reciprocal two-cycle pruning must not confuse parallel sister isolates with reciprocal parental illusions; `RhizAeon` explicitly protects co-recombinant sister events sharing identical Home and Donor assignments ($\text{Home}_1 = \text{Home}_2$ and $\text{Donor}_1 = \text{Donor}_2$) from mutual reflection pruning. All verified sister isolates are preserved and subsequently aggregated into the ancestral event roster $\mathcal{I} = \{s_1, \ldots, s_k\}$.

In surveillance cohorts containing multiple related recombinants, an isolate A may acquire a tract from an ancestral recombinant B before B further diverged or was sampled. Under these conditions, isolate A cites B as its donor or home. If B was itself introgressed from donor C, this dependency creates a transitive donor chain ($A \to B \to C$). `RhizAeon` iteratively traverses donor and home dependency chains across overlapping tracts sharing $\geq 50\%$ tract overlap. When isolate A cites B as donor, and B (or its co-recombinant cluster) cites external donor C with a compatible home lineage, the algorithm reassigns A donor to C, updating tract and flanking divergences $d(A, C \mid \text{tract})$ and $d(A, C \mid \text{flank})$. Symmetrically, if A cites B as home and B cites C as home with a compatible donor, A inherits home C. The chain traversal forbids cyclic attributions ($A \to B \to A$), resolving transitive co-recombinant cascades into primary evolutionary transfers.

When an ancestral recombination event precedes clonal expansion, multiple descendant isolates inherit the identical mosaic tract. Reporting each isolate as an independent recombination event inflates event counts and misrepresents epidemic transmission. `RhizAeon` clusters verified events into shared ancestral recombination events using an interval overlap criterion. Events sharing identical resolved Home and Donor lineages

whose breakpoint intervals overlap by $\geq 50\%$ of the shorter tract:

$$\frac{\min(u_2^{(1)}, u_2^{(2)}) - \max(u_1^{(1)}, u_1^{(2)}) + 1}{\min(u_2^{(1)} - u_1^{(1)} + 1,\, u_2^{(2)} - u_1^{(2)} + 1)} \geq 0.50, \tag{47}$$

are collapsed into a single ancestral event. The grouped event aggregates the complete roster of co-recombinant descendant isolates ($\mathcal{I} = \{s_1, \ldots, s_k\}$). The consensus breakpoint interval spans the outer envelope $[\min_k u_{1,k},\, \max_k u_{2,k}]$, while preserving the maximal statistical support across member isolates ($\min p_{\text{Fisher}}$, $\max Z_{\text{phys}}$, $\max S_{\text{informative}}$). Each event receives a persistent sequential identifier $(1, 2, \ldots)$, reporting both the count of unique evolutionary events and total recombinant isolates across the alignment inventory.

Outgroup-Invariant Statistical Due Diligence

In cohorts with $N \geq 4$ taxa, `RhizAeon` searches for an unrooted quartet outgroup O that maximizes divergence to candidate and home lineages ($d(O, H) + d(O, D)$), preferentially selecting non-candidate, non-rate-burst background leaves to prevent outgroup poisoning. If all external lineages have been admitted as candidates due to high sensitivity in low-divergence regimes ($d \leq 0.02$), the selector falls back to the most divergent available taxon outside $\{C, H, D\}$, ensuring quartet verification never starves. When an authentic external outgroup exists, the engine tests the unrooted Buneman four-point condition across tract and flank partitions:

$$S_1 = d(C, H) + d(D, O), \tag{48}$$

$$S_2 = d(C, D) + d(H, O), \tag{49}$$

$$S_3 = d(C, O) + d(H, D). \tag{50}$$

Under genuine recombination, the minimal parsimony split flips from S_1 (grouping C with H on flanks) to S_2 (grouping C with D in tract), with the topological difference exceeding the neutral binomial standard error ($z \geq z_{\text{crit}}$).

In two-clade epidemic expansions where all sampled sequences belong to parental clades or recombinant progeny, no authentic phylogenetic outgroup exists. Under this regime, `RhizAeon` falls back onto outgroup-invariant triplet criteria:

1. **Hypergeometric Contingency**: Fisher's exact test across informative sites within versus outside the tract, calibrated against the exact number of unordered candidate triplets:
$$p \leq p_{\text{crit}} = \frac{\alpha}{\binom{N}{3}} = \frac{6\alpha}{N(N-1)(N-2)}. \tag{51}$$
2. **Poisson Informative Floor**: A minimal count of distinguishing mutations donated by the minor parent ($s_{\text{donor}} \geq 3$).
3. **Directional Recoil Significance**: Standardized transition displacement along the parental dipole axis exceeding Gaussian extreme-value bounds ($\Delta_{\text{net}} \geq z_{\text{crit}} \cdot \text{SE}$).
4. **Private Autapomorphy Sieve**: Bounding private autapomorphic mutations along the candidate branch ($\rho_{\text{priv}} < 0.50$) to reject localized heterotachy spikes.

This invariant fallback guarantees rigorous false positive control without requiring external reference taxa or inducing false negatives in closed outbreak cohorts.

The Reticulation Fidelity Scoring Framework

A simulated ground-truth recombination event is formalized as an oriented tuple:

$$E = (C, H, D, \mathcal{I}), \tag{52}$$

where $C \in \{1, \ldots, N\}$ denotes the recombinant child isolate, $H \in \{1, \ldots, N\}$ denotes the major parental recipient lineage (home), $D \in \{1, \ldots, N\} \cup \{\text{Ghost}\}$ denotes the minor parental donor lineage, and $\mathcal{I} \subset [1, L]$ denotes the genomic interval of horizontal genetic exchange. An inferred event is denoted $\hat{E} = (\hat{C}, \hat{H}, \hat{D}, \hat{\mathcal{I}})$.

Let $\mathcal{S}_{\text{inf}}(H, D) \subset \{1, \ldots, L\}$ denote the set of informative polymorphisms distinguishing parental lineages H and D:

$$\mathcal{S}_{\text{inf}}(H, D) = \{s \in \{1, \ldots, L\} \mid H[s] \neq D[s]\}. \tag{53}$$

For true exchange interval $\mathcal{I}$ and inferred interval $\hat{\mathcal{I}}$, the sets of true and inferred transferred informative polymorphisms are:

$$\mathcal{S}_{\text{true}} = \{s \in \mathcal{S}_{\text{inf}}(H, D) \mid s \in \mathcal{I}\}, \qquad \mathcal{S}_{\text{inferred}} = \{s \in \mathcal{S}_{\text{inf}}(H, D) \mid s \in \hat{\mathcal{I}}\}. \tag{54}$$

The spatial accuracy score is formulated as the Jaccard similarity over informative sites:

$$J_{\text{SNP}}(\mathcal{I}, \hat{\mathcal{I}}) = \frac{|\mathcal{S}_{\text{true}} \cap \mathcal{S}_{\text{inferred}}|}{|\mathcal{S}_{\text{true}} \cup \mathcal{S}_{\text{inferred}}|}. \tag{55}$$

Alignments contain zero phylogenetic signal between adjacent informative sites. Consequently, the likelihood profile forms an exact mathematical plateau. Whenever predicted boundaries fall within the flanking uninformative plateaus of the true exchange ($E_{\text{SNP}} = 0$), the informative sets

coincide ($\mathcal{S}_{\text{inferred}} \equiv \mathcal{S}_{\text{true}}$), yielding $J_{\text{SNP}} = 1.00$. The metric naturally scales across evolutionary regimes, penalizing predictions that traverse contradictory mutations without imposing arbitrary nucleotide distance thresholds.

Lineage identification requires genealogical discipline. To evaluate parental attribution without penalizing close sister isolates or framing innocent taxa, we define genealogical kinship affinity. Let $d_{\text{tree}}(X, Y)$ denote the patristic distance between taxa X and Y on the reference phylogeny. The genealogical affinity kernel is defined as:

$$w(X, Y) = \exp\left(-\frac{d_{\text{tree}}(X, Y)}{\sigma_{\text{clade}}}\right), \tag{56}$$

where σ_{clade} denotes the characteristic divergence scale of the surrounding monophyletic sublineage. The directed triad attribution score is formulated as:

$$S_{\text{triad}}(C, H, D; \hat{C}, \hat{H}, \hat{D}) = w(C, \hat{C}) \times \left(\frac{w(H, \hat{H}) + w(D, \hat{D})}{2}\right). \tag{57}$$

The child affinity $w(C, \hat{C})$ operates as an outer multiplicative gate. If an algorithm misidentifies an unreticulated background isolate as the recombinant child, $w(C, \hat{C}) \approx 0$, collapsing the triad score. Symmetrically, if an algorithm inverts the reticulation direction by swapping child and donor ($\hat{C} \approx D$), the deep divergence between recombinant and donor lineages extinguishes $w(C, D)$, purging inverted calls without ad-hoc rules. When the true donor represents an unsampled ghost lineage ($D = \text{Ghost}$), identifying an unsampled donor awards full credit ($w(D, \hat{D}) = 1.00$ for $\hat{D} = \text{Ghost}$), while naming a sampled sister lineage receives partial credit proportional to its patristic distance to the introgression node.

For each pair of true event E_i and inferred event $\hat{E}_j$, the composite event affinity is the product of lineage fidelity and spatial accuracy:

$$s(E_i, \hat{E}_j) = S_{\text{triad}}(E_i, \hat{E}_j) \times J_{\text{SNP}}(E_i, \hat{E}_j). \tag{58}$$

To evaluate reticulations under strict cardinality discipline, we construct the pairwise cost matrix $C_{ij} = 1 - s(E_i, \hat{E}_j)$ for all $i \in \{1, \ldots, K\}$ and $j \in \{1, \ldots, \hat{K}\}$. We compute the optimal bijective bipartite matching $\mathcal{M}$ using the Hungarian algorithm (26). Every unmatched true event counts as a false negative, while every unmatched predicted event counts as an ungrounded false alarm.

Across the optimal matching $\mathcal{M}$, reticulation precision and recall are defined as:

$$\text{PPV}_{\mathcal{R}} = \frac{1}{\hat{K}} \sum_{(E_i, \hat{E}_j) \in \mathcal{M}} s(E_i, \hat{E}_j), \tag{59}$$

$$\text{TPR}_{\mathcal{R}} = \frac{1}{K} \sum_{(E_i, \hat{E}_j) \in \mathcal{M}} s(E_i, \hat{E}_j). \tag{60}$$

The harmonic mean of precision and recall defines the composite Reticulation Fidelity Index:

$$\mathcal{R}_{F_1} = \frac{2 \cdot \text{PPV}_{\mathcal{R}} \cdot \text{TPR}_{\mathcal{R}}}{\text{PPV}_{\mathcal{R}} + \text{TPR}_{\mathcal{R}}} = \frac{2 \sum_{(E_i, \hat{E}_j) \in \mathcal{M}} s(E_i, \hat{E}_j)}{K + \hat{K}}. \tag{61}$$

On non-recombinant clonal null alignments ($K = 0$), the index evaluates to $\mathcal{R}_{F_1} = 1.00$ if $\hat{K} = 0$, and $\mathcal{R}_{F_1} = 0.00$ if any false positive event is declared ($\hat{K} > 0$), with alignment-level false positive rate $\text{FPR} = \mathbb{I}(\hat{K} > 0)$. To summarize performance across all four biological axes, benchmarks report the multi-dimensional evaluation profile:

$$\mathbf{S}_{\mathcal{R}} = \left(\text{TPR}_{\mathcal{R}},\, \text{PPV}_{\mathcal{R}},\, \overline{S}_{\text{triad}},\, \overline{J}_{\text{SNP}}\right). \tag{62}$$

Synthetic Benchmark Protocols: Epidemic Expansions, Multi-Way Mosaics, Ghost Donors, and Micro-Conversions

To rigorously assess algorithmic fidelity under authentic evolutionary challenges without synthetic placeholders or parametric shortcuts, we implemented automated benchmark generators operating on raw sequence alignments under continuous-time Markov substitution dynamics. All evolutionary simulations were conducted under the HKY85 substitution model with transition/transversion bias $\kappa = 2.5$ and empirical base frequencies $\pi = [0.30, 0.20, 0.20, 0.30]$ using exact matrix exponentiation $P(t) = \exp(Qt)$.

Scenario 1.1: Epidemic Clonal Expansion Suite.

To assess event consolidation and parental attribution discipline under epidemic lineage growth, we simulated an ancestral recombinant node descending from two parental lineages separated by branch length $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ ($L = 5{,}000$ nt). The ancestral recombinant acquired a 1,500 nt cassette ($[1{,}500\text{–}3{,}000]$) from P_2 in a P_1 background. The recombinant node subsequently underwent epidemic transmission expansion, sprouting $K \in \{2, 5, 8, 10, 15\}$ contemporary descendant isolates, each accumulating independent transmission drift ($\delta = 0.002$). The alignment contained 40 strictly non-recombinant background parental isolates (20 in P_1 and 20 in P_2), yielding $N = 42$–55 taxa. Twenty independent replicates were generated for each of the 15 parameter conditions (300 alignments total).

Scenario 1.2: Multi-Way Mosaic Swarms Suite.

To evaluate algorithmic limits when the two-parent assumption is violated, we simulated 540 alignments ($L = 6,000$ nt; 20 replicates across 27 parameter conditions) across three distinct multi-parental topologies:

1. **Dual-Cassette Swarms**: A P_1 background acquiring two independent foreign cassettes from P_2 ($[1,200–2,400]$, length 1,201 nt) and P_3 ($[3,600–4,800]$, length 1,201 nt), leaving P_1 on flanks and interstitial spacer.
2. **Nested Russian Doll Swarms**: A P_1 background acquiring an outer introgression from P_2 ($[1,500–4,500]$, length 3,001 nt), within which an internal cassette ($[2,500–3,500]$, length 1,001 nt) is replaced by P_3, modeling secondary crossover within a recombinant lineage.
3. **Triple Crossover Cascades**: Sequential serial crossovers traversing the three parental lineages ($P_1[1–2,000] \rightarrow P_2[2,001–4,000] \rightarrow P_3[4,001–6,000]$).

Each topology was evaluated across divergence levels $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ and descendant swarm sizes $K \in \{1, 5, 10\}$ with 45 non-recombinant background taxa (15 per parental clade; $N = 46–55$ taxa). Performance was evaluated against 3SEQ using identical alignments, measuring background false positive rate ($\text{FPR}_{\text{parent}}$), precision, event cardinality ($\hat{K}_{\text{unique}}$), spatial Jaccard similarity, and execution runtime.

Scenario 1.3: Unsampled Ghost Lineage Introgression Suite.

To evaluate algorithmic behavior when introgressions originate from unsampled or extinct taxa, we simulated 540 alignments ($L = 5,000$ nt; 20 replicates across 27 parameter regimes) under three distinct donor availability regimes:

1. **Pure Ghost Donor**: The donor lineage P_2 is completely omitted from the alignment sample, leaving only P_1 background isolates and the recombinant progeny.
2. **Surrogate Sister Clade**: An unsampled donor lineage introgresses into P_1, while a related surrogate sister clade residing at an incremental divergence of $\Delta d_{\text{surr}} = 0.03$ is sampled in the alignment.
3. **Deep Reservoir Spillover**: The donor originates from a deeply divergent reservoir lineage separated by patristic divergence $d_{\text{res}} = 0.15$ from the recipient clade.

Each regime was crossed with parental divergences $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$ and recombinant swarm sizes $K \in \{2, 5, 10\}$ isolates with 30 non-recombinant background parental taxa (15 per clade; $N \in \{32, 35, 40\}$ taxa). Evaluated metrics scored ghost recall, parental attribution accuracy, background false positive rate, and spatial Jaccard similarity.

Scenario 1.4: Bacterial Micro-Conversion and Scale-Space Resolution Suite.

To systematically benchmark the resolution limits of continuous multiscale matched filtering against discrete triplet scanning on short non-reciprocal gene conversions, we simulated 900 alignments ($L = 5,000$ nt; 20 replicates across 45 parameter regimes) spanning five micro-conversion tract lengths: $L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nucleotides (1.0%, 2.0%, 4.0%, 8.0%, and 16.0% of chromosome length). The micro-conversion tract was centered at nucleotide 2,500 ($[2,500 - L_{\text{tract}}/2,\ 2,500 + L_{\text{tract}}/2 - 1]$), replacing recipient P_1 sequence with homologous donor P_2 sequence. The ancestral recombinant sprouted $K \in \{2, 5, 10\}$ contemporary descendant isolates accumulating independent transmission drift ($\delta = 0.002$), alongside 30 non-recombinant background parental taxa (15 in P_1, 15 in P_2; total $N \in \{32, 35, 40\}$). Each tract length was evaluated across three parental divergence tiers: $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$. Performance was evaluated under exact, un-faked scoring metrics: inferred event cardinality ($\hat{K}$), recombinant isolate recall and precision, background parental false positive rate ($\text{FPR}_{\text{parent}}$), spatial Jaccard overlap (J), and execution runtime in milliseconds.

References

1. Erin J. Adams and Peter Parham. Species-specific evolution of MHC class I genes in the higher primates. *Immunological Reviews*, 183(1):41–64, 2001.
2. Tommaso Alfonsi, Eleonora Cella, Giorgio Berno, Diana Fanti, Silvia Fabris, Marta Giovanetti, Massimo Ciccozzi, Alberto Riva, Gianguglielmo Zehender, and Allegra Via. Data-driven recombination detection in viral genomes. *Nature Communications*, 15:3110, 2024.
3. L. D. Barber, L. Percival, N. M. Valiante, L. Chen, C. Lee, J. E. Gumperz, J. H. Phillips, L. L. Lanier, J. C. Bigge, R. B. Parekh, and P. Parham. The inter-locus recombinant HLA-B*4601 has high selectivity in peptide binding and functions characteristic of HLA-C. *The Journal of Experimental Medicine*, 184(2):735–740, 1996.
4. Maciej F Boni, Philippe Lemey, Xiao-Wei Jiang, Tommy Tsan-Yuk Lam, Martha B Perry, Todd A Castoe, Andrew Rambaut, and David L Robertson. Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature Microbiology*, 5(11):1408–1417, 2020.
5. Maciej F Boni, David Posada, and Marcus W Feldman. An exact nonparametric method for inferring mosaic structure with application to genetics data and simulated data. *Genetics*, 176(2):1035–1047, 2007.
6. Trevor C Bruen, Hervé Philippe, and David Bryant. A simple and robust statistical test for detecting the presence of recombination. *Genetics*, 172(4):2665–2681, 2006.
7. Peter Buneman. A note on the metric properties of trees. *Journal of Combinatorial Theory, Series B*, 17(1):48–50, 1974.
8. Jean K Carr, Mika O Salminen, Jan Albert, Eric Sanders-Buell, Daniela Gotte, Deborah L Birx, and Francine E McCutchan. Full genome sequences of human immunodeficiency virus type 1 subtypes G and A/G intersubtype recombinants. *Virology*, 247(1):22–31,

1998.

9. Nicholas J Croucher, Simon R Harris, Christophe Fraser, Michael A Quail, John Burton, Mark van der Linden, Lesley McGee, Anne von Gottberg, Jae Hoon Song, Kwan Soo Ko, Bruno Pichon, Stephen Baker, Christopher M Parry, Lotte M Lambertsen, Dea Shahinas, Dylan R Pillai, Timothy J Mitchell, Gordon Dougan, Alexander Tomasz, Keith P Klugman, Julian Parkhill, William P Hanage, and Stephen D Bentley. Rapid pneumococcal evolution in response to clinical interventions. *Science*, 331(6016):430–434, 2011.
10. Nicholas J Croucher, Andrew J Page, Thomas R Connor, Aidan J Delaney, Jacqueline A Keane, Stephen D Bentley, Julian Parkhill, and Simon R Harris. Rapid phylogenetic analysis of large samples of recombinant bacterial whole genome sequences using Gubbins. *Nucleic Acids Research*, 43(3):e15, 2015.
11. Gilles Deleuze and Félix Guattari. *A Thousand Plateaus: Capitalism and Schizophrenia*. University of Minnesota Press, 1987.
12. Xavier Didelot and Daniel Falush. Inference of bacterial microevolution using multilocus sequence data. *Genetics*, 175(3):1251–1266, 2007.
13. Xavier Didelot and Martin C J Maiden. Impact of recombination on bacterial evolution. *Trends in Microbiology*, 18(7):315–322, 2010.
14. Xavier Didelot and Daniel J Wilson. ClonalFrameML: Efficient inference of recombination in whole bacterial genomes. *PLoS Computational Biology*, 11(2):e1004041, 2015.
15. Monroe D. Donsker. An invariance principle for certain probability limit theorems. *Memoirs of the American Mathematical Society*, 6:1–12, 1951.
16. Alyssa Lyn Fortier and Jonathan K. Pritchard. Ancient trans-species polymorphism at the Major Histocompatibility Complex in primates. *eLife*, 14:RP103547, 2025.
17. Ron Geller, Erin Adams, Lisbeth Guethlein, Ann-Margaret Little, Alejandro Madrigal, and Peter Parham. Linkage of Patr-AL to Patr-A and -B in the major histocompatibility complex of the common chimpanzee (*Pan troglodytes*). *Immunogenetics*, 54(6):441–444, 2002.
18. I Glais, M Tribodet, and C Kerlan. Genomic variability in Potato potyvirus Y (PVY): evidence that PVY(N)W and PVY(NTN) variants are single to multiple recombinants between PVY(O) and PVY(N) isolates. *Archives of Virology*, 147(2):363–378, 2002.
19. Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985.
20. J. C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3-4):325–338, 1966.
21. J. C. Gower. Adding a point to vector diagrams in multivariate analysis. *Biometrika*, 55(3):582–585, 1968.
22. Robert C Griffiths and Paul Marjoram. Ancestral inference from gene trees with recombination. *Theoretical Population Biology*, 50(3):341–377, 1996.
23. Richard R Hudson. Properties of a neutral allele model with intragenic recombination. *Genetical Research*, 41(3):261–269, 1983.
24. Melissa J Jansen van Rensburg, Duncan J Berger, Iman Yassine, David Shaw, Andy Fohrmann, James E Bray, Keith A Jolley, Martin C J Maiden, and Angela B Brueggemann. Development of the Pneumococcal Genome Library, a core genome multilocus sequence typing scheme, and a taxonomic life identification number barcoding system to investigate and define pneumococcal population structure. *Microbial Genomics*, 10(8):001280, 2024.
25. Sergei L Kosakovsky Pond, David Posada, Michael B Gravenor, Christopher H Woelk, and Simon D W Frost. Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution*, 23(10):1891–1901, 2006.
26. Harold W. Kuhn. The hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2):83–97, 1955.
27. H M Lam, O Ratmann, and M F Boni. Improved algorithmic complexity for the 3SEQ recombination detection algorithm. *Molecular Biology and Evolution*, 35(1):247–251, 2018.
28. Spyros Lytras, Joseph Hughes, Darren Martin, Phillip Swanepoel, Arné de Klerk, Rentia Lourens, Sergei L. Kosakovsky Pond, Wei Xia, Xiaowei Jiang, and David L. Robertson. Exploring the Natural Origins of SARS-CoV-2 in the Light of Recombination. *Genome Biology and Evolution*, 14(2):evac018, 2022.
29. Giuseppe Maccari, James Robinson, Keith Ballingall, Lisbeth A. Guethlein, Unni Grimholt, Jim Kaufman, Chak-Sum Ho, Natasja G. de Groot, Paul Flicek, Ronald E. Bontrop, John A. Hammond, and Steven G. E. Marsh. IPD-MHC 2.0: an improved inter-species database for the study of the major histocompatibility complex. *Nucleic Acids Research*, 45(D1):D860–D864, 2017.
30. Vladimir A. Marchenko and Leonid A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967.
31. Darren P Martin, Ben Murrell, Michael Golden, Arshan Khoosal, and Brejnev Muhire. RDP4: Detection and analysis of recombination patterns in virus genomes. *Virus Evolution*, 1(1):vev003, 2015.
32. Darren P Martin, Arvind Varsani, Philippe Roumagnac, Gonasagren Botha, Shailen Maslamoney, Travis Schwab, Zander Kelz, Sudhir Kumar, and Ben Murrell. RDP5: a computer program for analyzing recombination in, and removing signals of recombination from, nucleotide sequence datasets. *Virus Evolution*, 7(1):veaa087, 2021.
33. Gil A T McVean, Simon R Myers, Sarah Hunt, Panos Deloukas, David R Bentley, and Peter Donnelly. The fine-scale structure of recombination rate variation in the human genome. *Science*, 304(5670):581–584, 2004.
34. Bui Quang Minh, Heiko A Schmidt, Olga Chernomor, Dominik Schrempf, Michael D Woodhams, Arndt von Haeseler, and Robert Lanfear. IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution*, 37(5):1530–1534, 2020.
35. Peter Parham, Erin J. Adams, and Kelly L. Arnett. The origins of HLA-A,B,C polymorphism. *Immunological Reviews*, 143(1):141–180, 1995.

36. David Posada. Evaluation of methods for detecting recombination from DNA sequences: Empirical data. *Molecular Biology and Evolution*, 19(5):708–717, 2002.
37. David Posada and Keith A Crandall. Evaluation of methods for detecting recombination from DNA sequences: Computer simulations. *Proceedings of the National Academy of Sciences*, 98(24):13757–13762, 2001.
38. Loren H Rieseberg, Olivier Raymond, David M Rosenthal, Zhao Lai, Kevin Livingstone, Takuya Nakazato, Jennifer L Durphy, Andrea E Schwarzbach, Lisa A Donovan, and Christian Lexer. Major ecological transitions in wild sunflowers facilitated by hybridization. *Science*, 301(5637):1211–1216, 2003.
39. James Robinson, Dominic J. Barker, Xenia Georgiou, Michael A. Cooper, Paul Flicek, and Steven G. E. Marsh. IPD-IMGT/HLA database. *Nucleic Acids Research*, 48(D1):D948–D955, 2020.
40. Mikkel H Schierup and Jotun Hein. Consequences of recombination on traditional phylogenetic analysis. *Genetics*, 156(2):879–891, 2000.
41. Zbyněk Šidák. Rectangular confidence regions for the means of multivariate normal distributions. *Journal of the American Statistical Association*, 62(318):626–633, 1967.
42. Peter Simmonds and Jenny Welch. Frequency and dynamics of recombination within different species of human enteroviruses. *Journal of Virology*, 80(1):483–493, 2006.
43. Etienne Simon-Loriere and Edward C Holmes. Why do RNA viruses recombine? *Nature Reviews Microbiology*, 9(8):617–626, 2011.
44. Kyle Smith, Cheng Ye, and Yatish Turakhia. Tracking and curating putative SARS-CoV-2 recombinants with RIVET. *Bioinformatics*, 39(9):btad538, 2023.
45. Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. RoFormer: Enhanced transformer with Rotary Position Embedding. *Neurocomputing*, 568:127063, 2024.
46. Ling Su, Marcus Graf, Ying Zhang, Hagen von Briesen, Hui Xing, Josef Köstler, H Melzl, Hans Wolf, Yiming Shao, and Ralf Wagner. Characterization of a virtually full-length human immunodeficiency virus type 1 genome of a prevalent intersubtype (C/B′) recombinant strain in China. *Journal of Virology*, 74(23):11367–11376, 2000.
47. Koichiro Tamura and Masatoshi Nei. Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Molecular Biology and Evolution*, 10(3):512–526, 1993.
48. Tomokazu Tamura, Junpei Ito, Keiya Uriu, Jiri Zahradnik, Isaac Kida, Yuki Anraku, Hesham Nasser, Mai Toyoda, Keita Mizuma, Ryo Shimizu, et al. Virological characteristics of the SARS-CoV-2 XBB variant derived from recombination of two Omicron subvariants. *The Lancet Microbe*, 4(7):e529–e539, 2023.
49. Sarah Temmam, Khamsing Vongphayloth, Eduard Baquero, Sandie Munier, Massimiliano Bonomi, Béatrice Regnault, Bounsavane Douangboubpha, Yasaman Karami, Delphine Chrétien, Daosavanh Sanamxay, Vilakhan Xayaphet, Phetphoumin Paphaphanh, Vincent Lacoste, Somphavanh Somlor, Khaithong Lakeomany, Nothasin Phommavanh, Philippe Pérot, Océane Dehan, Faustine Amara, Flora Donati, Thomas Bigot, Michael Nilges, Félix A. Rey, Sylvie van der Werf, Paul T. Brey, and Marc Eloit. Bat coronaviruses related to SARS-CoV-2 and infectious for human cells. *Nature*, 604(7905):330–336, 2022.
50. Yatish Turakhia, Bryan Thornlow, Angie S Hinrichs, Nicola De Maio, Landen Gozashti, Robert Lanfear, David Haussler, and Russell Corbett-Detig. Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature*, 609(7929):994–997, 2022.
51. Michael Worobey, Andrew Rambaut, Oliver G. Pybus, and David L. Robertson. Questioning the evidence for genetic recombination in the 1918 "spanish flu" virus. *Science*, 296(5566):211, 2002.